EXERCISES COURSE

LING WANG

Only For Teaching! Continues Updating!
Personal website: https://lwmath.github.io

Contents

1.	Week 3 (9.19)	1
2.	Week 4 (9.26)	5
3.	Week 6 (10.10)	13
4.	Week 7 (10.17)	23
5.	Week 8 (10.24)	37
6.	Week 9 (10.31)	48
7.	Week 10 (11.7)	52
8.	Week 11 (11.14)	62
9.	Week 12 (11.21)	70
10.	Week 13 (11.28)	80
11.	Week 14 (12.5)	87
12.	Week 15 (12.12)	99

1. Week 3 (9.19)

Problem 1.1 (1.3)**.** *For* $\forall\, n \in \mathbb{N}$*, let* $A_n = \left[-1 + \frac{1}{2n}, 1 - \frac{1}{n}\right]$*, prove that* $\bigcup\limits_{n=1}^{\infty} A_n = (-1, 1)$*.*

Proof. Firstly, by definition we have that

$$\forall\, x \in \bigcup_{n=1}^{\infty} A_n, \quad \exists\, n_0 \in \mathbb{N}, \quad s.t.\ x \in A_{n_0} \subset (-1, 1).$$

Hence

$$x \in (-1, 1),$$

i.e.

$$\bigcup_{n=1}^{\infty} A_n \subset (-1, 1). \tag{1.1}$$

Date: November 30, 2022.

On the other hand, $\forall\, x \in (-1,1)$, set

$$N = \left[\max\left\{\frac{1}{2(x+1)}, \frac{1}{1-x}\right\}\right] + 1,$$

then there is $x \in A_N$, thus $x \in \bigcup\limits_{n=1}^{\infty} A_n$, which means

$$(-1,1) \subset \bigcup_{n=1}^{\infty} A_n. \tag{1.2}$$

Combining (1.1) and (1.2) yields

$$\bigcup_{n=1}^{\infty} A_n = (-1,1).$$

□

Problem 1.2 (1.5)**.** *Prove that* $A = \left\{n \sin \frac{n\pi}{2}; n \in \mathbb{Z}\right\}$ *is unbounded.*

Proof. **Definition:** A set X is unbounded if and only if $\forall\, M > 0$, there exists a $x \in X$, such that $|x| > M$. Then $\forall\, M > 0$, let

$$N = 2([M]+1)+1,$$

hence

$$\left|N \sin \frac{N\pi}{2}\right| = 2([M]+1)+1 > M,$$

which implies that A is unbounded. □

Problem 1.3 (1.23)**.** *Suppose that $f(x)$ is defined on E, where $|E| \geq 3$. Prove that $f(x)$ is strictly monotonic on E iff for $\forall\, x_1, x_2, x_3 \in E$, if $x_1 < x_2 < x_3$, then there must be*

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) > 0. \tag{1.3}$$

Proof. " only if ": Obviously.

" if ": Suppose that $f(x)$ is not a strictly monotonic function, hence there exist x_1, x_2, x_3, $x_4 \in E$ with $x_1 < x_2$ and $x_3 < x_4$ such that

$$f(x_1) \leq f(x_2) \quad \text{and} \quad f(x_3) \geq f(x_4).$$

If $x_1 < x_2 < x_3 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) \quad \text{and} \quad (f(x_2) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

If $x_1 < x_3 < x_2 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_4)) \quad \text{and} \quad (f(x_1) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

Similar for other cases. Then we obtain contradictions for all cases, wihch means that $f(x)$ is strictly monotonic on E. □

Problem 1.4 (1.25)**.** *Prove that* $\sin(x^2+x)$ *is not a periodic function.*

Proof. Suppose $T>0$ is the period of $f(x) := \sin(x^2+x)$, we have that

$$f(T) = f(-T) = f(0) = 0.$$

Hence there are $k_1, k_2 \in \mathbb{Z}$ such that

$$T^2 + T = k_1\pi \quad \text{and} \quad T^2 - T = k_2\pi,$$

which yields

$$T = \frac{k_1 - k_2}{2}\pi.$$

Then

$$\frac{(k_1-k_2)^2}{4}\pi + \frac{k_1-k_2}{2} = k_1,$$

i.e.

$$\pi = \frac{2(k_1+k_2)}{(k_1-k_2)^2} \in \mathbb{Q},$$

contradiction. □

Problem 1.5 (1.26)**.** *Suppose that* $f(x)$ *is defined on* $(0,+\infty)$*,* $x_1, x_2 > 0$*. Prove*

(1) *If* $\frac{f(x)}{x}$ *is decreasing, then* $f(x_1+x_2) \le f(x_1) + f(x_2)$*;*
(2) *If* $\frac{f(x)}{x}$ *is increasing, then* $f(x_1+x_2) \ge f(x_1) + f(x_2)$*.*

Proof. (1) Since $\frac{f(x)}{x}$ is decreasing, then $f(x_1+x_2) \le f(x_1) + f(x_2)$, we have

$$\frac{f(x_1+x_2)}{x_1+x_2} \le \frac{f(x_1)}{x_1} \quad \text{and} \quad \frac{f(x_1+x_2)}{x_1+x_2} \le \frac{f(x_2)}{x_2}.$$

Hence

$$f(x_1+x_2) = \frac{x_1}{x_1+x_2}f(x_1+x_2) + \frac{x_2}{x_1+x_2}f(x_1+x_2) \le f(x_1) + f(x_2).$$

Similar for (2). □

Problem 1.6 (1.27)**.** *Suppose* $f(x)$ *is defined on* $(-\infty,+\infty)$*, and* $f(f(x)) \equiv x$*.*

(1) *Is* $f(x)$ *unique? If not, please give an example;*
(2) *If* $f(x)$ *is strictly increasing, is it unique? Why?*

Solution. (1) $f(x)$ is not unique. For example $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}, x \neq 0; f_3(0) = 0$.

(2) We prove that $f(x) = x$. Suppose not, if there is a $x_0 \in \mathbb{R}$, such that $f(x_0) \neq x_0$. Without loss of generality, we assume that $f(x_0) > x_0$. Then $f(f(x_0)) > f(x_0)$ by f is strictly increasing. Hence

$$x_0 = f(f(x_0)) > f(x_0) > x_0,$$

contradiction. □

Problem 1.7. *Prove that* $f(x) = \sin x + \sin\sqrt{2}x$*,* $x \in \mathbb{R}$ *is not a periodic function.*

Proof. Note that

$$f''(x) = -\sin x - 2\sin\sqrt{2}x.$$

Hence we have

$$\begin{aligned} f(x) + f''(x) &= -\sin\sqrt{2}x, \\ 2f(x) + f''(x) &= \sin x. \end{aligned}$$

If $T > 0$ is the period of $f(x)$, then T is also the period of $f(x)+f''(x)$ and $2f(x)+f''(x)$. Hence

$$T = \frac{2k\pi}{\sqrt{2}} = 2m\pi, \quad \text{for some } k, m \in \mathbb{N},$$

which implies

$$\sqrt{2} = \frac{k}{m} \in \mathbb{Q},$$

contradiction. □

Problem 1.8. *Suppose $f(x)$ is an increasing function defined on closed interval $[a, b]$, and f satisfies $f(a) \geq a$, $f(b) \leq b$. Prove that there exists $x_0 \in [a, b]$, such that $f(x_0) = x_0$.*

Proof. Set

$$A = \{x \in [a, b] : f(x) \geq x\}.$$

Obviously, A is not empty since $f(a) \geq a$, and b is an upper bound of A. Hence $\sup A$ exists. We denote $x_0 := \sup A$. For $\forall\, x \in A$, there is $x \leq x_0$. Since f is increasing, we have $x \leq f(x) \leq f(x_0)$, i.e. $f(x_0)$ is an upper bound of A, thus $x_0 \leq f(x_0)$. If $x_0 < f(x_0)$, then $f(x_0) \leq f(f(x_0))$, which means $f(x_0) \in A$, contradiction. Hence $f(x_0) = x_0$. □

Problem 1.9. *Is there a function, whose period are all rational numbers but none of irrational number?*

Solution. Dirichlet function, i.e.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R}\backslash\mathbb{Q}. \end{cases}$$

□

2. Week 4 (9.26)

Problem 2.1 (2.3(5))**.** *Using definition to prove*

$$\lim_{n\to\infty} n^3 q^n = 0\,(|q| < 1).$$

Proof. It suffices to prove $\lim\limits_{n\to\infty} n^3|q|^n = 0$, and it is obviously valid when $|q| = 0$, hence we assume that $|q| \neq 0$ in the following. Let

$$\frac{1}{|q|} = 1 + \alpha.$$

By binomial theorem, we have

$$\frac{1}{|q|^n} = (1+\alpha)^n = \sum_{k=0}^{n} C_n^k \alpha^k \geq \frac{n(n-1)(n-2)(n-3)}{4!}\alpha^4$$

provided with $n \geq 4$. Hence for $n \geq 4$,

$$n^3|q|^n \leq \frac{24n^2}{(n-1)(n-2)(n-3)\alpha^4} < \frac{72}{\alpha^4}\frac{1}{n-3}.$$

Then $\forall\, \varepsilon > 0$, choosing $N = \left[\frac{72}{\alpha^4\varepsilon}\right] + 4$, for any $n > N$, there is $n^3|q|^n < \varepsilon$, i.e. $\lim\limits_{n\to\infty} n^3|q|^n = 0$. □

Problem 2.2 (2.4)**.** *Suppose for all $n \in \mathbb{N}$, there is $x_n \leq a \leq y_n$ and $\lim\limits_{n\to\infty}(y_n - x_n) = 0$. Prove*

$$\lim_{n\to\infty} x_n = a = \lim_{n\to\infty} y_n.$$

Proof. Since $x_n \leq a \leq y_n$, we know that

$$0 \leq a - x_n \leq y_n - x_n.$$

By Sandwich Theorem, we have

$$\lim_{n\to\infty} x_n = a.$$

Similar for y_n. □

Problem 2.3. *Suppose $\{a_n\}$ is monotonic increasing, $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$. Prove that $\lim\limits_{n\to\infty} a_n$, $\lim\limits_{n\to\infty} b_n$ exist, and they are equal.*

Proof. Assume $\lim\limits_{n\to\infty} a_n$ doesn't exist, since $\{a_n\}$ is monotonic increasing, we know there is $\lim\limits_{n\to\infty} a_n = +\infty$. By $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$, there must be $\lim\limits_{n\to\infty} b_n = -\infty$. Hence, for $\forall\, M > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N$, there are

$$a_n > M \quad \text{and} \quad b_n < -M.$$

Then

$$b_n - a_n < -2M,$$

which contradicts with $\lim_{n\to\infty}(b_n - a_n) = 0$. Thus, $\lim_{n\to\infty} a_n$ exists. It's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. $\square$

Problem 2.4 (2.6(3))**.** *$\{F_n\}$ is the Fibonacci sequence, defined by*

$$F_0 = F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}.$$

Prove $\lim_{n\to\infty} F_n = +\infty$ by definition.

Proof. **First way:** We can calculate the general terms of F_n by 'eigenvalue method' (See Problem 2.10 for details). Then we know

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right].$$

Since $\frac{1+\sqrt{5}}{2} > 1$ and $\frac{\sqrt{5}-1}{2} < 1$, we have $\lim_{n\to\infty} F_n = +\infty$.

Second way: We can prove by induction that $F_n \geq n$ for $n \geq 1$. Since $\lim_{n\to\infty} n = +\infty$, we know $\lim_{n\to\infty} F_n = +\infty$. $\square$

Problem 2.5 (2.10(6))**.** *Calculate*

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}).$$

Solution.

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}) = \lim_{n\to\infty} \frac{\sqrt[3]{n}}{(\sqrt[3]{n+1})^2 + \sqrt[3]{n+1}\sqrt[3]{n} + (\sqrt[3]{n})^2} = 0.$$

$\square$

Problem 2.6 (2.14)**.** *Calculate $\lim_{n\to\infty} x_n$.*

(1) $x_n = \dfrac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$;

(2) $x_n = \displaystyle\sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}}$;

(3) $x_n = \sqrt[n]{n\ln n}$.

Solution. (1)Note that

$$\frac{q}{p} < \frac{q+1}{p+1}, \quad \text{for } 0 < q < p.$$

Then

$$x_n = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$$

$$
\begin{aligned}
&= \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n} \\
&< \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{2n-2}{2n-1}\cdot\frac{2n}{2n+1} \\
&= \frac{2}{1}\cdot\frac{4}{3}\cdot\frac{6}{5}\cdots\frac{2n}{2n-1}\cdot\frac{1}{2n+1} \\
&= \frac{1}{x_n}\frac{1}{2n+1},
\end{aligned}
$$

which gives us

$$x_n < \frac{1}{\sqrt{2n+1}}.$$

Hence $\lim_{n\to\infty} x_n = 0$.

(2) Note that $(n+1)^2 - n^2 + 1 = 2(n+1)$, we have

$$2 = \frac{2(n+1)}{n+1} \le \sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}} \le \frac{2(n+1)}{n} \to 2, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 2$.

(3) When $n > 3$, there is $n < n\ln n < n^2$. Then

$$1 \leftarrow \sqrt[n]{n} < \sqrt[n]{n\ln n} < \sqrt[n]{n^2} \to 1, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 1$. □

Problem 2.7 (2.17)**.** *Sequence $\{q_n\}$ satisfies*

$$0 < q_n < 1, \quad (1-q_n)q_{n+1} > \frac{1}{4}, \quad \forall\, n \in \mathbb{N}. \tag{2.1}$$

Prove that $\{q_n\}$ is monotonic increasing and $\lim_{n\to\infty} q_n = \frac{1}{2}$.

Proof. By (2.1) and the mean value inequality, we have

$$\frac{q_{n+1}}{q_n} > \frac{1}{4q_n(1-q_n)} \ge \frac{1}{4\left(\frac{q_n+1-q_n}{2}\right)^2} = 1,$$

i.e. $q_{n+1} > q_n$. Hence $\{q_n\}$ is monotonic increasing. Then by the monotone bounded convergence theorem, we know $\lim_{n\to\infty} q_n$ exists. Denote $q := \lim_{n\to\infty} q_n$, by (2.1),

$$(1-q)q \ge \frac{1}{4},$$

i.e.

$$\left(q - \frac{1}{2}\right)^2 \le 0,$$

which means $q = \frac{1}{2}$. □

Problem 2.8 (2.19)**.** *Suppose that* $0 < a_1 < b_1$*, let*

$$a_{n+1} = \sqrt{a_n b_n}, \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*.*

Proof. By the mean value inequality, we have

$$a_{n+1} = \sqrt{a_n b_n} \le \frac{1}{2}(a_n + b_n) = b_{n+1} \quad \text{for } n \ge 1.$$

Hence

$$a_{n+1} = \sqrt{a_n b_n} \ge a_n \quad \text{and} \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \le b_n \quad \text{for } n \ge 2.$$

Then

$$a_2 \le \cdots \le a_n \le b_n \le \cdots \le b_2 \quad \text{for all } n \ge 2.$$

By the monotone bounded convergence theorem, we know $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, and it's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. □

Problem 2.9. *Suppose that* $a_1 > b_1 > 0$*, let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{2a_n b_n}{a_n + b_n} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$*.*

Proof. The existence is similar to Problem 2.8. Thus, we only need to prove $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$, and if we notice that

$$a_{n+1} b_{n+1} = a_n b_n = \cdots = a_1 b_1,$$

and

$$a_{n+1} \ge \sqrt{a_n b_n} \ge b_{n+1},$$

it's easy to obtain the conclusion. □

Problem 2.10. *Suppose that* $a_1 = \alpha$*,* $b_1 = \beta$*. Let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{a_{n+1} + b_n}{2} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist,* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*, and find the limitation.*

Proof. By $a_{n+1} = (a_n + b_n)/2$, we have $b_n = 2a_{n+1} - a_n$. Hence

$$2a_{n+2} - a_{n+1} = \frac{a_{n+1} + 2a_{n+1} - a_n}{2} = \frac{3}{2}a_{n+1} - \frac{1}{2}a_n,$$

which is

$$a_{n+2} = \frac{5}{4}a_{n+1} - \frac{1}{4}a_n. \tag{2.2}$$

Then the characteristic equation of (2.2) is

$$x^2 - \frac{5}{4}x + \frac{1}{4} = 0.$$

We find that $x = 1$ and $x = \frac{1}{4}$ are the solution. Hence the general form of a_n is

$$a_n = A + B\left(\frac{1}{4}\right)^n.$$

By $a_1 = \alpha$, $a_2 = (a_1 + b_1)/2 = (\alpha + \beta)/2$, we have

$$a_n = \frac{1}{3}\alpha + \frac{2}{3}\beta + \frac{2}{3}(\alpha - \beta)\left(\frac{1}{4}\right)^{n-1}.$$

Hence $\lim\limits_{n\to\infty} a_n$ and $\lim\limits_{n\to\infty} b_n$ exist, and $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \frac{1}{3}\alpha + \frac{2}{3}\beta$. □

Problem 2.11 (2.20)**.** *Calculate following limitations.*

(2) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n$;

(3) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2}$.

Solution. (2)We first note that

$$\left(1 + \frac{1}{n^2}\right)^{n^2} < e.$$

(This can be proved by definition of e.) Then we have

$$1 < \left(1 + \frac{1}{n^2}\right)^n < e^{\frac{1}{n}} \to 1 \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem

$$\lim_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n = 1.$$

(3)**First way:** By binomial theorem, we know

$$\left(1 + \frac{1}{n}\right)^{n^2} = 1 + n^2\frac{1}{n} + \cdots \geq n.$$

Hence

$$\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2} = +\infty.$$

Second way: By the definition of e, we know there exists a $N \in \mathbb{N}$, such that for $n > N$, there is

$$\frac{e}{2} < \left(1 + \frac{1}{n}\right)^n < e.$$

(Via choosing $\varepsilon = \frac{e}{2}$.) Hence we have

$$\left(1+\frac{1}{n}\right)^{n^2} > \left(\frac{e}{2}\right)^n \to +\infty \quad \text{as } n \to \infty,$$

i.e.

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = +\infty.$$

□

Problem 2.12. *Using* $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$ *to prove* $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e$.

Proof. By binomial theorem, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^n &= \sum_{k=0}^{n} C_n^k \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \\
&\le \sum_{k=0}^{n} \frac{1}{k!}.
\end{aligned}$$

From the above calculation, we know for $\forall\, m > n$, there is

$$\left(1+\frac{1}{m}\right)^m \ge \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{m}\right)\cdots\left(1-\frac{k-1}{m}\right).$$

Let $m \to +\infty$, there is

$$e \ge \sum_{k=0}^{n} \frac{1}{k!} \ge \left(1+\frac{1}{n}\right)^n.$$

Hence

$$\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e.$$

□

Problem 2.13. *Calculate the limitation of*

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right).$$

Solution.

$$\lim_{n\to\infty}\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\lim_{n\to\infty}\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdots\frac{n-1}{n}\cdot\frac{n+1}{n}=\frac{1}{2}.$$

□

Problem 2.14. *Suppose* $\lim_{n\to\infty} a_n = a$. *Prove*

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=a,$$

where $p_k>0$ *and* $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$.

Proof. Without loss of generality, we can assume $a=0$. (Otherwise, we can consider a_n-a instead.) Since $\lim_{n\to\infty} a_n=0$, we know for $\forall\,\varepsilon>0$, there exists $N_1\in\mathbb{N}$, such that for all $n>N_1$, there is $|a_n|<\varepsilon/2$. Then

$$\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|\le\frac{p_1+\cdots+p_{n-N_1}}{p_1+p_2+\cdots+p_n}\cdot\frac{\varepsilon}{2}<\frac{\varepsilon}{2}.$$

Since a_n is convergence, we know $\{a_n\}$ is bounded. We assume $|a_n|\le M$ for some $M>0$. By $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$, we have

$$\begin{aligned}\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left(\frac{p_{n-N_1+1}+\cdots+p_n}{p_1+p_2+\cdots+p_n}\right)M\\&<M\sum_{k=1}^{N_1}\frac{p_{n-N_1+k}}{p_1+p_2+\cdots+p_{n-N_1+k}}\\&\to 0\quad\text{as } n\to\infty.\end{aligned}$$

Hence there exists a $N_2\in\mathbb{N}$ such that for all $n>N_2$, there is

$$\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|<\frac{\varepsilon}{2}.$$

Then, set $N=\max\{N_1,N_2\}$, we know for all $n>N$, there is

$$\begin{aligned}\left|\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|+\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|\\&<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon,\end{aligned}$$

i.e.

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=0.$$

□

Problem 2.15. *Define sequence* $\{a_n\}$ *by* $a_{n+1}=2a_n-a_n^2$, *where* a_0 *is given. Discuss the convergence and divergence of* $\{a_n\}$ *regards the choice of* a_0.

Solution. Firstly, we have

$$1 - a_{n+1} = 1 - 2a_n + a_n^2 = (1 - a_n)^2.$$

By induction, we have

$$1 - a_{n+1} = (1 - a_n)^2 = \cdots = (1 - a_0)^{2^{n+1}},$$

i.e.

$$a_n = 1 - (1 - a_0)^{2^n} \quad \text{for } n \geq 1.$$

Hence,

When $|1 - a_0| < 1$, i.e. $0 < a_0 < 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 1$.

When $|1 - a_0| = 1$, i.e. $a_0 = 0$ or $a_0 = 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 0$.

When $|1 - a_0| > 1$, i.e. $a_0 < 0$ or $a_0 > 2$, $\{a_n\}$ is divergent, $\lim\limits_{n\to\infty} a_n = -\infty$. $\square$

3. Week 6 (10.10)

Problem 3.1 (2.21)**.** *Suppose that* $\{b_n\}$ *is strictly increasing and* $\lim_{n\to\infty} b_n = +\infty$*. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.1}$$

where A *is finite or* $\pm\infty$*. Then*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We first prove the case for A is finite. By (3.1), we know $\forall\, \varepsilon > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left| \frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A \right| < \varepsilon.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n-1}) < a_n - a_{n-1} < (A + \varepsilon)(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_{N_1}) < a_n - a_{N_1} < (A + \varepsilon)(b_n - b_{N_1}),$$

i.e.

$$\left| \frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A \right| < \varepsilon.$$

Note the identity

$$\frac{a_n}{b_n} - A = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A\right) + \frac{a_{N_1} - Ab_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$0 < 1 - \frac{b_{N_1}}{b_n} < 2 \quad \text{and} \quad \left| \frac{a_{N_1} - Ab_{N_1}}{b_n} \right| < \varepsilon.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\left| \frac{a_n}{b_n} - A \right| < 3\varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Next, we prove the case for $A = +\infty$. By (3.1), we know $\forall\, M > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\frac{a_n - a_{n-1}}{b_n - b_{n-1}} > 3M.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$a_n - a_{n-1} > 3M(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$a_n - a_{N_1} > 3M(b_n - b_{N_1}),$$

i.e.

$$\frac{a_n - a_{N_1}}{b_n - b_{N_1}} > 3M.$$

Note the identity

$$\frac{a_n}{b_n} = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}}\right) + \frac{a_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$\frac{1}{2} < 1 - \frac{b_{N_1}}{b_n} \quad \text{and} \quad \left|\frac{a_{N_1}}{b_n}\right| < \frac{1}{2}M.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\frac{a_n}{b_n} > \frac{3}{2}M - \frac{1}{2}M = M,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = +\infty.$$

Similar for $A = -\infty$. □

Problem 3.2 ($\frac{0}{0}$ type of Stolz theorem)**.** *Suppose that* $\lim_{n\to\infty} a_n = 0$, $\lim_{n\to\infty} b_n = 0$, *and* $\{b_n\}$ *is strictly decreasing. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.2}$$

where A is finite or $\pm\infty$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We only prove the case for A is finite. By (3.2), we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left|\frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A\right| < \varepsilon.$$

Since $b_n > b_{n+1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n+1}) < a_n - a_{n+1} < (A + \varepsilon)(b_n - b_{n+1}).$$

For any $m > n$, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_m) < a_n - a_m < (A + \varepsilon)(b_n - b_m),$$

i.e.

$$\left|\frac{a_n - a_m}{b_n - b_m} - A\right| < \varepsilon.$$

Let $m \to \infty$, and combining $\lim\limits_{n\to\infty} a_n = 0$, $\lim\limits_{n\to\infty} b_n = 0$, we know for $\forall\, n > N$, there is

$$\left|\frac{a_n}{b_n} - A\right| \le \varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

□

Remark 3.3. *If $A = \infty$ in Problem 3.1 and Problem 3.2, the conclusion is not correct. For example $a_n = (-1)^n n$, $b_n = n$ in Problem 3.1 and $a_n = (-1)^n \frac{1}{n}$, $b_n = \frac{1}{n}$ Problem 3.2.*

Problem 3.4. *Using Stolz theorem to prove*

(1) *If $\lim\limits_{n\to\infty} a_n = a$, then $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k$ exists and $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$.*

(2) *If $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$, and $\lim\limits_{n\to\infty} n(a_n - a_{n-1}) = 0$, then $\lim\limits_{n\to\infty} a_n$ exists and $\lim\limits_{n\to\infty} a_n = a$.*

Proof. (1) By Stolz theorem,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{\sum\limits_{k=1}^{n} a_k - \sum\limits_{k=1}^{n-1} a_k}{n-(n-1)} = \lim_{n\to\infty} a_n = a.$$

(2) Assume $a_0 = 0$. Let $A_n := a_n - a_{n-1}$, then $a_n = \sum\limits_{k=1}^{n} A_k$. The conditions become

$$a = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n},$$

and

$$0 = \lim_{n\to\infty} n(a_n - a_{n-1}) = \lim_{n\to\infty} nA_n.$$

Hence

$$\begin{aligned}
\lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{na_n}{n} = \lim_{n\to\infty} \frac{n\sum\limits_{k=1}^{n} A_k}{n} \\
&= \lim_{n\to\infty} \frac{nA_1 + nA_2 + \cdots + nA_n}{n} \\
&= \lim_{n\to\infty} \left(\frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \frac{A_2 + \cdots + (n-1)A_n}{n}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{(n-1)A_n}{n-(n-1)} \qquad \text{(Stolz theorem)} \\
&= a + \lim_{n\to\infty} \frac{n-1}{n} \cdot \lim_{n\to\infty} nA_n \\
&= a.
\end{aligned}
$$

□

Problem 3.5. *Suppose that* $x_{n+1} = x_n(1-x_n)$, $n = 1, 2, \cdots$, $0 < x_1 < 1$. *Prove that* $\lim_{n\to\infty} x_n = 0$ *and* $\lim_{n\to\infty} nx_n = 1$.

Proof. Firstly. note that

$$x_{n+1} - x_n = -x_n^2 \leq 0.$$

Then $\{x_n\}$ is monotonic decreasing. Since $0 < x_1 < 1$, we can prove $0 < x_n < 1$ by induction. Hence by monotone bounded convergence theorem, we know $\lim_{n\to\infty} x_n$ exists and $\lim_{n\to\infty} x_n = 0$. And by Stolz theorem, we have

$$
\begin{aligned}
\lim_{n\to\infty} nx_n &= \lim_{n\to\infty} \frac{n}{\frac{1}{x_n}} = \lim_{n\to\infty} \frac{n-(n-1)}{\frac{1}{x_n} - \frac{1}{x_{n-1}}} \\
&= \lim_{n\to\infty} \frac{x_{n+1}x_n}{x_n - x_{n+1}} = \lim_{n\to\infty} \frac{x_{n+1}}{x_n} \\
&= \lim_{n\to\infty} (1 - x_n) = 1.
\end{aligned}
$$

□

Remark 3.6. *If we replace* $x_{n+1} = x_n(1-x_n)$ *by* $x_{n+1} = \ln(1+x_n)$, *we can have a similar conclusion.*

Problem 3.7 (2.23)**.** *Suppose* f *is defined on* (a,b), *and for* $\forall\, \xi \in (a,b)$, *there exists a* $\delta > 0$, *such that for* $x \in (\xi - \delta, \xi + \delta) \cap (a,b)$,

(1) *If* $x < \xi$, *there is* $f(x) < f(\xi)$;
(2) *If* $x > \xi$, *there is* $f(x) > f(\xi)$.

Prove that f *is strictly increasing in* (a,b).

Proof. For $\forall\, x_1, x_2 \in (a,b)$ with $x_1 < x_2$, we want to show $f(x_1) < f(x_2)$. By assumption, $\forall\, \xi \in [x_1, x_2]$, there exists a $\delta = \delta(\xi)$ (If necessary, we can shrink δ so that $U(\xi,\delta) \subset (a,b)$), such that for $x \in U(\xi,\delta)$,

(1) If $x < \xi$, there is $f(x) < f(\xi)$;
(2) If $x > \xi$, there is $f(x) > f(\xi)$.

Then we know

$$[x_1, x_2] \subset \bigcup_{\xi \in [x_1, x_2]} U(\xi, \delta(\xi))$$

is an open covering. By Heine–Borel theorem, there exists a finite subcovering, i.e. there are $\xi_1, \cdots, \xi_n \in [x_1, x_2]$ and $\delta_1, \cdots, \delta_n > 0$, such that

$$[x_1, x_2] \subset \bigcup_{i=1}^{n} U(\xi_i, \delta_i).$$

Without loss of generality, we assume that

$$\xi_1 < \xi_2 < \cdots < \xi_n \quad \text{and} \quad U(\xi_i, \delta_i) \cap U(\xi_{i+1}, \delta_{i+1}) \neq \emptyset, \ i = 1, 2, \cdots, n-1.$$

Hence, we have $f(x_1) \leq f(\xi_1) < f(\xi_2) < \cdots < f(\xi_n) \leq f(x_2)$, i.e. $f(x_1) < f(x_2)$ valids for any $x_1 < x_2$. Thus, f is strictly increasing in (a, b). □

Problem 3.8 (2.25)**.** *Using supremum and infimum principle to prove accumulation point principle.*

Proof. **First way:** Suppose S is a bounded set consists of an infinite number of elements. By supremum and infimum principle, we know that $\sup S$ and $\inf S$ exist. If one of them is not a isolated point of S, it's obviously a accumulation point. Now, we assume none of them is the accumulation point of S. Set

$$E := \{x \in \mathbb{R} \,|\text{There are only a finite number of elments in } S \text{ that are less than } x\}.$$

Then E is nonempty and has an upper bound. Let $\eta := \sup E$, we prove η is a accumulation point of S. Indeed, by the construction of E, we know that $\forall\, \varepsilon > 0$, there must be $\eta + \varepsilon \notin E$, i.e. there are an infinite number of elements in S that are less that $\eta + \varepsilon$. Since there eixsts a $x_0 \in E$ such that $\eta - \varepsilon < x_0$, we know that there are only a finite number of elments in S that are less than $\eta - \varepsilon$. Then $(\eta - \varepsilon, \eta + \varepsilon)$ contains an infinite number of elements in S, which means that η is a accumulation point of S.

Second way: We first claim that any sequence in $\mathbb{R}$ has at least a monotonic subsequence (either increasing or decreasing). Indeed, if there is no increasing subsequence in $\{x_n\}$, we know there exists a $n_1 > 0$, such that $\forall\, n > n_1$, there is $x_n < x_{n_1}$. Similarly, there is no increasing subsequence in $\{x_n\}_{n > n_1}$, we know there exists a $n_2 > n_1$, such that $\forall\, n > n_2$, there is $x_n < x_{n_2} < x_{n_1}$. Proceeding like this, we can find a strictly decreasing subsequence $\{x_{n_k}\}$. Then it's easy to see that accumulation point principle is valid. □

Problem 3.9 (2.34)**.** *Prove that if $x_n > 0$ and*

$$\overline{\lim_{n\to\infty}}\, x_n \cdot \overline{\lim_{n\to\infty}}\, \frac{1}{x_n} = 1,$$

then the sequence $\{x_n\}$ is convergent.

Proof. Suppose there is a subsequence $\{x_{n_k}\} \subset \{x_n\}$, such that

$$\lim_{k\to\infty} x_{n_k} = \underline{\lim_{n\to\infty}}\, x_n.$$

By the definition of limit superior and $x_n > 0$, we know

$$\frac{1}{x_{n_k}} \leq \sup_{n \geq n_k} \frac{1}{x_n}.$$

Hence

$$\varliminf_{n\to\infty} x_n = \lim_{k\to\infty} x_{n_k} \geq \lim_{k\to\infty} \frac{1}{\sup\limits_{n\geq n_k} \frac{1}{x_n}} = \frac{1}{\varlimsup\limits_{n\to\infty} \frac{1}{x_n}} = \varlimsup_{n\to\infty} x_n,$$

i.e.

$$\varliminf_{n\to\infty} x_n = \varlimsup_{n\to\infty} x_n.$$

Hence the sequence $\{x_n\}$ is convergent. □

Problem 3.10 (2.35)**.** *Suppose that $\{x_n\}$ is bounded, and*

$$\lim_{n\to\infty} (x_{n+1} - x_n) = 0. \tag{3.3}$$

Denote $l = \varliminf\limits_{n\to\infty} x_n$ and $L = \varlimsup\limits_{n\to\infty} x_n$. Prove that any number in $[l, L]$ is the limitation of a subsequence of $\{x_n\}$.

Proof. **First way:** By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. We first prove the claim that for any given $\varepsilon > 0$ and $N \in \mathbb{N}$, there must exist a $\bar{n} > N$ such that

$$|x_{\bar{n}} - a| < \varepsilon.$$

By (3.3), we know there exists a $N' \in \mathbb{N}$, such that for $\forall\, n > N'$, there is

$$|x_{n+1} - x_n| < \varepsilon.$$

Let $N_0 := \max\{N, N'\}$. We know there must exist at least two points $x_{n'}$, $x_{n''}$ in $\{x_n\}_{n\geq N_0}$ such that $x_{n'} < a$, $x_{n''} > a$ (otherwise, if there is no points that are less than a, we must have $\varliminf\limits_{n\to\infty} x_n \geq a$, contradicts with $l < a$; if there is no points that are large than a, we must have $\varlimsup\limits_{n\to\infty} x_n \leq a$, contradicts with $a < L$). Without loss of generality, we assume $n' < n''$. Let $\bar{n}$ be the maximal integer which satisfies $n' \leq n \leq n''$ and $x_n \leq a$. Clearly, $\bar{n} \leq n'' - 1$ and $x_{\bar{n}} \leq a$, $x_{\bar{n}+1} > a$. Hence $\bar{n} > N$, and

$$|x_{\bar{n}} - a| \leq x_{\bar{n}+1} - x_{\bar{n}} < \varepsilon.$$

The claim is proved.

Now choosing $\varepsilon_1 = 1$, $N_1 = 1$, there exists a x_{n_1} $(n_1 > 1)$ such that $|x_{n_1} - a| < 1$. Then choosing $\varepsilon_2 = \frac{1}{2}$, $N_2 = n_1$, there exists a x_{n_2} $(n_2 > n_1)$ such that $|x_{n_2} - a| < \frac{1}{2}$. Next, choosing $\varepsilon_3 = \frac{1}{3}$, $N_3 = n_2$, there exists a x_{n_3} $(n_3 > n_2)$ such that $|x_{n_3} - a| < \frac{1}{3}$. Proceeding like this, we obtain a subsequence $\{x_{n_k}\} \subset \{x_n\}$ satisfies

$$|x_{n_k} - a| < \frac{1}{k}.$$

Hence $\lim\limits_{k\to\infty} x_{n_k} = a$, i.e. a is a accumulation point of $\{x_n\}$.

Second way: By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. Let $\delta := \min\{L - a, a - l\}$. For $\forall\, i \in \mathbb{N}$, there exists a $k_i \in \mathbb{N}$, when $n \geq k_i$, we have $|x_{n+1} - x_n| < \frac{\delta}{2^i}$.

By definition of l, L, we can choose $l_1 \geq k_1$, such that $x_{l_1} > L - \frac{\delta}{2}$, and choose $m_1 > l_1$, such that $x_{m_1} < l + \frac{\delta}{2}$. Again, choosing $l_2 > \max\{k_2, m_1\}$, such that $x_{l_2} > L - \frac{\delta}{2^2}$, and choose $m_2 > l_2$, such that $x_{m_2} < l + \frac{\delta}{2^2}$. Proceeding like this, we can choose $l_i > \max\{k_i, m_{i-1}\}$, such that $x_{l_i} > L - \frac{\delta}{2^i}$, and choose $m_i > l_i$, such that $x_{m_1} < l + \frac{\delta}{2^i}$. Hence, we know there exists n_i, $l_i < n_i < m_i$ $(i \in \mathbb{N})$, such that

$$x_{n_i} < l + \delta \leq a, \quad x_{n_i - 1} \geq a \quad (i \in \mathbb{N}).$$

Then

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| \quad (i \in \mathbb{N}).$$

By $n_i > l_i > k_i$, we know $n_i - 1 \geq k_i$. Hence

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| < \frac{\delta}{2^i}.$$

Hence $\lim\limits_{i \to \infty} x_{n_i} = a$, i.e. a is a accumulation point of $\{x_n\}$. □

Problem 3.11 (2.36)**.** *Suppose $\{x_n\}$ and $\{y_n\}$ satisfy*

$$x_{n+1} = y_n + q x_n \, (0 < q < 1), \quad n = 1, 2, \cdots.$$

Prove $\{y_n\}$ converges iff $\{x_n\}$ converges.

Proof. "if": Note that $y_n = x_{n+1} - q x_n$, it's easy to show $\{y_n\}$ converges if $\{x_n\}$ is convergent.

"only if": We first prove that $\{x_n\}$ is bounded. Since $\{y_n\}$ is convergent, we know it is bounded. Assume that $|y_n| \leq M$ for some $M > 0$. Then

$$\begin{aligned}
|x_{n+1}| &= |y_n + q x_n| = |y_n + q(y_{n-1} + q x_{n-1})| \\
&= \left|y_n + q y_{n-1} + q^2 x_{n-1}\right| \\
&= \left|y_n + q y_{n-1} + \cdots + q^{n-1} y_1 + q^n x_1\right| \\
&\leq M\left(1 + q + \cdots + q^{n-1}\right) + q^n |x_1| \\
&< \frac{M}{1-q} + |x_1|.
\end{aligned}$$

Hence the upper limit and lower limit of $\{x_n\}$ exist. We have from $x_{n+1} = y_n + q x_n$ that

$$\overline{\lim_{n\to\infty}}\, x_n = \overline{\lim_{n\to\infty}}\,(y_{n-1} + q x_{n-1}) = \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_{n-1} \leq \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_n,$$

which yields

$$\overline{\lim_{n\to\infty}}\, x_n \leq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Similarly, we have

$$\underline{\lim_{n\to\infty}}\, x_n \geq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Hence

$$\overline{\lim_{n\to\infty}}\, x_n = \underline{\lim_{n\to\infty}}\, x_n = (1-q)^{-1} \lim_{n\to\infty} y_n,$$

i.e. $\{x_n\}$ converges. $\square$

Problem 3.12.

(1) *(2.37). Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n + x_m.$$

Prove that $\left\{\dfrac{x_n}{n}\right\}$ has a limitation.

(2) *Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n \cdot x_m.$$

Prove that $\left\{x_n^{\frac{1}{n}}\right\}$ has a limitation.

Proof. (1) By

$$x_n \le x_{n-1} + x_1 \le x_{n-2} + 2x_1 \le \cdots \le nx_1,$$

we have

$$0 \le \frac{x_n}{n} \le x_1.$$

Hence $\left\{\dfrac{x_n}{n}\right\}$ is bounded. Denote $\underline{\lim_{n\to\infty}}\, \dfrac{x_n}{n} = l$, then $0 \le l \le x_1$. By definition, for $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that

$$\frac{x_N}{N} < l + \varepsilon.$$

For any $n > N$, we choose $q \in \mathbb{N}$ and $0 \le r < N$, such that $n = qN + r$. Then

$$x_n = x_{qN+r} \le x_{qN} + x_r \le qx_N + rx_1 \le qx_N + Nx_1.$$

Hence

$$\frac{x_n}{n} \le \frac{qx_N}{n} + \frac{Nx_1}{n} \le \frac{x_N}{N} + \frac{Nx_1}{n} < l + \varepsilon + \frac{Nx_1}{n}.$$

Thus

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l + \varepsilon.$$

Let $\varepsilon \to 0$, we have

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n}.$$

Hence

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n},$$

i.e. $\lim\limits_{n\to\infty} \dfrac{x_n}{n}$ exists.

(2) The proof is similar to (1), we leave it to readers. $\square$

Problem 3.13 (3.8)**.** *Calculate the following limitations.*

(6) $\lim\limits_{t\to 1}(1-t)\tan\dfrac{\pi t}{2}$;

(8) $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$;

(9) $\lim\limits_{x\to 0}\dfrac{\cos(n\arccos x)}{x}$ $\quad$ (n *is odd*).

Solution. (6)

$$\lim_{t\to 1}(1-t)\tan\frac{\pi t}{2}=\lim_{t\to 1}\frac{1-t}{\cos\frac{\pi t}{2}}=\lim_{t\to 1}\frac{1-t}{\sin\frac{\pi}{2}(1-t)}=\frac{2}{\pi}.$$

(8)

$$\lim_{x\to\frac{\pi}{4}}\frac{\tan x-1}{x-\frac{\pi}{4}}=\lim_{x\to\frac{\pi}{4}}\frac{1}{\cos x}\cdot\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}=2\lim_{x\to\frac{\pi}{4}}\frac{\sin(x-\frac{\pi}{4})}{x-\frac{\pi}{4}}=2.$$

(9)

$$\begin{aligned}\lim_{x\to 0}\frac{\cos(n\arccos x)}{x}&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin(\frac{n\pi}{2}-n\arccos x)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\big(n(\frac{\pi}{2}-\arccos x)\big)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\big(n(\arcsin x)\big)}{x}\\&=(-1)^{\frac{n-1}{2}}n,\end{aligned}$$

where we have used $\arcsin x+\arccos x=\dfrac{\pi}{2}$. $\square$

Problem 3.14 (3.9(4))**.** *Calculate the following limitation.*

$$\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}\qquad(a\neq 0).$$

Solution. Note that

$$\begin{aligned}\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}&=\lim_{x\to 0}(\cos ax)^{\frac{1}{x^2}}=\lim_{x\to 0}(1+\cos ax-1)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}\right)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}.\end{aligned}$$

Since

$$\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}=e,$$

we know that for $\forall\, \varepsilon > 0$, there exists a $\delta > 0$, such that $\forall\, x : |x| < \delta$, there is

$$e - \varepsilon < \left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} < e + \varepsilon.$$

Hence, we have for $|x| < \delta$ that

$$(e+\varepsilon)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} < \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} < (e-\varepsilon)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}}.$$

Then

$$\begin{aligned}
(e+\varepsilon)^{\frac{-a^2}{2}} &\leq \varliminf_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} \\
&\leq \varlimsup_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} \\
&\leq (e-\varepsilon)^{\frac{-a^2}{2}}.
\end{aligned}$$

Since ε is arbitrary, we have

$$\varliminf_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = \varlimsup_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}},$$

i.e.

$$\lim_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}}.$$

$\square$

4. Week 7 (10.17)

Problem 4.1.

(1) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)\geq 1.$$

(2) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n\geq e.$$

Proof. (1) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Then we have

$$\frac{1}{n+1}<\frac{x_n}{n}-\frac{x_{n+1}}{n+1},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}<\frac{x_N}{N}-\frac{x_{n+1}}{n+1}\leq\frac{x_N}{N}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}\right)=+\infty$, which makes a contradiction.

(2) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e<\left(1+\frac{1}{n-1}\right)^n.$$

Then we have

$$\frac{1}{n}<\frac{x_n}{n-1}-\frac{x_{n+1}}{n},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}<\frac{x_N}{N-1}-\frac{x_{n+1}}{n}\leq\frac{x_N}{N-1}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}\right)=+\infty$, which makes a contradiction. $\square$

Remark 4.2. *Both constants in Problem 4.1 are optimal. Indeed, we can choose* $x_n = n\ln n$.

Exercise 4.3 (Leave to readers)**.** *Let* $\{a_n\}_{n=1}^{\infty}$ *be a sequence of positive numbers. Prove that*

$$\varliminf_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)\le\frac{1}{4}.$$

Hint: It suffices to prove the conclusion for $a_n\in(0,1)$. Assume by contradiction that $\varliminf\limits_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)>\dfrac{1}{4}$. Then for $\varliminf\limits_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)>l>\dfrac{1}{4}$, there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$(without loss of generality, we can assume that $N=1$), there is

$$n^2\left(4a_n(1-a_{n-1})-1\right)>l.$$

Firstly, to prove that $\{a_n\}$ is monotonic increasing and $\lim\limits_{n\to\infty}a_n=\frac{1}{2}$ (by the monotone bounded convergence theorem). Secondly, let $a_n:=\frac{1}{2}-b_n$, where $b_n\ge 0$ and $\lim\limits_{n\to\infty}b_n=0$. Note that

$$b_{n-1}-b_n-2b_nb_{n-1}>\frac{l}{2n^2}.$$

Next, to prove that $\{nb_n\}$ is monotonic decreasing for large enough n. Indeed, there is

$$\begin{aligned}
nb_n-(n-1)b_{n-1}&<\frac{nb_{n-1}-\frac{l}{2n}}{1+2b_{n-1}}-(n-1)b_{n-1}\\
&=\frac{nb_{n-1}-\frac{l}{2n}-(n-1)b_{n-1}-2(n-1)b_{n-1}^2}{1+2b_{n-1}}\\
&=\frac{-2(n-1)\left(b_{n-1}-\frac{1}{4(n-1)}\right)^2+\frac{1}{8(n-1)}-\frac{l}{2n}}{1+2b_{n-1}}\\
&=\frac{-2(n-1)\left(b_{n-1}-\frac{1}{4(n-1)}\right)^2+\frac{(1-4l)n+4l}{8n(n-1)}}{1+2b_{n-1}}\\
&\le 0,
\end{aligned}$$

provided with $n\ge\frac{4l}{4l-1}$. Hence $\lim\limits_{n\to\infty}nb_n$ exists. Denote that $\lim\limits_{n\to\infty}nb_n=A$. By Stolz theorem (for lower limit), we have

$$\begin{aligned}
A=\lim_{n\to\infty}nb_n&=\varliminf_{n\to\infty}\frac{b_n}{\frac{1}{n}}\\
&\ge\varliminf_{n\to\infty}\frac{b_n-b_{n-1}}{\frac{1}{n}-\frac{1}{n-1}}
\end{aligned}$$

$$
\begin{aligned}
&\geq \lim_{n\to\infty} n(n-1)\left(\frac{l}{2n^2}+2b_nb_{n-1}\right)\\
&=\frac{l}{2}+2\lim_{n\to\infty} n(n-1)b_nb_{n-1}\\
&=\frac{l}{2}+2A^2,
\end{aligned}
$$

i.e.

$$2A^2-A+\frac{l}{2}\leq 0.$$

However, $\Delta = 1^2-4\times 2\times \frac{l}{2} = 1-4l<0$, which makes a contradiction.

From the above proof, we can see that $\frac{1}{4}$ is the optimal constant. Actually, we can choose $a_n=\frac{1}{2}-\frac{1}{4n}$. □

Problem 4.4. *Suppose that $x_n>0$. Prove that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$ iff $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0,\ \forall\, l>1$.*

Proof. "if": If $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0$, we know there exists a $N\in\mathbb{N}$ such that $\forall\, n>N$, there is $x_n<\frac{1}{2}l^n$. Hence $\sqrt[n]{x_n}<\left(\frac{1}{2}\right)^{\frac{1}{n}}l$, which means $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq l$. Since $l>1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$.

"only if": For $\forall\, l>1$, by $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$ we have there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$, there is

$$\sqrt[n]{x_n}<1+\frac{l-1}{2}=\frac{l+1}{2}.$$

Then

$$\frac{x_n}{l^n}<\left(\frac{l+1}{2l}\right)^n\to 0\quad \text{as } n\to\infty,$$

since $\frac{l+1}{2l}<1$. Thus, $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0,\ \forall\, l>1$. □

Problem 4.5. *Suppose that $x_n>0$. If $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$, prove that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{\sum\limits_{k=1}^{n}x_k}=1$.*

Proof. It is clear that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{\sum\limits_{k=1}^{n}x_k}\geq\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$. Hence, we only need to prove the inverse inequality. Since $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$, we know for any $l>1$, there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$, there is $\sqrt[n]{x_n}<l$. Then

$$\sum_{k=1}^{n}x_k\leq\sum_{k=1}^{N}x_k+\sum_{k=N+1}^{n}l^k$$

$$
\begin{aligned}
&\leq \sum_{k=1}^{N} x_k + \sum_{k=1}^{n} l^k \\
&< \sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}.
\end{aligned}
$$

Hence

$$
\overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{n} x_k} \leq \overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}} = l.
$$

Since $l > 1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{\sum\limits_{k=1}^{n} x_k} \leq 1$. Then we are done. $\square$

Problem 4.6. *Suppose that* $x_n > 0$, $\lim\limits_{n\to\infty} \frac{x_n}{n} = 0$ *and* $\overline{\lim\limits_{n\to\infty}} \frac{1}{n} \sum\limits_{k=1}^{n} x_k < +\infty$. *Prove that* $\overline{\lim\limits_{n\to\infty}} \frac{1}{n^2} \sum\limits_{k=1}^{n} x_k^2 = 0$.

Proof. Since $\lim\limits_{n\to\infty} \frac{x_n}{n} = 0$ and $L := \overline{\lim\limits_{n\to\infty}} \frac{1}{n} \sum\limits_{k=1}^{n} x_k < +\infty$, we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that $\forall\, n > N$, there is

$$
\frac{x_n}{n} < \varepsilon \quad \text{and} \quad \frac{1}{n} \sum_{k=1}^{n} x_k < L + 1.
$$

Then for $\forall\, n > N$,

$$
\begin{aligned}
\frac{1}{n^2} \sum_{k=1}^{n} x_k^2 &= \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \frac{1}{n^2} \sum_{k=N+1}^{n} x_k^2 \\
&< \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \frac{1}{n^2} \sum_{k=N+1}^{n} x_k \cdot k\varepsilon \\
&\leq \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \left(\frac{1}{n} \sum_{k=N+1}^{n} x_k \right) \varepsilon \\
&< \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon.
\end{aligned}
$$

Hence

$$
\overline{\lim_{n\to\infty}} \frac{1}{n^2} \sum_{k=1}^{n} x_k^2 \leq \overline{\lim_{n\to\infty}} \left(\frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon \right) = (L+1)\varepsilon.
$$

Sicne $\varepsilon > 0$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \frac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$, i.e. $\lim\limits_{n\to\infty} \frac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$. $\square$

Problem 4.7. *Calculate the following limitations.*

(1) $\lim\limits_{x\to 0} \dfrac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x}$, *where* $m, n \in \mathbb{N}$, $\alpha, \beta \in \mathbb{R}$ *are constants;*

(2) $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \sin\left(\dfrac{2k-1}{n^2}x\right)$, *where* $x \in \mathbb{R}$ *is a constant;*

(3) $\lim\limits_{x\to+\infty} \sin\sqrt{x+1} - \sin\sqrt{x}$;

(4) $\lim\limits_{x\to 0} (x+e^x)^{\frac{1}{x}}$;

(5) $\lim\limits_{x\to+\infty} \dfrac{[xf(x)]}{x}$, *where* $\lim\limits_{x\to+\infty} f(x) = 1$;

(6) $\lim\limits_{x\to+\infty} \left(\dfrac{1}{p}\sum\limits_{k=1}^{p} a_k^x\right)^{\frac{1}{x}}$, *where* $a_1, \cdots, a_p$ $(p \geq 2)$ *are positive.*

(7) $\lim\limits_{x\to 0+} \left(\dfrac{1}{p}\sum\limits_{k=1}^{p} a_k^x\right)^{\frac{1}{x}}$, *where* $a_1, \cdots, a_p$ $(p \geq 2)$ *are positive.*

Solution. (1) By replacement with equivalent infinitesimal, we have

$$\begin{aligned}\lim_{x\to 0} \frac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x} &= \lim_{x\to 0} \frac{\sqrt[m]{1+\alpha x} - 1}{x} + \lim_{x\to 0} \frac{1 - \sqrt[n]{1+\beta x}}{x} \\ &= \frac{\alpha}{m} - \frac{\beta}{n}.\end{aligned}$$

(2) It's clear that the limitation is 0 when $x = 0$. Next, we assume that $x \neq 0$. By the Prosthaphaeresis formula, we have

$$\begin{aligned}\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\frac{2k-1}{n^2}x\right) &= \lim_{n\to\infty} \frac{1}{2\sin\frac{x}{n^2}} \sum_{k=1}^{n} 2\sin\frac{x}{n^2}\sin\left(\frac{2k-1}{n^2}x\right) \\ &= \lim_{n\to\infty} \frac{1}{2\sin\frac{x}{n^2}} \sum_{k=1}^{n} \left[\cos\left(\frac{2k-2}{n^2}x\right) - \cos\left(\frac{2k}{n^2}x\right)\right] \\ &= \lim_{n\to\infty} \frac{1-\cos\left(\frac{2}{n}x\right)}{2\sin\frac{x}{n^2}} \\ &= x.\end{aligned}$$

(3) By the Prosthaphaeresis formula, we have

$$\lim_{x\to+\infty} \sin\sqrt{x+1} - \sin\sqrt{x} = \lim_{x\to+\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right)\sin\left(\frac{\sqrt{x+1}-\sqrt{x}}{2}\right)$$

$$
\begin{aligned}
&= \lim_{x\to+\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right)\sin\left(\frac{1}{2(\sqrt{x+1}+\sqrt{x})}\right)\\
&= 0.
\end{aligned}
$$

(4) By replacement with equivalent infinitesimal, we have

$$
\begin{aligned}
\lim_{x\to 0}(x+e^x)^{\frac{1}{x}} &= \lim_{x\to 0} e^{\frac{\ln(x+e^x)}{x}}\\
&= \lim_{x\to 0} e^{\frac{\ln(1+x+e^x-1)}{x}}\\
&= \lim_{x\to 0} e^{\frac{x+e^x-1}{x}}\\
&= e^2.
\end{aligned}
$$

(5) Note that

$$
f(x)-\frac{1}{x} < \frac{[xf(x)]}{x} \le f(x).
$$

It's easy to obtain $\lim\limits_{x\to+\infty}\dfrac{[xf(x)]}{x}=1$ by the Sandwich Theorem.

(6) Note that

$$
\frac{\max\{a_1,\cdots,a_p\}}{p^{\frac{1}{x}}} \le \left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} \le \max\{a_1,\cdots,a_p\}.
$$

By the Sandwich Theorem, we have

$$
\lim_{x\to+\infty}\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} = \max\{a_1,\cdots,a_p\}.
$$

(7) Note that

$$
\begin{aligned}
\frac{1}{x}\ln\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right) &= \frac{1}{x}\ln\left(1+\frac{1}{p}\sum_{k=1}^{p}(a_k^x-1)\right)\\
&\sim \frac{1}{p}\sum_{k=1}^{p}\frac{a_k^x-1}{x} \quad \text{as } x\to 0+\\
&\to \frac{1}{p}\sum_{k=1}^{p}\ln a_k \quad \text{as } x\to 0+.
\end{aligned}
$$

Hence

$$
\lim_{x\to 0+}\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} = e^{\frac{1}{p}\sum\limits_{k=1}^{p}\ln a_k} = \sqrt[p]{a_1a_2\cdots a_p}.
$$

□

Problem 4.8. *Calculate* $\lim\limits_{n\to\infty}\prod\limits_{k=1}^{n}\cos\dfrac{x}{2^k}$, *and prove the Viète formula*

$$\frac{\pi}{2}=\frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}.$$

Proof. It's clear that $\lim\limits_{n\to\infty}\prod\limits_{k=1}^{n}\cos\dfrac{x}{2^k}=1$ when $x=0$. Next, we assume that $x\neq 0$. By $\sin 2x=2\sin x\cos x$, we have

$$\lim_{n\to\infty}\prod_{k=1}^{n}\cos\frac{x}{2^k}=\lim_{n\to\infty}\frac{\cos\frac{x}{2}\cos\frac{x}{2^2}\cdots\cos\frac{x}{2^n}\sin\frac{x}{2^n}}{\sin\frac{x}{2^n}}=\lim_{n\to\infty}\frac{\sin x}{2^n\sin\frac{x}{2^n}}=\frac{\sin x}{x}.$$

Note that $\cos 2\theta=2\cos^2\theta-1$, we know that $\cos\theta=\sqrt{\dfrac{1}{2}+\dfrac{1}{2}\cos 2\theta}$. Choosing $x=\frac{\pi}{2}$, we know that $\cos\frac{x}{2}=\sqrt{\frac{1}{2}}$, $\cos\frac{x}{4}=\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}$. Hence

$$\frac{\sin x}{x}=\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots,$$

i.e.

$$\frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}=\frac{\frac{\pi}{2}}{\sin\frac{\pi}{2}}=\frac{\pi}{2}.$$

□

Problem 4.9. *Suppose that* $f(x)$, $g(x)$ *are periodic function defined on* $\mathbb{R}$, *and satisfy* $\lim\limits_{x\to+\infty}(f(x)-g(x))=0$. *Prove that* $f(x)=g(x)$, $\forall\, x\in\mathbb{R}$.

Proof. Denote T_1 as the period of $f(x)$ and T_2 as the period of $g(x)$. We first note that by $\lim\limits_{x\to+\infty}(f(x)-g(x))=0$, there is

$$f(x)-g(x+nT_1)=f(x+nT_1)-g(x+nT_1)\to 0\quad\text{as } n\to\infty.$$

Similarly, we have

$$g(x)=\lim_{n\to\infty}f(x+nT_2),\quad\forall\, x\in\mathbb{R}.$$

Hence we have

$$\begin{aligned}f(x)-g(x)&=\lim_{n\to\infty}(g(x+nT_1)-f(x+nT_2))\\&=\lim_{n\to\infty}(g(x+nT_1+nT_2)-f(x+nT_2+nT_1))\\&=0,\quad\forall\, x\in\mathbb{R}.\end{aligned}$$

$\square$

Problem 4.10. *Suppose that $f(x)$ does not have an upper bound in (a,b). Prove that there exists a sequence $\{x_n\} \subset (a,b)$ such that $\lim\limits_{n\to\infty} f(x_n) = +\infty$.*

Proof. Since $f(x)$ has no upper bound, we know that for any $n \in \mathbb{N}$, there exists a $x_n \in (a,b)$ such that $f(x_n) > n$. Then we have that $\lim\limits_{n\to\infty} f(x_n) = +\infty$. $\square$

Problem 4.11. *Suppose $f(x)$ is defined on $(0,1)$, and $\lim\limits_{x\to 0} f(x) = 0$, $\lim\limits_{x\to 0} \dfrac{f(x) - f\left(\frac{x}{2}\right)}{x} = 0$. Prove that $\lim\limits_{x\to 0} \dfrac{f(x)}{x} = 0$.*

Proof. By $\lim\limits_{x\to 0} \dfrac{f(x) - f\left(\frac{x}{2}\right)}{x} = 0$, we know that $\forall\, \varepsilon > 0$, there exists a $\delta > 0$, such that $\forall\, x : 0 < x < \delta$, there is

$$\frac{\left|f(x) - f\left(\frac{x}{2}\right)\right|}{x} < \varepsilon,$$

i.e.

$$\left|f(x) - f\left(\frac{x}{2}\right)\right| < \varepsilon x.$$

Then for $0 < x < \delta$, there is

$$\begin{aligned}
|f(x)| &= \left|f(x) - f\left(\frac{x}{2}\right) + f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right) + \cdots + f\left(\frac{x}{2^n}\right)\right| \\
&\le \left|f(x) - f\left(\frac{x}{2}\right)\right| + \left|f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right)\right| + \cdots + \left|f\left(\frac{x}{2^n}\right)\right| \\
&< \varepsilon\left(x + \frac{x}{2} + \cdots + \frac{x}{2^n}\right) + \left|f\left(\frac{x}{2^n}\right)\right| \\
&< 2\varepsilon x + \left|f\left(\frac{x}{2^n}\right)\right|.
\end{aligned}$$

Since $\lim\limits_{x\to 0} f(x) = 0$, we have by letting $n \to \infty$ that

$$|f(x)| \le 2\varepsilon x, \quad 0 < x < \delta.$$

Thus,

$$\lim_{x\to 0} \frac{f(x)}{x} = 0.$$

$\square$

Problem 4.12. *Suppose $f(x) > 0$ is an increasing function defined on $(0,+\infty)$ satisfies $\lim\limits_{x\to+\infty} \dfrac{f(2x)}{f(x)} = 1$. Prove that $\lim\limits_{x\to+\infty} \dfrac{f(ax)}{f(x)} = 1$, $\forall\, a \in (0,+\infty)$.*

Proof. It suffices to prove $\lim\limits_{x\to+\infty}\dfrac{f(ax)}{f(x)}=1$ for all $a>1$. Indeed, for any $0<a<1$, we have

$$\lim_{x\to+\infty}\frac{f(ax)}{f(x)}=\lim_{x\to+\infty}\frac{f(ax)}{f(\frac{1}{a}ax)}=\frac{1}{\lim\limits_{ax\to+\infty}\frac{f(\frac{1}{a}ax)}{f(ax)}}=1.$$

For $a>1$, we know there exist some $N\in\mathbb{N}$ such that $2^N\le a<2^{N+1}$. Since $f(x)$ is increasing, we know

$$\frac{f(2^Nx)}{f(x)}\le\frac{f(ax)}{f(x)}\le\frac{f(2^{N+1}x)}{f(x)}.$$

Then the problem is reduced to prove $\lim\limits_{x\to+\infty}\dfrac{f(2^Kx)}{f(x)}=1$ for all $K\in\mathbb{N}$, and it is clear since

$$\lim_{x\to+\infty}\frac{f(2^Kx)}{f(x)}=\lim_{x\to+\infty}\frac{f(2^Kx)}{f(2^{K-1}x)}\cdot\lim_{x\to+\infty}\frac{f(2^{K-1}x)}{f(2^{K-2}x)}\cdots\lim_{x\to+\infty}\frac{f(2x)}{f(x)}=1.$$

□

Problem 4.13. *Suppose $f:(0,+\infty)\to\mathbb{R}$ satisfies $\forall\, a>0$, f is bounded on $(0,a)$. Prove that if $\lim\limits_{x\to+\infty}[f(x+1)-f(x)]=l$, then $\lim\limits_{x\to+\infty}\dfrac{f(x)}{x}=l$.*

Proof. Without loss of generality, we can assume that $l=0$ (otherwise, we replace $f(x)$ by $f(x)-lx$). Since $\lim\limits_{x\to+\infty}[f(x+1)-f(x)]=0$, we know $\forall\,\varepsilon>0$, there exists a $X>0$, such that $\forall\, x>X$, there is

$$-\varepsilon<f(x+1)-f(x)<\varepsilon.$$

Summing all inequalities, we have

$$-\left([x-X]+1\right)\varepsilon<f(x+1)-f\left(x-[x-X]\right)<\left([x-X]+1\right)\varepsilon.$$

Note that $\lim\limits_{x\to+\infty}\dfrac{f\left(x-[x-X]\right)}{x+1}=0$ since f is bounded on $(0,X+1)$. We have

$$-\varepsilon\le\varliminf_{x\to+\infty}\frac{f(x+1)}{x+1}\le\varlimsup_{x\to+\infty}\frac{f(x+1)}{x+1}\le\varepsilon.$$

Since ε is arbitrary, we obtain $\lim\limits_{x\to+\infty}\dfrac{f(x)}{x}=0$. □

Problem 4.14. *Suppose $f,g:(a,+\infty)\to\mathbb{R}$ satisfy $\forall\, b>a$, f,g are bounded on (a,b); g is strictly increasing, and $\lim\limits_{x\to+\infty}g(x)=+\infty$. Prove that if $\lim\limits_{x\to+\infty}\dfrac{f(x+1)-f(x)}{g(x+1)-g(x)}=l$, then $\lim\limits_{x\to+\infty}\dfrac{f(x)}{g(x)}=l$.*

Proof. The proof is very similar to Problem 4.13, we omit details here and leave it to readers. □

Remark 4.15. *(1) Problem 4.14 is called the function version of Stolz theorem.*
(2) We can change the constant "1" to any positive constant "T".
(3) l can be chosen as $\pm\infty$, but it's not correct for ∞.

Problem 4.16. *Suppose that f is defined on $\mathbb{R}$, and f is bounded in some neighborhood of $x = 0$. If there exist $a > 1$, $b > 1$, such that $f(ax) = bf(x)$, prove that $f(x)$ is continuous at $x = 0$.*

Proof. First, by $f(ax) = bf(x)$, it's easy to know $f(0) = 0$. Assume that there exist $M > 0$, $\delta_0 > 0$, such that $\forall\, x : |x| < \delta_0$, there is $|f(x)| \leq M$. For $\forall\, \varepsilon > 0$, choosing $\delta = \delta_0/a^{N+1}$, where N satisfies $M/b^N < \varepsilon$. Then for $\forall\, x : |x| < \delta$, there is

$$|f(x)| = \frac{1}{b^N}|f(a^N x)| \leq \frac{M}{b^N} < \varepsilon.$$

Hence

$$\lim_{x\to 0} f(x) = 0 = f(0),$$

i.e. $f(x)$ is continuous at $x = 0$. □

Problem 4.17. *Suppose that $f(x) \in C[0, +\infty)$ is bounded, and $\lim\limits_{x\to+\infty} f(x)$ does not exist. Prove there exists $t \in \mathbb{R}$ such that $f(x) = t$ has an infinite number of solutions.*

Proof. Since $f(x) \in C[0, +\infty)$ is bounded, and $\lim\limits_{x\to+\infty} f(x)$ does not exist, we know

$$-\infty < \varliminf_{x\to+\infty} f(x) < \varlimsup_{x\to+\infty} f(x) < +\infty.$$

Denote $l := \varliminf\limits_{x\to+\infty} f(x)$ and $\varlimsup\limits_{x\to+\infty} f(x)$. Let $t = \frac{l+L}{2}$, we will show $f(x) = t$ has an infinite number of solutions. Indeed, by the definition of upper limit and lower limit, we know that for $X_1 = 1$, there exist $x_1, y_1 > X_1$, such that $f(x_1) < t$, $f(y_1) > t$. Hence there exists z_1 between x_1 and y_1 such that $f(z_1) = t$ by $f(x)$ is continuous. For $X_2 = \max\{2, x_1, y_1\}$, there exist $x_2, y_2 > X_2$, such that $f(x_2) < t$, $f(y_2) > t$. Hence there exists z_2 between x_2 and y_2 such that $f(z_2) = t$ by $f(x)$ is continuous. Proceeding like this, we can find an infinite number of z_n such that $f(z_n) = t$, i.e. $f(x) = t$ has an infinite number of solutions. □

Problem 4.18. *Suppose that $f(x)$ is uniformly continuous on $[0, +\infty)$, and $\forall\, h > 0$, $\lim\limits_{n\to\infty} f(nh)$ exists. Prove that $\lim\limits_{x\to+\infty} f(x)$ exists.*

Proof. **First method: Find the limitation** Pick $h = 1$, assume that the sequence $\{f(n)\}_{n=1}^{\infty}$ converges to L. Then for each $m \in \mathbb{N}$, let $h = 1/m$. Then $\{f(n/m\}_{n=1}^{\infty}$ contains the subsequence $\{f(n)\}_{n=1}^{\infty}$. Thus $\{f(n/m\}_{n=1}^{\infty}$ converges to L for all m. Now we show that $\lim\limits_{x\to\infty} f(x) = L$.

Let $\varepsilon > 0$. Since f is uniformly continuous, there is $\delta > 0$ so that if $x, y \in \mathbb{R}_+$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\varepsilon}{2}.$$

Now let $m \in \mathbb{N}$ so that $1/m < \delta$. Since $\{f(n/m\}_{n=1}^{\infty}$ converges to L, there is $N \in \mathbb{N}$ so that

$$\left| f\left(\frac{n}{m}\right) - L \right| < \frac{\varepsilon}{2}$$

for all $n \geq N$. Let $M = N/m$. Then if $x \geq M$, there is $n \geq N$ so that $|x - n/m| < \delta$ (we used $1/m < \delta$ here). Then $|f(x) - f(n/m)| < \varepsilon/2$ and thus

$$|f(x) - L| \leq \left| f(x) - f\left(\frac{n}{m}\right) \right| + \left| f\left(\frac{n}{m}\right) - L \right| < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary we conclude $\lim\limits_{x\to\infty} f(x) = L$.

Second method: Cauchy principle Let $\varepsilon > 0$. Since f is uniformly continuous, there is $\delta > 0$ so that if $x, y \in \mathbb{R}_+$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\varepsilon}{3}.$$

For ε, δ ginven as above, since $\lim\limits_{n\to\infty} f(n\delta)$ exists, we know there exists $N \in \mathbb{N}$, such that $\forall\, m, n > N$, there is

$$|f(m\delta) - f(n\delta)| < \frac{\varepsilon}{3}.$$

Then, let $X = (N+1)\delta$, we know that $\forall\, x_1, x_2 > X$, there are

$$\left[\frac{x_i}{\delta}\right] > \frac{x_i}{\delta} - 1 > N \quad (i = 1, 2)$$

and

$$\left| x_i - \left[\frac{x_i}{\delta}\right]\delta \right| = \delta \left| \frac{x_i}{\delta} - \left[\frac{x_i}{\delta}\right] \right| < \delta \quad (i = 1, 2).$$

Hence

$$\begin{aligned}
|f(x_1) - f(x_2)| \leq & \left| f(x_1) - f\left(\left[\frac{x_1}{\delta}\right]\delta\right) \right| \\
& + \left| f\left(\left[\frac{x_1}{\delta}\right]\delta\right) - f\left(\left[\frac{x_2}{\delta}\right]\delta\right) \right| \\
& + \left| f\left(\left[\frac{x_2}{\delta}\right]\delta\right) - f\left(x_2\right) \right| \\
< & \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\
= & \varepsilon.
\end{aligned}$$

□

Remark 4.19. *We can make a weak assumption that $f(x)$ is just continuous on $[0, +\infty)$, but to prove this conclusion is so difficult, we omit the detail here and leave it to someone who interested.*

Problem 4.20. *Suppose that $f(x)$ is uniformly continuous on $[0, +\infty)$, and $\forall\, x > 0$, $\lim\limits_{n\to\infty} f(x + n) = 0$. Prove that $\lim\limits_{x\to+\infty} f(x) = 0$.*

Proof. By $f(x)$ is uniformly continuous on $[0,+\infty)$, we know $\forall\,\varepsilon>0$, there is $\delta>0$, such that $\forall\,x,y\geq 0:|x-y|<\delta$, there is

$$|f(x)-f(y)|<\frac{\varepsilon}{2}.$$

Take $k>\frac{1}{\delta}$, and cut $[0,1]$ uniformly into k pieces. Let $x_i=\frac{i}{k}$ $(i=1,2,\cdots,k)$ be the cut points. Note that $x_i-x_{i-1}=\frac{1}{k}<\delta$. Since for every x_i, there is $\lim\limits_{n\to\infty}f(x_i+n)=0$. We know there exists $N_i>0$, such that $\forall\,n>N_i$, there is $|f(x_i+n)|<\dfrac{\varepsilon}{2}$. Let $N=\max\{N_1,N_2,\cdots,N_k\}$, then $\forall\,n>N$, there is

$$|f(x_i+n)|<\frac{\varepsilon}{2}\quad(i=1,2,\cdots,k).$$

Choose $X=N+1$, for $\forall\,x>X$, there is $[x]>N$. Since $x-[x]\in[0,1)$, we know there exists $i\in\{1,2,\cdots,k\}$, such that $|(x-[x])-x_i|<\delta$, i.e. $|x-(x_i+[x])|<\delta$. Hence, there is

$$\begin{aligned}|f(x)|\leq&|f(x)-f(x_i+[x])|+|f(x_i+[x])|\\<&\frac{\varepsilon}{2}+\frac{\varepsilon}{2}\\=&\varepsilon.\end{aligned}$$

i.e. $\lim\limits_{x\to+\infty}f(x)=0$. □

Remark 4.21. *(1) We can use the same method to prove that if $f(x)$ is uniformly continuous on $[0,+\infty)$, and $\forall\,x>0$, $\lim\limits_{n\to\infty}f(x+n)=A$, then $\lim\limits_{x\to+\infty}f(x)=A$.*
(2) From the proof, we can see that the conclusion is still right for a weak assumption, i.e. $\forall\,x\in[0,1]$, $\lim\limits_{n\to\infty}f(x+n)=0$.
(3) Find a counterexample if we do not assume $f(x)$ is uniformly continuous, but just continuous. Indeed, define

$$f_n(x)=\begin{cases}2nx, & 0\leq x\leq\dfrac{1}{2n},\\ 2-2nx, & \dfrac{1}{2n}<x\leq\dfrac{1}{n},\\ 0, & \textit{otherwise}.\end{cases}$$

Let $f(x)=\sum\limits_{n=1}^{\infty}f_n(x-n)$, then $f(x)$ is a counterexample.

Exercise 4.22 (Leave to readers)**.** *Find x such that $\lim\limits_{m\to\infty}\sqrt{1+\sqrt{x+\sqrt{x^2+\cdots+\sqrt{x^m}}}}=2$.*

Hint: Let me describe a sketch of proof that $x=4$.

A. Observe that if $f(x) = \lim_{n\to\infty}\sqrt{1+\sqrt{x+\sqrt{x^2+\cdots\sqrt{x^n}}}}$, then f is strictly increasing.

B. We shall show that $f(4) = 2$, and hence $x = 4$ is the unique answer.

B_1. Fix $m \in \mathbb{N}$ and show that, for $n = m, m-1, m-2, \cdots$ (induction backwards)

$$2^n < \sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} < 2^n+1,$$

while

$$\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}} = 2^n+1.$$

B_2. Next estimate the difference

$$\begin{aligned}
&(2^n+1)-\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}\\
=&\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}}-\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}\\
=&\frac{\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}-\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}{\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}}+\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}}\\
<&\frac{\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}-\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}{2\cdot 2^n}\\
<&\cdots<\frac{(\sqrt{4^m}+1)-\sqrt{4^m}}{2^{m-n}\cdots 2^{n+(n+1)+\cdots+(m-1)}}\\
=&2^{-\frac{(m-n)(n+m+1)}{2}}.
\end{aligned}$$

Thus

$$\lim_{m\to\infty}\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} = 2^n+1.$$

For $n = 0$ we have

$$\lim_{m\to\infty}\sqrt{1+\sqrt{4+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} = 2^0+1 = 2.$$

□

A similar but easier question: Find x in:

$$\sqrt{x^2+\sqrt{4x^2+\sqrt{16x^2+\sqrt{64x^2+\cdots}}}} = 5.$$

Hint: $x+1=\sqrt{x^2+2x+1}=\sqrt{x^2+\sqrt{4x^2+4x+1}}=\cdots$.

Exercise 4.23 (Challenge!)**.** *Assume $f \in C[0,+\infty)$, and for all $a>0$, we have*

$$\lim_{x\to+\infty}(f(x+a)-f(x))=0.$$

Prove that $f(x)$ is uniformly continuous.

Hint: Fix $\varepsilon>0$, we want to find $\delta>0$ such that

$$|x-y|<\delta \Rightarrow |f(x)-f(y)|<\varepsilon. \tag{4.1}$$

For every $N\in\mathbb{N}$, let $E_N := \{a \mid x\geq N \Rightarrow |f(x+a)-f(x)|\leq \varepsilon/4\}$. E_N is closed (by continuity of f) and $\bigcup_{N\in\mathbb{N}} E_N=[0,\infty)$. By Baire Category Theorem, at least one of them, say, E_N contains a closed interval $[b,c]$. For $x,y\geq N+c$, without loss of generality, say $y\geq x$, if $|y-x|<c-b$, there always exists $z\geq N$ such that $[x,y]\subset[z+b,z+c]$. Then $|f(x)-f(y)|\leq|f(x)-f(z)|+|f(y)-f(z)|=|f(z+d)-f(z)|+|f(z+e)-f(z)|\leq\varepsilon/2$ where $d,e\in[b,c]$. For $x,y\leq N+c$, as $[0,N+c]$ is compact, f restricted to $[0,N+c]$ is uniformly continuous, hence there exists $\delta'>0$ satisfing the requirements in (4.1). Let $\delta=\min(c-b,\delta')$, then we are done. $\square$

5. Week 8 (10.24)

Problem 5.1. *Suppose that $f(x)$ is uniformly continuous on $\mathbb{R}$. Prove that there exist $a > 0$, $b > 0$ such that $|f(x)| \le a|x| + b$, $\forall\, x \in \mathbb{R}$.*

Proof. Since $f(x)$ is uniformly continuous on $\mathbb{R}$, we have that $\forall\, \varepsilon > 0$, there exists $\delta > 0$, such that $\forall\, x, y \in \mathbb{R} : |x - y| < \delta$, there is $|f(x) - f(y)| < \varepsilon$. Now, fix ε and δ. For $\forall\, x \in \mathbb{R}$, there exists $n \in \mathbb{Z}$, such that $x = n\delta + x_0$, where $x_0 \in (-\delta, \delta)$. Note that $f(x)$ is bounded on $[-\delta, \delta]$, i.e. $\exists M > 0$, such that $|f(x)| \le M$ $(\forall\, x \in)[-\delta, \delta]$. Hence,

$$\begin{aligned}
|f(x)| &= \left| \sum_{k=1}^{n} [f(k\delta + x_0) - f((k-1)\delta + x_0)] + f(x_0) \right| \\
&\le \sum_{k=1}^{n} |f(k\delta + x_0) - f((k-1)\delta + x_0)| + |f(x_0)| \\
&\le |n|\varepsilon + M \\
&= \frac{\varepsilon}{\delta}|x - x_0| + M \qquad \left(\text{since } \left|\frac{x - x_0}{\delta}\right| = |n|\right) \\
&\le \frac{\varepsilon}{\delta}|x| + \left(M + \frac{\varepsilon}{\delta}|x_0|\right) \\
&\le \frac{\varepsilon}{\delta}|x| + (M + \varepsilon).
\end{aligned}$$

Denote $a = \varepsilon/\delta$, $b = M + \varepsilon$, hence

$$|f(x)| \le a|x| + b \quad (\forall\, x \in (-\infty, +\infty)).$$

□

Problem 5.2 (4.24)**.** *Prove that at any point of the curve*

$$\begin{cases} x = a(\cos t + t\sin t), \\ y = a(\sin t - t\cos t), \end{cases} \quad (a > 0),$$

the distance of the normal line to the origin is equal to a.

Proof. Differentiating respect to t yields

$$\begin{cases} dx = at\cos t\, dt, \\ dy = at\sin t\, dt, \end{cases} \quad (a > 0)$$

Then the normal line at point (x, y), $t \ne k\pi$ is

$$Y - a(\sin t - t\cos t) = -\frac{dx}{dy}(X - a(\cos t + t\sin t)),$$

i.e.

$$Y - a(\sin t - t\cos t) = -\cot t(X - a(\cos t + t\sin t)).$$

Hence

$$\begin{aligned} d &= \frac{|a\cot t(\cos t + t\sin t) + a(\sin t - t\cos t)|}{\sqrt{1+\cot^2 t}} \\ &= |a\cos^2 t + at\sin t\cos t + a\sin^2 t - at\sin t\cos t| \\ &= a. \end{aligned}$$

When $t = k\pi$, we know the normal line is $x = a(-1)^k$, thus $d = a$.

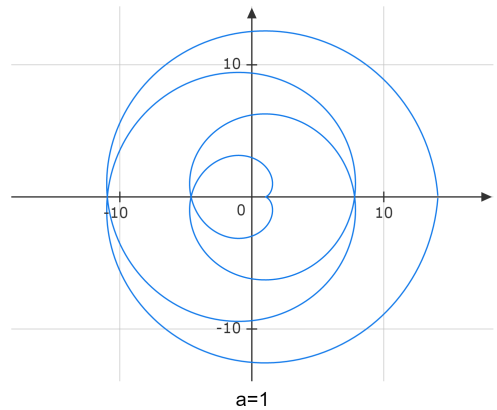

FIGURE 1. Graph of the curve for $a = 1$

□

Problem 5.3. *Calculate the derivative of* $f(x) = x^{\arcsin x}$.

Solution. Note that

$$f(x) = x^{\arcsin x} = e^{\ln x \cdot \arcsin x}.$$

Denote $g(x) = \ln x \cdot \arcsin x$, we have

$$g'(x) = \frac{\arcsin x}{x} + \frac{\ln x}{\sqrt{1-x^2}}.$$

Hence

$$f'(x) = \left(e^{g(x)}\right)' = e^{g(x)}g'(x) = x^{\arcsin x}\left(\frac{\arcsin x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right).$$

□

Problem 5.4. *Calculate the left right derivative of* $f(x) = \begin{cases} \dfrac{x}{e^{1/x}+1}, & x \neq 0, \\ 0, & x = 0, \end{cases}$ *at* $x = 0$, *and determine whether* $f(x)$ *is differentiable at* $x = 0$.

Solution. Left derivative:

$$f'_-(0) = \lim_{x\to 0-0} \frac{\frac{x}{e^{1/x}+1} - 0}{x - 0} = \lim_{x\to 0-0} \frac{1}{e^{1/x}+1} = 1.$$

Right derivative:

$$f'_+(0) = \lim_{x\to 0+0} \frac{\frac{x}{e^{1/x}+1} - 0}{x - 0} = \lim_{x\to 0+0} \frac{1}{e^{1/x}+1} = 0.$$

Hence $f(x)$ is not differentiable at $x = 0$. The graph of $f(x)$ is as follows:

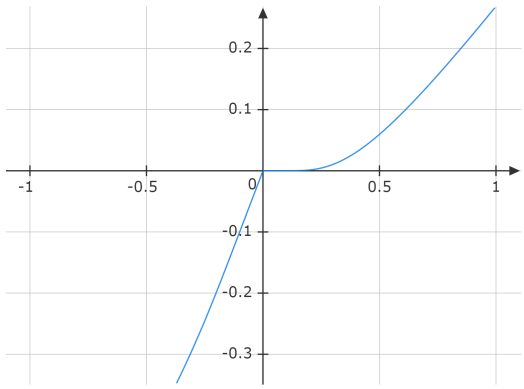

FIGURE 2. Graph of $f(x)$

□

Problem 5.5. *Suppose that $f(x)$ is differentiable on $\mathbb{R}$ and satisfies $f(x) \geq x$, $f(x) \geq 1 - x$, $\forall\, x \in \mathbb{R}$. Prove that $f(\frac{1}{2}) > \frac{1}{2}$.*

Proof. Suppose that $f(\frac{1}{2}) \leq \frac{1}{2}$. Then we have

$$\begin{aligned} f'_-\left(\frac{1}{2}\right) &= \lim_{x\to\frac{1}{2}-0} \frac{f(x) - f(\frac{1}{2})}{x - \frac{1}{2}} \\ &= \lim_{x\to\frac{1}{2}-0} \frac{f(\frac{1}{2}) - f(x)}{\frac{1}{2} - x} \\ &\leq \lim_{x\to\frac{1}{2}-0} \frac{\frac{1}{2} - (1 - x)}{\frac{1}{2} - x} \\ &= -1, \end{aligned}$$

and

$$\begin{aligned} f'_+\left(\frac{1}{2}\right) &= \lim_{x\to\frac{1}{2}+0} \frac{f(x) - f(\frac{1}{2})}{x - \frac{1}{2}} \\ &\geq \lim_{x\to\frac{1}{2}+0} \frac{x - \frac{1}{2}}{x - \frac{1}{2}} \\ &= 1, \end{aligned}$$

contradicts with $f(x)$ is differentiable at $x = \frac{1}{2}$. Hence $f(\frac{1}{2}) > \frac{1}{2}$. □

Remark 5.6. *From the proof of Problem 5.5, we know that it only needs to assume $f(x)$ is differentiable at $x = \frac{1}{2}$.*

Problem 5.7. *Suppose that $f(x)$ is differentiable. Prove that $F(x) = f(x)(1 + |\sin x|)$ is differentiable at $x = 0$ if and only if $f(0) = 0$.*

Proof. Calculating the left derivative of F at $x = 0$ yields

$$\begin{aligned} F'_-(0) &= \lim_{x\to 0-0} \frac{f(x)(1+|\sin x|) - f(0)}{x} \\ &= \lim_{x\to 0-0} \frac{f(x) - f(0)}{x} + \lim_{x\to 0-0} \frac{f(0)|\sin x|}{x} \\ &= f'(0) - f(0). \end{aligned}$$

Similarly, the right left derivative of F at $x = 0$ is

$$\begin{aligned} F'_+(0) &= \lim_{x\to 0+0} \frac{f(x)(1+|\sin x|) - f(0)}{x} \\ &= \lim_{x\to 0+0} \frac{f(x) - f(0)}{x} + \lim_{x\to 0+0} \frac{f(0)|\sin x|}{x} \\ &= f'(0) + f(0). \end{aligned}$$

Hence $F(x)$ is differentiable at $x = 0$ if and only if $F'_-(0) = F'_+(0)$, if and only if $f'(0) - f(0) = f'(0) + f(0)$, if and only if $f(0) = 0$. □

Problem 5.8. *Suppose that $y = y(x)$ is determined by parametric equation $\begin{cases} x = 2t + |t| \\ y = t^2 + 2t|t| \end{cases}$, $t \in \mathbb{R}$. Prove that $y(x)$ is differentiable at $x = 0$, and find $y'(0)$.*

Proof. **First way:** By $x = 2t + |t|$, we have

$$t = \begin{cases} \dfrac{x}{3}, & x \geq 0, \\ x, & x < 0. \end{cases}$$

Hence by $y = t^2 + 2t|t|$, we know

$$y = \begin{cases} \dfrac{1}{3}x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

It's easy to see that $y(x)$ is differentiable at $x = 0$, and $y'(0) = 0$.

Second way: A direct differentiating yields

$$\begin{cases} dx = \left(2 + \dfrac{t}{|t|}\right) dt \\ dy = (2t + 4|t|)\, dt \end{cases}$$

Hence

$$\frac{dy}{dx} = \frac{2t + 4|t|}{2 + \frac{t}{|t|}} = 2|t|.$$

It's easy to see that $y(x)$ is differentiable at $x = 0$, and $y'(0) = 0$. $\square$

Problem 5.9. *Suppose that* $x^2y^2+x^2+y^2 = 1$, $(xy > 0)$. *Prove that* $\dfrac{dx}{\sqrt{1-x^4}} + \dfrac{dy}{\sqrt{1-y^4}}$.

Proof. A direct differentiating yields

$$x(1 + y^2)dx + y(1 + x^2)dy = 0.$$

Note by $x^2y^2 + x^2 + y^2 = 1$ that

$$x^2(1 + y^2)^2 = 1 - y^4,$$

and

$$y^2(1 + x^2)^2 = 1 - x^4.$$

Since $xy > 0$, we know $x(1 + y^2)$ and $y(1 + x^2)$ have the same sign. Then

$$\sqrt{1 - y^4}dx + \sqrt{1 - x^4}dy = 0,$$

i.e.

$$\frac{dx}{\sqrt{1-x^4}} + \frac{dy}{\sqrt{1-y^4}}.$$

The graph of the curve is as follows:

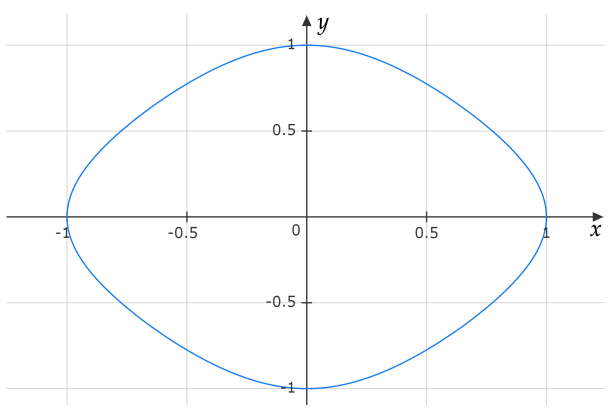

FIGURE 3. Graph of the curve

$\square$

Problem 5.10. *Prove the following identities:*

(1) $\displaystyle\sum_{k=1}^{n} kC_n^k = n2^{n-1}$, $n \in \mathbb{N}_+$;

(2) $\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}$, $n \in \mathbb{N}_+$.

Proof. (1) Note that

$$\sum_{k=1}^{n} kC_n^k x^{k-1} = \left(\sum_{k=1}^{n} C_n^k x^k\right)' = ((1+x)^n - 1)' = n(1+x)^{n-1}.$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} kC_n^k = n2^{n-1}.$$

(2) **First way:** Note that

$$\begin{aligned}
\sum_{k=1}^{n} k^2 C_n^k x^{k-1} &= \left(\sum_{k=1}^{n} kC_n^k x^k\right)' \\
&= \left(x\sum_{k=1}^{n} kC_n^k x^{k-1}\right)' \\
&= \left(nx(1+x)^{n-1}\right)' \\
&= n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2}.
\end{aligned}$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}.$$

Second way: Firstly, we have

$$k^2 C_n^k = k^2 \frac{n!}{k!(n-k)!} = nkC_{n-1}^{k-1}.$$

Then

$$\begin{aligned}
\sum_{k=1}^{n} k^2 C_n^k x^{k-1} &= n\sum_{k=1}^{n} kC_{n-1}^{k-1} x^{k-1} \\
&= \left(n\sum_{k=1}^{n} C_{n-1}^{k-1} x^k\right)' \\
&= \left(nx(1+x)^{n-1}\right)' \\
&= n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2}.
\end{aligned}$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}.$$

□

Problem 5.11. *Suppose that $f : [a, b] \to [a, b]$ satisfies*

$$|f(x) - f(y)| \le |x - y|, \quad \forall\, x, y \in [a, b].$$

Define $x_{n+1} = \frac{1}{2}(x_n + f(x_n))$ for any given $x_1 \in [a, b]$. Prove that $\lim\limits_{n\to\infty} x_n$ exists.

Proof. Note that

$$\begin{aligned} x_{n+1} - x_n &= \frac{1}{2}(x_n + f(x_n)) - \frac{1}{2}(x_{n-1} + f(x_{n-1})) \\ &= \frac{1}{2}(f(x_n) - f(x_{n-1})) + \frac{1}{2}(x_n - x_{n-1}). \end{aligned}$$

Then we have

$$\begin{aligned} (x_{n+1} - x_n)(x_n - x_{n-1}) &= \frac{1}{2}(f(x_n) - f(x_{n-1}))(x_n - x_{n-1}) + \frac{1}{2}(x_n - x_{n-1})^2 \\ &\ge -\frac{1}{2}|x_n - x_{n-1}|^2 + \frac{1}{2}(x_n - x_{n-1})^2 = 0, \end{aligned}$$

since

$$|f(x_n) - f(x_{n-1})||x_n - x_{n-1}| \le |x_n - x_{n-1}|^2.$$

Hence, $\{x_n\}$ is monotonic. Clearly, $\{x_n\}$ is bounded, it's easy to know that $\lim\limits_{n\to\infty} x_n$ exists. □

Problem 5.12. *Suppose that $f(x)$ is differentiable on $[0, 1]$, and $\{x \in [0, 1] \big| f(x) = 0,\ f'(x) = 0\} = \emptyset$. Prove that f has a finite number of zero pionts in $[0, 1]$.*

Proof. Assume that f has an infinite number of zero points. Let $Z := \{x \in [0, 1] \big| f(x) = 0\}$. Since Z is a bounded set with an infinite number of elements, we know by the Bolzano-Weierstrass theorem that there is a sequence $\{x_n\} \subset Z$ converges, say $\lim\limits_{n\to\infty} x_n = x_0$. By the continuity of f, we know that $f(x_0) = \lim\limits_{n\to\infty} f(x_n) = 0$. Since $f(x)$ is differentiable, we have

$$f'(x_0) = \lim_{n\to\infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} = 0,$$

i.e. $x_0 \in \{x \in [0, 1] \big| f(x) = 0,\ f'(x) = 0\}$, contradiction. □

Problem 5.13.

(1) *Suppose that $f \in C[0, 1]$, $f(0) = f(1)$. Prove that for $0 < \alpha < 1$, if $\frac{1}{\alpha} \in \mathbb{N}$, then there exists $\xi \in [0, 1 - \alpha]$ such that $f(\xi) = f(\xi + \alpha)$;*

(2) *Prove that for $0 < \alpha < 1$, $\frac{1}{\alpha} \notin \mathbb{N}$, there always exists $f \in C[0, 1]$, $f(0) = f(1)$ such that $\forall\, x \in [0, 1 - \alpha]$, there is $f(x) \ne f(x + \alpha)$.*

Proof. (1) Let $g(x) = f(x) - f(x + \alpha)$. Since $\frac{1}{\alpha} \in \mathbb{N}$, we know

$$\sum_{k=0}^{\frac{1}{\alpha}-1} g(k\alpha) = f(0) - f(1) = 0.$$

Then there must be $i, j \in \{0, 1, \cdots, \frac{1}{\alpha} - 1\}$, $i \neq j$, such that $g(i\alpha)g(j\alpha) \leq 0$. Hence there exists $\xi \in [i\alpha, j\alpha] \subset [0, 1-\alpha]$ such that $f(\xi) = f(\xi + \alpha)$.

(2) For $0 < \alpha < 1$, we define

$$f(x) = \sin^2\left(\frac{\pi x}{\alpha}\right) - x\sin^2\left(\frac{\pi}{\alpha}\right).$$

Clearly, f is continuous and $f(0) = 0 = f(1)$. If there is some $x_0 \in [0, 1-\alpha]$, such that $f(x_0) = f(x_0 + \alpha)$, we know there is $\alpha \sin^2\left(\frac{\pi}{\alpha}\right) = 0$. However, since $\frac{1}{\alpha} \notin \mathbb{N}$, we know it's impossible for $\sin^2\left(\frac{\pi}{\alpha}\right) = 0$. Hence, we have that $\forall\, x \in [0, 1-\alpha]$, there is $f(x) \neq f(x + \alpha)$. □

Problem 5.14. *Define $f \in C(\mathbb{R})$ satisfying*

$$f(f(x)) = -x^3 + \sin(x^2 + \ln(1 + |x|)). \tag{5.1}$$

Prove that this equation has no continuous solution.

Proof. Assume by contradiction that there is a continuous function f satisfies (5.1). Then we know that

$$\lim_{x\to+\infty} f(f(x)) = -\infty \quad \text{and} \quad \lim_{x\to-\infty} f(f(x)) = +\infty.$$

We conclude that there must be $\lim\limits_{x\to+\infty} f(x) = -\infty$. Indeed, if there exists a sequence $\{x_n\}$ satisfying $\lim\limits_{n\to\infty} x_n = +\infty$ but $f(x_n)$ is bounded, we know that $f(f(x_n))$ must be bounded since f is continuous, which contradicts with $\lim\limits_{x\to+\infty} f(f(x)) = -\infty$. Hence by the continuity of f, we have that $\lim\limits_{x\to+\infty} f(x) = +\infty$ or $\lim\limits_{x\to+\infty} f(x) = -\infty$. If $\lim\limits_{x\to+\infty} f(x) = +\infty$, then we'll get $\lim\limits_{x\to+\infty} f(f(x)) = +\infty$ which generates contradiction. So we must have $\lim\limits_{x\to+\infty} f(x) = -\infty$. Similarly we must have $\lim\limits_{x\to-\infty} f(x) = +\infty$. But using $\lim\limits_{x\to+\infty} f(x) = -\infty$ and $\lim\limits_{x\to-\infty} f(x) = +\infty$ we have $\lim\limits_{x\to+\infty} f(f(x)) = +\infty$, which also generates contradiction. Hence, (5.1) has no continuous solution. □

Remark 5.15. *We can prove a general conclusion: For any $f \in C(\mathbb{R})$, $\lim\limits_{x\to+\infty} f(f(x)) = -\infty$ and $\lim\limits_{x\to-\infty} f(f(x)) = +\infty$ cannot be simultaneously true.*

Problem 5.16. *Suppose that $g(x)$ is defined on $[0, 1]$, and $g(0) = 1$, $g(1) = 0$. If there exists a continuous function $h(x)$ such that $g(x) + h(x)$ is monotonic increasing on $[0, 1]$, prove that $[0, 1] \subset g([0, 1])$.*

Proof. Denote that $f = g + h$. Since f is monotonic increasing, we have that $f(x-0) \leq f(x) \leq f(x+0)$, $(0 < x < 1)$. Since h is continuous, we know that $g(x-0) \leq g(x) \leq g(x+0)$, $(0 < x < 1)$. For $\forall\, y \in (0, 1)$, we define

$$E_y = \{t \in [0, 1] \big| g(x) > y, \forall\, x \in [0, t]\}.$$

Since $g(0) = 1 > y$, we know that E_y is not empty. Then the supremum of E_y exists. Let $x_0 = \sup E_y$, then $g(x_0 - 0) \geq y \geq g(x_0 + 0)$. Combining above, we obtain that

$g(x_0-0)=g(x_0)=g(x_0+0)=y$, i.e. $y\in g([0,1])$. Hence $(0,1)\subset g([0,1])$. It is clear that $0,1\in g([0,1])$, we have that $[0,1]\subset g([0,1])$. $\square$

Exercise 5.17. *Let $f(x)$ be continuous on $\mathbb{R}$. Suppose that f is periodic with the minimal postive period $\mu>0$, μ is irrational. Show that $\lim\limits_{n\to\infty} f(n)$ does not exist.*

Hint: By Kronecker's Approximation Theorem, we know that the sequence of numbers $\{n\mu-[n\mu]\}$ is dense in the unit interval. Hence we know that for any $x_0\in[0,1]$, there exists a sequence $\{n_j\mu\}$ such that $n_j\mu-[n_j\mu]\to x_0$, $j\to\infty$. Then

$$f([n_j\mu])=f([n_j\mu]-n_j\mu)\to f(-x_0),\quad j\to\infty,$$

which means $\{f(n)\}$ does not converge. $\square$

Exercise 5.18. *Suppose that $x_0=1$, $x_n=x_{n-1}+\cos x_{n-1}$, $(n=1,2,\cdots)$. Prove that $x_n-\dfrac{\pi}{2}=o\left(\dfrac{1}{n^n}\right)$ as $n\to\infty$.*

Hint: Let $y_n=\frac{\pi}{2}-x_n$, $n=0,1,2,\cdots$. Then $y_n=y_{n-1}-\sin y_{n-1}$. By the inequality

$$x-\frac{x^3}{6}<\sin x<x,\quad x\in(0,+\infty),$$

we have

$$0<y_n=y_{n-1}-\sin y_{n-1}<\frac{y_{n-1}^3}{6}<y_{n-1}^3,\quad n\in\mathbb{N}_+.$$

Hence

$$y_n<y_0^{3^n}.$$

Note that $0<y_0<1$. We konw that there is $N\in\mathbb{N}$, such that $\forall\, n>N$,

$$0<y_nn^n<y_0^{3^n}n^n<n^ny_0^{n^2}=\left(\frac{n}{\left(\frac{1}{y_0}\right)^n}\right)^n<\frac{1}{2^n}.$$

Hence, we have

$$y_n=o\left(\frac{1}{n^n}\right),\quad n\to\infty.$$

$\square$

Exercise 5.19. *Suppose that $f\in C[0,1]$, $\lim\limits_{x\to0+0}\dfrac{f(x)-f(0)}{x}=\alpha<\beta=\lim\limits_{x\to1-0}\dfrac{f(x)-f(1)}{x-1}$. Prove that $\forall\,\lambda\in(\alpha,\beta)$, $\exists\, x_1,x_2\in[0,1]$, such that $\lambda=\dfrac{f(x_1)-f(x_2)}{x_1-x_2}$.*

Hint: Let $g(x)=f(x)-\lambda x$. Since

$$\lim_{x\to0+0}\frac{g(x)-g(0)}{x}=\alpha-\lambda<0,$$

we know there exists $\delta_1 > 0$, such that $0 < x < \delta_1$, there is $g(x) < g(0)$. Similarly, since

$$\lim_{x \to 1-0} \frac{g(x) - g(1)}{x - 1} = \beta - \lambda > 0,$$

we know there exists $\delta_2 > 0$, such that $0 < x < \delta_2$, there is $g(x) < g(1)$. Then we know that the minimum of g is achieved in $(0, 1)$. Assume that $g(x_0) = \min_{x \in [0,1]} g(x)$, then $g(x_0) < g(0)$, $g(x_0) < g(1)$. Note that if there exist $x_1, x_2 \in [0, 1]$, $x_1 \neq x_2$, such that $g(x_1) = g(x_2)$, then we have $f(x_1) - \lambda x_1 = f(x_2) - \lambda x_2$, i.e.

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} = \lambda.$$

Indeed, if $g(0) = g(1)$, we are done. Next, we assume that $g(0) \neq g(1)$. Without loss of generality, we assume that $g(0) < g(1)$, then there is $g(x_0) < g(0) < g(1)$. Since $g(x)$ is continuous on $[x_0, 1]$, we know there exists $\xi \in (x_0, 1)$ such that $g(\xi) = g(0)$, then we are done. □

Exercise 5.20. *Let a, b be two nonzero real numbers and a function $f : \mathbb{R} \longrightarrow [0, \infty)$ satisfying the functional equation*

$$f(x + a + b) + f(x) = f(x + a) + f(x + b). \tag{5.2}$$

(1) *Prove that f is periodic if a/b is rational.*
(2) *If a/b is not rational, could f be nonperiodic?*

Hint: (1) From (5.2), easy induction gives

$$f(x + a + nb) - f(x + nb) = f(x + a) - f(x), \quad \forall\, n \in \mathbb{Z}.$$

From $f(x + a + nb) - f(x + a) = f(x + nb) - f(x)$, easy induction gives

$$f(x + ma + nb) - f(x + ma) = f(x + nb) - f(x), \quad \forall\, m \in \mathbb{Z}.$$

So

$$f(x + ma + nb) + f(x) = f(x + ma) + f(x + nb), \quad \forall\, x \in \mathbb{R} \text{ and } \forall\, m, n \in \mathbb{Z}. \tag{5.3}$$

If $\frac{a}{b}$ is rational, we can choose m, n such that $ma + nb = 0$. Let then $u = |ma| > 0$ and (5.3) becomes

$$2f(x) = f(x + u) + f(x - u),$$

or also

$$f(x + u) - f(x) = f(x) - f(x - u).$$

From there, we easily get

$$f(x + nu) = f(x) + n(f(x + u) - f(x)), \quad \forall\, x \in \mathbb{R},\ \forall\, n \in \mathbb{Z}. \tag{5.4}$$

Then, if $f(x + u) - f(x) \neq 0$, setting $n \to +\infty$ or $n \to -\infty$ (depending on sign), (5.4) implies $f(x + nu) < 0$, contradicts with $f(x) \geq 0$. Hence

$$f(x + u) = f(x), \quad \forall\, x \in \mathbb{R} \text{ and for some } u > 0.$$

(2) Define

$$f(x) = \frac{1}{2}\left(\cos\frac{2\pi x}{a} + \cos\frac{2\pi x}{b}\right) + 1.$$

It's easy to verify that f satisfies (5.3). But f is periodic if and only if a/b is rational. This can be seen by assuming $f(T) = f(0) = 2$, which implies both $\cos\frac{2\pi T}{a}$ and $\cos\frac{2\pi T}{b}$ have to be 1, i.e. T/a and T/b have to be integers. $\square$

Remark 5.21. *From* (5.4) *in the proof of Exercise 5.20, we can see that f is only needed to be bounded above or below.*

Exercise 5.22. *Suppose that $f(x)$ is a uniformly continuous function on $[1, +\infty)$. Prove that $\overline{\lim\limits_{x\to+\infty}} \dfrac{f(x)}{x} < +\infty$.*

Hint: By Problem 5.1, we know that there exist $a, b > 0$ such that $|f(x)| \le ax+b$. Hence

$$\overline{\lim_{x\to+\infty}} \frac{f(x)}{x} \le \overline{\lim_{x\to+\infty}} \frac{|f(x)|}{x} \le \overline{\lim_{x\to+\infty}} \left(a + \frac{b}{x}\right) = a < +\infty.$$

$\square$

6. Week 9 (10.31)

Problem 6.1 (Mid 1)**.** *Calculate limitations.*

(1) $\lim\limits_{n\to\infty}[\sin(\ln(n+1))-\sin(\ln n)]$;

(2) $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+x\sin x}-1}{\arctan x^2}$;

(3) $\lim\limits_{x\to 0}(1+2x)^{\frac{(x+1)^2}{x}}$;

(4) $\lim\limits_{n\to\infty}[(n+\ln n)^a-n^a]$, *where* $0<a<1$;

(5) $\lim\limits_{x\to+\infty}\left(\dfrac{2^{1/x}+8^{1/x}}{2}\right)^x$.

Solution. (1)

$$\begin{aligned}\lim_{n\to\infty}|\sin(\ln(n+1))-\sin(\ln n)| &= \lim_{n\to\infty}\left|2\cos\left(\frac{\ln(n+1)+\ln n}{2}\right)\sin\left(\frac{\ln(n+1)-\ln n}{2}\right)\right| \\ &\le \lim_{n\to\infty}(\ln(n+1)-\ln n)=0.\end{aligned}$$

(2)

$$\lim_{x\to 0}\frac{\sqrt[3]{1+x\sin x}-1}{\arctan x^2}=\lim_{x\to 0}\frac{1+\frac{1}{3}x\sin x-1}{x^2}=\frac{1}{3}.$$

(3)

$$\lim_{x\to 0}(1+2x)^{\frac{(x+1)^2}{x}}=\lim_{x\to 0}(1+2x)^{x+\frac{1}{x}+2}=e^2.$$

(4)

$$\begin{aligned}\lim_{n\to\infty}[(n+\ln n)^a-n^a] &= \lim_{n\to\infty}n^a\left[\left(1+\frac{\ln n}{n}\right)^a-1\right] \\ &= \lim_{n\to\infty}\frac{a\ln n}{n^{1-a}}=0.\end{aligned}$$

(5) 4. (Problem 4.7 (7)). □

Problem 6.2 (Mid 2)**.** *Discuss the continuity of the following functions.*

(1) $f(x)=[|\cos x|]$;

(2) $f(x)=\dfrac{1}{1-e^{\frac{x}{1-x}}}$.

Solution. (1) It is easy to see that

$$0\le|\cos x|<1,\quad \forall\, x\in(n\pi,(n+1)\pi),\ n\in\mathbb{Z}.,$$

and

$$|\cos x|=1,\quad x=n\pi,\ n\in\mathbb{Z}.$$

Then f is discontinuous at $x = n\pi$, $\forall\, n \in \mathbb{Z}$, and it's the removable discontinuity.

(2) Note that

$$\lim_{x\to 1-0} \frac{1}{1-e^{\frac{x}{1-x}}} = 0, \qquad \lim_{x\to 1+0} \frac{1}{1-e^{\frac{x}{1-x}}} = 1,$$

and

$$\lim_{x\to 0-0} \frac{1}{1-e^{\frac{x}{1-x}}} = +\infty, \qquad \lim_{x\to 0+0} \frac{1}{1-e^{\frac{x}{1-x}}} = -\infty,$$

we know that $x = 1$ is the jump discontinuity and $x = 0$ is the discontinuity of second kind. The graph of $f(x)$ is as follows: $\square$

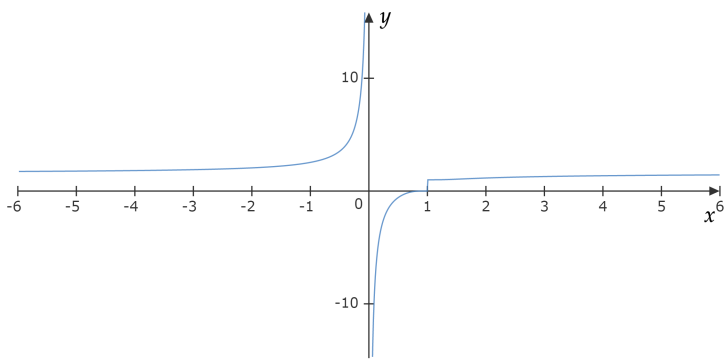

FIGURE 4. Graph of $f(x)$

Problem 6.3 (Mid 3)**.** *Find $n \in \mathbb{N}_+$, such that when $x \to 0$, $e^{x^n} - 1$ is a infinitesimals whose order is lower than $x(\cos\sqrt{x} - 1)(\sqrt[3]{x+1} - 1)$ but higher than $\sqrt{x}\ln(1 + \sqrt[3]{x})$.*

Solution. Note that when $x \to 0$, there are

$$x(\cos\sqrt{x} - 1)(\sqrt[3]{x+1} - 1) \sim -\frac{1}{6}x^3,$$

and

$$\sqrt{x}\ln(1 + \sqrt[3]{x}) \sim x^{\frac{5}{6}}.$$

Hence $n = 1, 2$. $\square$

Problem 6.4 (Mid 5)**.** *Suppose that $f(x)$ satisfying*

$$\lim_{x\to 0} \left(1 + x + \frac{f(x)}{x}\right)^{\frac{1}{x}} = e^3.$$

Prove that $\lim\limits_{x\to 0} \dfrac{f(x)}{x^2}$ exists, and find the limitation.

Proof. By $\lim_{x\to 0}\left(1+x+\frac{f(x)}{x}\right)^{\frac{1}{x}}=e^3$, we know that there must be $\lim_{x\to 0}\left(x+\frac{f(x)}{x}\right)=0$. What's more, we have

$$\lim_{x\to 0}\frac{1}{x}\ln\left(1+x+\frac{f(x)}{x}\right)=3.$$

Hence

$$\begin{aligned}\lim_{x\to 0}\frac{f(x)}{x^2}&=\lim_{x\to 0}\frac{1}{x}\left(x+\frac{f(x)}{x}\right)-1\\&=\lim_{x\to 0}\frac{1}{x}\ln\left(1+x+\frac{f(x)}{x}\right)\cdot\lim_{x\to 0}\frac{x+\frac{f(x)}{x}}{\ln\left(1+x+\frac{f(x)}{x}\right)}-1\\&=3-1=2.\end{aligned}$$

□

Problem 6.5 (Mid 6)**.** *Suppose sequence* $\{x_n\}$ *satisfies* $x_0=0$, $x_{2k}=\frac{x_{2k-1}}{2}$ *and* $x_{2k+1}=x_{2k}+\frac{1}{2}$. *Find the upper and lower limit of* $\{x_n\}$.

Solution. Firstly, we have

$$x_{2k+1}=x_{2k}+\frac{1}{2}=\frac{x_{2k-1}}{2}+\frac{1}{2},$$

which gives us

$$x_{2k+1}=\frac{2^{k+1}-1}{2^{k+1}}\quad\text{and}\quad x_{2k}=\frac{2^k-1}{2^{k+1}}.$$

Hence, we know

$$\varliminf_{n\to\infty}x_n=\frac{1}{2},\qquad\varlimsup_{n\to\infty}x_n=1.$$

□

Problem 6.6 (Mid 7)**.** *Soppose that* $x_0=1$, $x_{n+1}=\frac{1}{x_n^3+4}$. *Prove that* $\{x_n\}$ *converges to the unique positive zero point of equation* $x^4+4x-1=0$.

Proof. Let

$$g(x)=x^4+4x-1.$$

It's easy to see that $g(x)$ is increasing on $[0,+\infty)$. Since $g(0)=-1<0$, $g(1)=4>0$, we know that $g(x)$ has only one zero point in $[0,+\infty)$, say α. In particular, $0<\alpha<1$. Since $g(\alpha)=0$, we have that $\alpha=\frac{1}{\alpha^3+4}$. By induction, we know that $0<x_n<1$, $n\geq 1$. Hence, there is

$$|x_{n+1}-\alpha|=\left|\frac{1}{x_n^3+4}-\frac{1}{\alpha^3+4}\right|$$

$$
\begin{aligned}
&= \frac{|x_n^3 - \alpha^3|}{(x_n^3+4)(\alpha^3+4)} \\
&= \frac{|x_n - \alpha||x_n^2 + \alpha x_n + \alpha^2|}{(x_n^3+4)(\alpha^3+4)} \\
&\le \frac{3}{16}|x_n - \alpha| \le \cdots \\
&\le \left(\frac{3}{16}\right)^n |x_1 - \alpha| \\
&\to 0, \quad \text{as } n \to \infty.
\end{aligned}
$$

□

Problem 6.7 (Mid 10)**.** *Suppose that $f(x)$ is Lipschitz on $[1, +\infty)$, i.e. there exists a constant $C > 0$, such that $\forall\, x, y \in [1, +\infty)$, there is $|f(x) - f(y)| \le C|x - y|$. Prove that $\dfrac{f(x)}{x}$ is uniformly continuous on $[1, +\infty)$.*

Proof. **First way:** By Exercise 5.22, we know that $\dfrac{f(x)}{x}$ is bounded on $[1, +\infty)$, i.e. there exists $M > 0$, such that $\left|\dfrac{f(x)}{x}\right| \le M$. Note that $\forall\, x, y \in [1, +\infty)$, there is

$$
\begin{aligned}
\left|\frac{f(x)}{x} - \frac{f(y)}{y}\right| &= \left|\frac{f(x)}{x} - \frac{f(y)}{x} + \frac{f(y)}{x} - \frac{f(y)}{y}\right| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{1}{x}\left|\frac{f(y)}{y}\right||x - y| \\
&\le (C + M)|x - y|.
\end{aligned}
$$

i.e. $\dfrac{f(x)}{x}$ is also Lipschitz. Thus, it is uniformly continuous.

Second way: We will show $\dfrac{f(x)}{x}$ is Lipschitz, directly. Indeed,

$$
\begin{aligned}
\left|\frac{f(x)}{x} - \frac{f(y)}{y}\right| &= \left|\frac{f(x)}{x} - \frac{f(y)}{x} + \frac{f(y)}{x} - \frac{f(y)}{y}\right| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{|f(y)|}{xy}|x - y| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{|f(y) - f(1)| + |f(1)|}{xy}|x - y| \\
&\le \frac{C|x - y|}{x} + \frac{C(y-1) + |f(1)|}{xy}|x - y| \\
&\le (2C + |f(1)|)|x - y|.
\end{aligned}
$$

□

7. Week 10 (11.7)

Problem 7.1 (4.36)**.**

(4) $y = \sin^3 x$;

(6) $y = \dfrac{x^n}{1-x}$;

(8) $\dfrac{\ln x}{x}$.

Solution. (4) Note that

$$\begin{aligned}\sin^3 x &= \sin x(1-\cos^x) = \sin x - \sin x \cos^2 x \\ &= \frac{1}{2}\sin x - \frac{1}{2}\sin x \cos 2x \\ &= \frac{1}{2}\sin x - \frac{1}{4}\sin 3x + \frac{1}{4}\sin x \\ &= \frac{3}{4}\sin x - \frac{1}{4}\sin 3x.\end{aligned}$$

Hence

$$y^{(n)} = \frac{3}{4}\sin\left(x + \frac{n\pi}{2}\right) - \frac{3^n}{4}\sin\left(3x + \frac{n\pi}{2}\right).$$

(6)**First way:** Note that

$$x^n - 1 = (x-1)(x^{n-1} + x^{n-1} + \cdots + 1).$$

Second way: Note that

$$x^n = (x-1+1)^n = 1 + \sum_{k=1}^{n} C_n^k (x-1)^k.$$

(8) By the Leibniz formula, we have

$$\begin{aligned}y^{(n)} &= \left(\frac{\ln x}{x}\right)^{(n)} \\ &= \ln x \left(\frac{1}{x}\right)^{(n)} + \sum_{k=1}^{n} C_n^k (\ln x)^{(k)} \left(\frac{1}{x}\right)^{(n-k)} \\ &= \frac{(-1)^n n!}{x^{n+1}} \ln x + \sum_{k=1}^{n} \left(\frac{n!}{k!(n-k)!} \frac{(-1)^{k-1}(k-1)!}{x^k} \frac{(-1)^{n-k}(n-k)!}{x^{n-k+1}}\right) \\ &= \frac{(-1)^n n!}{x^{n+1}} \left(\ln x - \sum_{k=1}^{n} \frac{1}{k}\right).\end{aligned}$$

□

Problem 7.2 (4.41(3))**.** *If* $y = x^{n-1}e^{\frac{1}{x}}$*, then* $y^{(n)} = \dfrac{(-1)^n}{x^{n+1}}e^{\frac{1}{x}}$.

Proof. Prove by induction. Let $y_n := x^{n-1}e^{\frac{1}{x}}$. Then $y_1 = e^{\frac{1}{x}}$, we have

$$y_1' = \frac{-1}{x^2}e^{\frac{1}{x}}.$$

Suppose that $y_n^{(n)} = \dfrac{(-1)^n}{x^{n+1}}e^{\frac{1}{x}}$, we calculate $y_{n+1}^{(n+1)}$. By the Leibniz formula, we have

$$\begin{aligned}
y_{n+1}^{(n+1)} &= (y_{n+1}^{(n)})' = ((xy_n)^{(n)}) \\
&= (xy_n^{(n)} + ny_n^{(n-1)})' \\
&= xy_n^{(n+1)} + (n+1)y_n^{(n)} \\
&= \frac{(-1)^{n+1}}{x^{n+2}}e^{\frac{1}{x}} + \frac{(-1)^{n+1}(n+1)}{x^{n+1}}e^{\frac{1}{x}} + \frac{(-1)^n(n+1)}{x^{n+1}}e^{\frac{1}{x}} \\
&= \frac{(-1)^{n+1}}{x^{n+2}}e^{\frac{1}{x}}.
\end{aligned}$$

□

Problem 7.3 (4.44)**.** *Suppose that* $f(x)$ *is continuous at* $x = 0$*, and satisfying*

$$\lim_{x\to 0}\frac{f(2x) - f(x)}{x} = m.$$

Prove that $f'(0) = m$.

Proof. The proof is very similar to that of Problem 4.11, we omit the detail here. □

Problem 7.4. *Suppose that* $f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ *Prove that* $f^{(n)}(0) = 0$, $\forall\, n \in \mathbb{N}_+$.

Proof. We firstly calculate $f'(0)$. By definition, there is

$$f'(0) = \lim_{x\to 0}\frac{1}{x}\cdot e^{-\frac{1}{x^2}} = \lim_{y\to\infty} ye^{-y^2} = 0.$$

Next, a direct calculation yields the formula of $f'(x)$ when $x \neq 0$

$$f'(x) = \frac{2}{x^3}e^{-\frac{1}{x^2}}.$$

By definition, again, we have

$$f''(0) = \lim_{x\to 0}\frac{f'(x) - f'(0)}{x} = \lim_{x\to 0}\frac{2}{x^4}\cdot e^{-\frac{1}{x^2}} = 0.$$

Then, a direct calculation yields the formula of $f''(x)$ when $x \neq 0$

$$f''(x) = \left(\frac{4}{x^6} - \frac{6}{x^4}\right)e^{-\frac{1}{x^2}}.$$

Hence, we claim that the nth-order derivative of f when $x \neq 0$ is

$$f^{(n)}(x) = P_n\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}}, \tag{7.1}$$

where $P_n(y)$ is a polynomial of degree n. We prove this claim by induction. We already know that (7.1) is true for $n = 1, 2$. We suppose that (7.1) is valid for $f^{(k)}(x)$, we calculate $f^{(k+1)}(x)$ in the following.

$$\begin{aligned} f^{(k+1)}(x) &= (f^{(k)}(x))' \\ &= \left[P_k\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}}\right]' \\ &= P_k'\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) \cdot e^{-\frac{1}{x^2}} + P_k\left(\frac{1}{x}\right) \cdot \frac{2}{x^3} \cdot e^{-\frac{1}{x^2}} \\ &= \left[P_k\left(y\right)\left(-y^2\right) + P_k(y)(2y^3)\right]\Big|_{y=\frac{1}{x}} \cdot e^{-\frac{1}{x^2}}. \end{aligned}$$

Denote $P_{k+1}\left(\frac{1}{x}\right) := \left[P_k\left(y\right)\left(-y^2\right) + P_k(y)(2y^3)\right]\Big|_{y=\frac{1}{x}}$, then we obtain (7.1).

Finally, we prove $f^{(n)}(0) = 0$ by induction. It is true for $n = 1, 2$. We suppose that there is $f^{(k)}(0) = 0$, we prove that $f^{(k+1)}(0) = 0$. Indeed,

$$\begin{aligned} f^{(k+1)}(0) &= \lim_{x\to 0} \frac{f^{(k)}(x) - f^{(k)}(0)}{x} = \lim_{x\to 0} \frac{f^{(k)}(x)}{x} \\ &= \lim_{x\to 0} \frac{1}{x} \cdot P_k\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}} = 0. \end{aligned}$$

The graph of $f(x)$ is as follows:

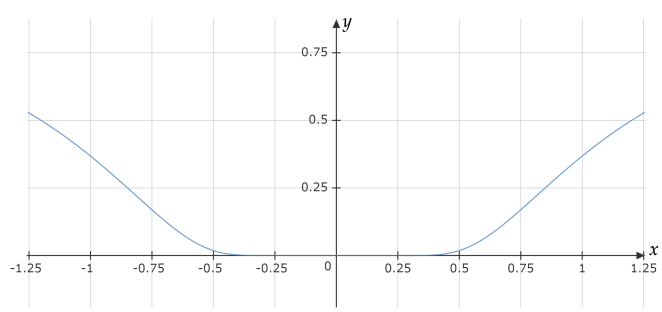

FIGURE 5. Graph of $f(x)$

□

Exercise 7.5. *Prove that there exists a smooth function* $f : \mathbb{R} \longrightarrow [0, 1]$ *such that* $f\big|_{(-\infty,0]} = 0$ *and* $f\big|_{[1,+\infty)} = 1$.

Hint: Firstly, define

$$g(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

Next, let

$$f(x) = \frac{g(x)}{g(x) + g(1-x)}.$$

It's easy to verify that f satisfies the condition. □

Problem 7.6. *Suppose that f is a polynomial of degree 7. If $f(x)+1$ is divisible by $(x-1)^4$ and $f(x)-1$ is divisible by $(x+1)^4$. Find f by the method of derivatives.*

Proof. Note that

$$f(x)+1 = p(x)(x-1)^4 \Rightarrow f'(x) = p'(x)(x-1)^4 + 4p(x)(x-1)^3 \Rightarrow f'(1) = 0,$$
$$f(x)-1 = q(x)(x+1)^4 \Rightarrow f'(x) = q'(x)(x+1)^4 + 4q(x)(x+1)^3 \Rightarrow f'(-1) = 0.$$

Hence we have

$$f'(x) = a(x-1)^3(x+1)^3.$$

Then

$$f(x) = \frac{a}{7}x^7 - \frac{3a}{5}x^5 + ax^3 - ax + b.$$

Since $f(1) = -1, f(-1) = 1$, we have that

$$-\frac{16}{35}a + b = -1, \quad \frac{16}{35}a + b = 1,$$

i.e.

$$a = \frac{35}{16}, \quad b = 0.$$

Hence

$$f(x) = \frac{1}{16}x\left(5x^6 - 21x^4 + 35x^2 - 35\right).$$

□

Problem 7.7. *Suppose that $f(x) = ax^2 + bx + c$, and $|f(x)| \le 1$, when $|x| \le 1$. Prove that $|f'(x)| \le 4$, when $|x| \le 1$.*

Proof. By $|f(-1)| = |a-b+c| \le 1$, $|f(0)| = |c| \le 1$, $|f(1)| = |a+b+c| \le 1$, there is

$$\begin{aligned} |2a+b| &= \left|\frac{1}{2}(a-b+c) - 2c + \frac{3}{2}(a+b+c)\right| \\ &\le \frac{1}{2}|a-b+c| + 2|c| + \frac{3}{2}|a+b+c| \\ &\le 4. \end{aligned}$$

Similarly, we have

$$|-2a+b| = \left|-\frac{3}{2}(a-b+c) + 2c - \frac{1}{2}(a+b+c)\right| \le 4.$$

Since the maximum of linear functions is achieved at endpoints, we know that

$$|f'(x)| = |2ax + b| \leq \max\{|2a + b|, |-2a + b|\} \leq 4, \quad \forall\, x \in [-1, 1].$$

□

Problem 7.8. *Prove that for every $n \in \mathbb{N}_+$ there is*

$$\sum_{k=0}^{n} (-1)^k C_n^k k^m = \begin{cases} 0, & 0 \leq m \leq n-1, \\ (-1)^n n!, & m = n. \end{cases}$$

Proof. Let $S_n^m = \sum_{k=0}^{n} (-1)^k C_n^k k^m$. We show that $S_n^m = 0$ if $0 \leq m \leq n-1$. Firstly, we have

$$S_n^0 = \sum_{k=0}^{n} (-1)^k C_n^k = (1-1)^n = 0.$$

In particular, $S_1^0 = 0$. By

$$kC_n^k = k\frac{n!}{k!(n-k)!} = nC_{n-1}^{k-1},$$

we know that when $1 \leq m \leq n-1$, there is

$$\begin{aligned} S_n^m &= n\sum_{k=1}^{n} (-1)^k C_{n-1}^{k-1} k^{m-1} \\ &= -n\sum_{k=0}^{n-1} (-1)^k C_{n-1}^k (k+1)^{m-1} \\ &= -n\sum_{k=0}^{n-1}\sum_{l=0}^{m-1} (-1)^k C_{n-1}^k C_{m-1}^l k^l \\ &= -n\sum_{l=0}^{m-1}\sum_{k=0}^{n-1} (-1)^k C_{n-1}^k C_{m-1}^l k^l \\ &= -n\sum_{l=0}^{m-1} C_{m-1}^l S_{n-1}^l. \end{aligned} \tag{7.2}$$

Hence, by induction, we have that $S_n^m = 0$ for $0 \leq m \leq n-1$. What's more, by (7.2), we have

$$S_n^n = -n\sum_{l=0}^{n-1} C_{n-1}^l S_{n-1}^l = -nS_{n-1}^{n-1} = (-1)^n n!.$$

Then the result follows. □

Exercise 7.9. *Given a positive integer n. Find*

$$S = \sum_{k=0}^{n} (-1)^k \binom{n}{k} k^{n+2},$$

and

$$T = \sum_{k=0}^{n} (-1)^k \binom{n}{k} k^{n+3}.$$

Hint: Using (7.2) and induction, we have

$$S = \frac{(-1)^n n(3n+1)(n+2)!}{24},$$

and

$$T = \frac{(-1)^n n^2(n+1)(n+3)!}{48}.$$

□

Problem 7.10. *Suppose that $f(x) = x^n \ln x$, $n \in \mathbb{N}_+$. Calculate* $\lim\limits_{n\to\infty} \dfrac{f^{(n)}(1/n)}{n!}$.

Proof. Denote that $f_n(x) = x^n \ln x$. Then

$$f_n'(x) = nx^{n-1} \ln x + x^{n-1} = nf_{n-1}(x) + x^{n-1}.$$

Hence, we have

$$f_n^{(n)}(x) = nf_{n-1}^{(n-1)}(x) + (n-1)!,$$

which gives us

$$\frac{f_n^{(n)}(x)}{n!} = \frac{f_{n-1}^{(n-1)}(x)}{(n-1)!} + \frac{1}{n} = \cdots = \ln x + 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

Then take $x = \frac{1}{n}$, ther is

$$\lim_{n\to\infty} \frac{f_n^{(n)}(1/n)}{n!} = \lim_{n\to\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n\right) = c,$$

where c is the Euler constant. □

Problem 7.11 (5.1)**.** *Prove the generalized Rolle's theorem, i.e. suppose that $f(x)$ is differentiable on (a, b), and $f(a+0) = f(b-0) = A$. Then there exists $\xi \in (a, b)$, such that $f'(\xi) = 0$, where a can be $-\infty$, b can be $+\infty$, A can be $+\infty$ or $-\infty$.*

Proof. We only prove the case that a, b and A are finite, others are similar and we leave them to the reader. The conclusion is clear if $f(x) = A$, $\forall\, x \in (a, b)$. Hence, without loss of generality, we may assume that there is at least a $x_0 \in (a, b)$ such that $f(x_0) > A$. By the definition of limits, we have there is a small $\delta > 0$ such that

$$f(x) < f(x_0), \quad \forall\, x \in (a, a+\delta) \cup (b-\delta, b).$$

Hence, we know that the maximum of $f(x)$ is achieved on $[a+\delta, b-\delta]$, thus there exists $\xi \in (a,b)$, such that $f'(\xi) = 0$. $\square$

Problem 7.12 (5.5)**.** *Prove that the Chebyshev-Laguerre polynomial*

$$L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x})$$

has n different zero points.

Proof. By Leibniz formula, it's easy to see that $L_n(x)$ is a polynomial of degree n. Hence, it has at most n zero points. The conclusion is clear for $n = 0, 1$, we show it's true for $n \geq 2$ in the following. Denote $g(x) = x^n e^{-x}$, then $L_n(x) = e^x g^{(n)}(x)$. It suffices to find all zero points of $g^{(n)}(x)$. Note that

$$g^{(l)}(x) = \sum_{k=0}^{l} C_l^k (x^n)^{(k)} (e^{-x})^{(l-k)} = \sum_{k=0}^{l} C_l^k n(n-1)\cdots(n-k+1)x^{n-k}(-1)^{l-k}e^{-x}.$$

Hence for $l < n$, there are always $g^{(l)}(0) = 0$ and $\lim\limits_{x\to+\infty} g^{(l)}(x) = 0$. Hence, by Rolle's theorem (Problem 7.11) and induction, we know that there is at least $n-1$ zero points of $g^{(l)}(x)$ between $(0, +\infty)$.By Rolle's theorem again, we have that $g^{(n)}(x)$ has at least n zero points in $(0, +\infty)$. Therefore, we know that $L_n(x)$ has n different zero points. $\square$

Problem 7.13. *Suppose that f is continuous on $[x_1, x_2]$, differentiable on (x_1, x_2). Show that there exists $\xi \in (x_1, x_2)$, such that* $\dfrac{1}{x_1 - x_2}\begin{vmatrix} x_1 & x_2 \\ f(x_1) & f(x_2) \end{vmatrix} = f(\xi) - \xi f'(\xi)$.

Proof. Note that

$$\frac{1}{x_1 - x_2}\begin{vmatrix} x_1 & x_2 \\ f(x_1) & f(x_2) \end{vmatrix} = \frac{x_1 f(x_2) - x_2 f(x_1)}{x_1 - x_2} = \frac{\frac{f(x_2)}{x_2} - \frac{f(x_1)}{x_1}}{\frac{1}{x_2} - \frac{1}{x_1}}.$$

By the Cauchy mean value theorem, we know that there exists $\xi \in (x_1, x_2)$, such that

$$\frac{\frac{f(x_2)}{x_2} - \frac{f(x_1)}{x_1}}{\frac{1}{x_2} - \frac{1}{x_1}} = \frac{\frac{\xi f'(\xi) - f()\xi}{\xi^2}}{-\frac{1}{\xi^2}} = f(\xi) - \xi f'(\xi).$$

Then the result follows. $\square$

Problem 7.14. *Suppose that $f(x)$ is differentiable on (a,b), $b < +\infty$, and $\lim\limits_{x\to b-0} f(x) = +\infty$. Prove that $\overline{\lim\limits_{x\to b-0}} f'(x) = +\infty$.*

Proof. Prove by contradiction. Assume that $\overline{\lim\limits_{x\to b-0}} f'(x) < +\infty$. Then there exist $M \in \mathbb{R}$, $\delta > 0$, such that $\forall\, x \in (b-\delta, b)$, there is $f'(x) \leq M$. Hence by the Lagrange mean value theorem, we know that $\forall\, x, y \in (b-\delta, b)$, $x > y$, there exist $\xi \in (b-\delta, b)$ such that

$$f(x) - f(y) = f'(\xi)(x-y) \leq M(x-y).$$

Let $x \to b-0$, and by $\lim\limits_{x\to b-0} f(x) = +\infty$, we have that

$$+\infty \le M(b-y),$$

contradiction. $\square$

Problem 7.15. *Suppose that $f(x)$ is continuous on $[a,b]$, differentiable on (a,b), and f is not a linear function. Prove that there exists $\xi \in (a,b)$, such that $f'(\xi) > \dfrac{f(b)-f(a)}{b-a}$.*

Proof. Prove by contradiction. Assume that $f'(x) \le \dfrac{f(b)-f(a)}{b-a}$, $\forall\, x \in (a,b)$. Define

$$F(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a).$$

Note that $F(a) = F(b) = 0$, and

$$F'(x) = f'(x) - \frac{f(b)-f(a)}{b-a} \le 0.$$

Hence

$$0 = F(a) \ge F(x) \ge F(b) = 0, \quad \forall\, x \in (a,b),$$

i.e. $F(x) = 0$, $\forall\, x \in (a,b)$. Therefore, we have

$$f(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a), \quad \forall\, x \in (a,b),$$

which implies that f is a linear function, contradiction. $\square$

Exercise 7.16. *Suppose that $f(x)$ is continuous on $[0,1]$, differentiable on $(0,1)$, and f is not a constant function. If $f(0) = 0$, prove that there exists $\xi \in (0,1)$ such that $f(\xi)f'(\xi) > 0$.*

Hint: Consider the function $F(x) := f^2(x)$. $\square$

Problem 7.17. *Suppose that $f(x)$ is differentiable on $[0,1]$, $f(0) = 0$, $f(1) = 1$, $k_1, \cdots, k_n$ are positive numbers. Prove that there are $x_1, \cdots, x_n \in [0,1]$, $x_i \ne x_j$, such that*

$$\sum_{i=1}^{n} \frac{k_i}{f'(x_i)} = \sum_{i=1}^{n} k_i.$$

Proof. Denote that $m = \sum\limits_{i=1}^{n} k_i$, $\lambda_i = \dfrac{k_i}{m}$. Then $0 < \lambda_i < 1$, $\lambda_1 + \cdots + \lambda_n = 1$. Since $f(0) = 0$, $f(1) = 1$ and $f(x)$ is continuous on $[0,1]$, we know that there exists $c_1 \in (0,1)$ such that $f(c_1) = \lambda_1$. Again, we know that there exists $c_2 \in (c_1, 1)$ such that $f(c_2) = \lambda_1 + \lambda_2$ since $\lambda_1 < \lambda_1 + \lambda_2 < 1$. Proceeding like this, we can find

$$0 < c_1 < c_2 < \cdots < c_n = 1,$$

such that

$$f(c_i) = \sum_{k=1}^{i} \lambda_k \quad (i = 1, 2, \cdots, n).$$

By the Lagrange mean value theorem, we have $x_i \in (c_{i-1}, c_i)$ $(c_0 = 0)$, such that

$$f'(x_i) = \frac{f(c_i) - f(c_{i-1})}{c_i - c_{i-1}} = \frac{\lambda_i}{c_i - c_{i-1}},$$

i.e.

$$\frac{\lambda_i}{f'(x_i)} = c_i - c_{i-1} \quad (i = 1, 2, \cdots, n).$$

Hence, we have

$$\sum_{i=1}^{n} \frac{\lambda_i}{f'(x_i)} = \sum_{i=1}^{n} (c_i - c_{i-1}) = c_n - c_0 = 1.$$

Recall that $\lambda_i = \dfrac{k_i}{m}$, we obtain

$$\sum_{i=1}^{n} \frac{k_i}{f'(x_i)} = \sum_{i=1}^{n} k_i.$$

□

Problem 7.18. *Suppose that $f(x)$ is differentiable on (a, b). Prove that the points in (a, b) are either the continuous point of $f'(x)$, or the discontinuous point of secong kind. i.e. $f'(x)$ has no discontinous points of first kind.*

Proof. Since $f(x)$ is differentiable on (a, b), we know that $\forall\, x_0 \in (a, b)$, there is

$$\begin{aligned} f'(x_0) = f'_+(x_0) &= \lim_{x \to x_0 + 0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{x \to x_0 + 0} \frac{f'(\xi)(x - x_0)}{x - x_0} \\ &= \lim_{x \to x_0 + 0} f'(\xi) \quad (x_0 < \xi < x). \end{aligned}$$

Hence, if $\lim\limits_{x \to x_0 + 0} f'(x)$ exists, there must be

$$f'(x_0) = \lim_{\xi \to x_0 + 0} f'(\xi) = f'(x_0 + 0).$$

Similarly, if $\lim\limits_{x \to x_0 - 0} f'(x)$ exists, there must be

$$f'(x_0) = f'(x_0 - 0).$$

Therefore $f'(x)$ is continuous at $x = x_0$ unless at least one of $\lim\limits_{x \to x_0 + 0} f'(x)$, $\lim\limits_{x \to x_0 - 0} f'(x)$ does not exist. □

Exercise 7.19 (Darboux Theorem)**.** *Suppose that $f(x)$ is differentiable on $[a, b]$, and $f'(a) < f'(b)$. Then $\forall\, c : f'(a) < c < f'(b)$, there exists $\xi \in (a, b)$, such that $f'(\xi) = c$.*

Hint: Define

$$g(x) = f(x) - cx, \quad \forall\, x \in [a, b].$$

Hence, $g(x)$ is differentiable on $[a, b]$. What's more, there are $g'(a) = f'(a) - c < 0$, $g'(b) = f'(b) - c > 0$. Note that

$$\lim_{x \to a+0} \frac{g(x) - g(a)}{x - a} = g'(a) < 0,$$

which implies that there is $\delta > 0$, such that $\forall\, x \in (a, a+\delta)$, there is $g(x) < g(a)$. Similarly, there is $\delta > 0$, such that $\forall\, x \in (b - \delta, b)$, there is $g(x) < g(b)$. Hence, the minimum of $g(x)$ is achieved on $[a + \delta, b - \delta]$, which yields that there is $\xi \in (a, b)$, such that $g'(\xi) = 0$, i.e. $f'(\xi) = c$. □

Exercise 7.20. *Suppuse that $f(x)$ is continuous on $[a, b]$, and differentiable on (a, c), (c, b). Prove that there exists $\xi \in (a, b)$ such that $|f(b) - f(a)| \leq |f'(\xi)||b - a|$.*

Hint: Using the Lagrange mean value theorem on (a, c), (c, b), respectively, and taking the maximum of intermediate points. □

Exercise 7.21. *Suppuse that $f(x)$ is continuous on $[0, 1]$, differentiable on $(0, 1)$, and $|f'(x)| < 1$. If $f(0) = f(1)$, prove that for any $x_1, x_2 \in (0, 1)$, there is*

$$|f(x_1) - f(x_2)| < \frac{1}{2}.$$

Hint: Consider $|x_1 - x_2| < \dfrac{1}{2}$ and $|x_1 - x_2| \geq \dfrac{1}{2}$, respectively. □

Exercise 7.22 (Challenge!)**.** *Suppose that $f(x)$, $f'_+(x) \in C(\mathbb{R})$. Prove that $f(x)$ is differentiable on $\mathbb{R}$.*

Hint: Firstly, prove the two lemmas:

Suppose that $f(x) \in C[a, b]$, $f(a) = f(b)$, $f'_+(x)$ exists on $[a, b)$. Prove that there exist $c, d \in [a, b)$ such that $f'_+(c) \leq 0$, $f'_+(d) \geq 0$.

Suppose that $f(x) \in C[a, b]$, $f'_+(x)$ exists on $[a, b)$. Prove that there exist $c, d \in [a, b)$ such that $f'_+(c) \leq \dfrac{f(b) - f(a)}{b - a} \leq f'_+(d)$.

Finally, using the continuity of $f'_+(x)$ to prove that $\forall\, x_0 \in \mathbb{R}$, there is

$$f'_-(x_0) = \lim_{x \to x_0 - 0} \frac{f(x) - f(x_0)}{x - x_0} = f'_+(x_0).$$

□

8. Week 11 (11.14)

Problem 8.1. *Calculate the following limitations.*

(1) $\lim\limits_{x\to 0+0} x^x$;

(2) $\lim\limits_{x\to 0+0} x^{x^x-1}$;

(3) $\lim\limits_{x\to +\infty} \left(\sqrt[3]{x^3-3x}-\sqrt{x^2-2x}\right)$;

(4) $\lim\limits_{x\to 0+0} \dfrac{x^x-(\sin x)^x}{x^2\ln(1+x)}$.

Solution. (1) By L'Hospital's rule, we have

$$\begin{aligned}\lim_{x\to 0+0} x^x &= \lim_{x\to 0+0} e^{x\ln x}\\ &= \exp\left\{\lim_{x\to 0+0}\frac{\ln x}{\frac{1}{x}}\right\}\\ &= \exp\left\{\lim_{x\to 0+0}\frac{\frac{1}{x}}{-\frac{1}{x^2}}\right\}\\ &= \exp\left\{\lim_{x\to 0+0} -x\right\}\\ &= 1.\end{aligned}$$

(2) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to 0+0} x^{x^x-1} &= e^{\lim\limits_{x\to 0+0}(x^x-1)\ln x}\\ &= e^{\lim\limits_{x\to 0+0} x(\ln x)^2} = 1.\end{aligned}$$

(3) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to +\infty}\left(\sqrt[3]{x^3-3x}-\sqrt{x^2-2x}\right) &= \lim_{x\to +\infty} x\left(\sqrt[3]{1-3x^{-2}}-\sqrt{1-2x^{-1}}\right)\\ &= \lim_{x\to +\infty} x(1-x^{-2}-1+x^{-1})\\ &= 1.\end{aligned}$$

(4) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to 0+0}\frac{x^x-(\sin x)^x}{x^2\ln(1+x)} &= \lim_{x\to 0+0}\frac{e^{x\ln x}-e^{x\ln\sin x}}{x^3}\\ &= \lim_{x\to 0+0} e^{x\ln x}\frac{1-e^{x\ln\frac{\sin x}{x}}}{x^3}\\ &= \lim_{x\to 0+0}\frac{1-e^{-\frac{1}{6}x^3}}{x^3}\end{aligned}$$

$$= \frac{1}{6}.$$

□

Exercise 8.2. *Prove that*

$$\left(1+\frac{1}{n}\right)^n = e - \frac{e}{2n} + \frac{11e}{24n^2} + o\left(\frac{1}{n^2}\right), \quad (n \to \infty).$$

Hint: By Taylor's formula, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^n &= e^{n\ln\left(1+\frac{1}{n}\right)} \\
&= e^{1-\frac{1}{2n}+\frac{1}{3n^2}+o\left(\frac{1}{n^2}\right)} \\
&= e\left(1-\frac{1}{2n}+\frac{1}{3n^2}+\frac{1}{2}\left(-\frac{1}{2n}+\frac{1}{3n^2}\right)^2\right)+o\left(\frac{1}{n^2}\right) \\
&= e - \frac{e}{2n} + \frac{11e}{24n^2} + o\left(\frac{1}{n^2}\right), \quad (n \to \infty).
\end{aligned}$$

□

Remark 8.3. *Similarly, for* $(1+x)^{\frac{1}{x}}$*, there is*

$$(1+x)^{\frac{1}{x}} = e - \frac{e}{2}x + \frac{11e}{24}x^2 + o\left(x^2\right), \quad (x \to 0).$$

Problem 8.4. *Suppose that* $f(x)$ *is twice differentiable on* $[0,1]$*,* $f(0) = f(1) = 0$*,* $\max\limits_{x\in[0,1]} f(x) = 2$*. Prove that* $\inf\limits_{x\in[0,1]} f''(x) \le -16$*.*

Proof. Since $f(x)$ is continuous on $[0,1]$, $f(0) = f(1) = 0$, and $\max\limits_{x\in[0,1]} f(x) = 2$, we know that there is $x_0 \in (0,1)$ such that

$$f(x_0) = \max_{x\in[0,1]} f(x).$$

Hence, by Fermat's theorem, there is

$$f'(x_0) = 0.$$

By Taylor's formula at $x = x_0$, we have

$$\begin{aligned}
0 = f(0) &= f(x_0) + \frac{1}{2}f''(\xi)(0-x_0)^2 = 2 + \frac{1}{2}f''(\xi)x_0^2, \\
0 = f(1) &= f(x_0) + \frac{1}{2}f''(\eta)(1-x_0)^2 = 2 + \frac{1}{2}f''(\eta)(1-x_0)^2.
\end{aligned}$$

Hence, we know

$$\inf_{x\in[0,1]} f''(x) \le \min\left\{f''(\xi), f''(\eta)\right\} = \min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\}.$$

Note that

$$\min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\} = -\frac{4}{(1-x_0)^2} \le -16, \quad x_0 \in \left[\frac{1}{2}, 1\right],$$
$$\min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\} = -\frac{4}{x_0^2} \le -16, \quad x_0 \in \left[0, \frac{1}{2}\right].$$

Therefore, we obtain that

$$\inf_{x\in[0,1]} f''(x) \le -16.$$

□

Remark 8.5. *Use the same method, we can prove that if $f(x)$ is twice differentiable on $[0,1]$, $f(0) = f(1) = 0$, $\min\limits_{x\in[0,1]} f(x) = -1$, then $\sup\limits_{x\in[0,1]} f''(x) \ge 8$.*

Problem 8.6 (5.16)**.** *Suppose that $f(x)$ is three times differentiable on $[a,b]$. Prove that there is $\xi \in (a,b)$, such that*

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 f^{(3)}(\xi).$$

Proof. Firstly, we choose M such that

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 M.$$

Define

$$F(x) = f(x) - f(a) - \frac{1}{2}(x-a)(f'(x) + f'(a)) + \frac{1}{12}(x-a)^3 M.$$

Hence, there is $F(a) = F(b) = 0$. By Rolle's theorem, we know that there exists $\eta \in (a,b)$ such that $F'(\eta) = 0$. Note that

$$F'(x) = f'(x) - \frac{1}{2}(f'(x) + f'(a)) - \frac{1}{2}(x-a)f''(x) + \frac{1}{4}(x-a)^2 M.$$

Hence, there is $F'(a) = F'(\eta) = 0$. By Rolle's theorem again, we know that there exists $\xi \in (a,\eta)$ such that $F''(\xi) = 0$. Note that

$$F''(x) = -\frac{1}{2}(x-a)f^{(3)}(x) + \frac{1}{2}(x-a)M.$$

Therefore, we have $M = f^{(3)}(\xi)$, i.e.

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 f^{(3)}(\xi).$$

□

Problem 8.7. *Suppose that $f(x)$ is twice differentiable on $[-1,1]$, $f(0) = f'(0) = 0$. Assume that $|f''(x)| \le |f(x)| + |f'(x)|$, $\forall\, x \in [-1,1]$. Prove that there exists $\delta > 0$, such that $f(x) = 0$, $\forall\, x \in (-\delta, \delta)$.*

Proof. Choose $\delta = 1/4$. Since $|f(x)| + |f'(x)|$ is continuous on $[-1/4, 1/4]$, we know that there is $x_0 \in [-1/4, 1/4]$ such that

$$|f(x_0)| + |f'(x_0)| = \max_{x\in[-1/4,1/4]} |f(x)| + |f'(x)| =: M.$$

By Taylor's formula, we have

$$\begin{aligned} f(x_0) &= f(0) + f'(0)x_0 + \frac{f''(\xi)}{2}x_0^2 = \frac{f''(\xi)}{2}x_0^2, \\ f'(x_0) &= f''(\eta)x_0. \end{aligned}$$

Hence, there is

$$\begin{aligned} |f(x_0)| + |f'(x_0)| &= \left|\frac{f''(\xi)}{2}x_0^2\right| + |f''(\eta)x_0| \\ &\le \frac{1}{4}(|f(\xi)| + |f'(\xi)|) + \frac{1}{4}(|f(\eta)| + |f'(\eta)|) \\ &\le \frac{1}{2}M, \end{aligned}$$

which implies that $M = 0$, i.e. $f(x) = 0$, $\forall\, x \in [-1/4, 1/4]$. □

Problem 8.8. *Suppose that $f(x)$ is twice differentiable on $\mathbb{R}$, and $f(x)$ is also a bounded function. Prove that there is $\xi \in \mathbb{R}$ such that $f''(\xi) = 0$.*

Proof. Assume that $f''(x) \neq 0$, $\forall\, x \in \mathbb{R}$. By Darboux's theorem (Exercise 7.19), we know that $f''(x)$ does not change sign. Without loss of generality, we may assume that $f''(x) > 0$, $\forall\, x \in \mathbb{R}$. Choosing $x_0 \in \mathbb{R}$ such that $f'(x_0) \neq 0$. By Taylor' formula, we have

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2}(x - x_0)^2 \ge f(x_0) + f'(x_0)(x - x_0),$$

which contradicts with f is bounded on $\mathbb{R}$. □

Problem 8.9. *Suppose that $f \in C^n(\mathbb{R})$, and there exist constants M_0, M_1 such that $|f(x)| \le M_0$, $|f^{(n)}(x)| \le M_1$, $\forall\, x \in \mathbb{R}$. Prove that there is $M > 0$ such that $|f^{(j)}(x)| \le M$, $j = 1, 2, \cdots, n-1$, $\forall\, x \in \mathbb{R}$.*

Proof. By Taylor's formula, we have

$$f(x+m) = f(x) + mf'(x) + \frac{m^2}{2!}f''(x) + \cdots + \frac{m^{n-1}}{(n-1)!}f^{(n-1)}(x) + \frac{m^n}{n!}f^{(n)}(\xi_m),$$

where $x < \xi_m < x+m$, $m = 1, 2, \cdots, n$. This is a linear system of equations about $f'(x)$, $f''(x)$, $\cdots$, $f^{(n-1)}(x)$, and the determinant of its coefficients is

$$\begin{vmatrix} 1 & 1 & \frac{1}{2!} & \cdots & \frac{1}{(n-1)!} \\ 1 & 2 & \frac{2^2}{2!} & \cdots & \frac{2^{n-1}}{(n-1)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & \frac{n^2}{2!} & \cdots & \frac{n^{n-1}}{(n-1)!} \end{vmatrix} = \frac{1}{1!2!\cdots(n-1)!} \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & \cdots & 2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{n-1} \end{vmatrix} = 1.$$

Hence, we know that $f'(x)$, $f''(x)$, $\cdots$, $f^{(n-1)}(x)$ can be written as linear combinations of $f(x+m)$ and $f^{(n)}(\xi_m)$, $m = 1, 2, \cdots, n$. Since $|f(x)| \le M_0$, $|f^{(n)}(x)| \le M_1$, $\forall\, x \in \mathbb{R}$, we have that there exists $M > 0$ such that $|f^{(j)}(x)| \le M$, $j = 1, 2, \cdots, n-1$, $\forall\, x \in \mathbb{R}$. □

Problem 8.10. *Suppose that $f(x)$ is bounded on $\mathbb{R}$ and $f'(x)$ is uniformly continuous on $\mathbb{R}$. Prove that $f'(x)$ is also bounded.*

Proof. Prove by contradiction. Without loss of generality, we may assume that $f'(x)$ has no upper bound. Hence, we know that $\forall\, n \in \mathbb{N}$, there exists $x_n \in \mathbb{R}$, such that $f'(x_n) > n$. Since $f'(x)$ is uniformly continuous on $\mathbb{R}$, we have that there exists $\delta > 0$ such that $\forall\, x, y : |x - y| < \delta$, there is

$$|f(x) - f(y)| < 1.$$

Then, there is $f(x) > f(x_n) - 1 > n - 1$, $\forall\, x \in (x_n, x_n + \delta)$. By Taylor's formula, we have

$$2 \sup_{x \in \mathbb{R}} |f(x)| \ge |f(x_n + \delta) - f(x_n)| = |f'(\xi_n)\delta| > (n-1)\delta \to +\infty, \quad n \to \infty,$$

which contradicts with $f(x)$ is bounded on $\mathbb{R}$. □

Remark 8.11. *Note that $\lim\limits_{x \to +\infty} f'(x)$ may not exist, for example, consider $f(x) = \sin x$. But if $\lim\limits_{x \to +\infty} f'(x)$ exists, there must be $\lim\limits_{x \to +\infty} f'(x) = 0$.*

Problem 8.12. *Suppose that $f \in C^3(\mathbb{R})$, and there exists $\theta \in (0, 1)$ such that*

$$f(x+h) = f(x) + hf'(x + \theta h), \quad \forall\, h \in \mathbb{R}. \tag{8.1}$$

Prove that f is a linear function or a quadratic function.

Proof. Differentiating (8.1) respect to h, we have

$$f'(x+h) = f'(x + \theta h) + \theta h f''(x + \theta h). \tag{8.2}$$

Hence, there is

$$\frac{f'(x+h) - f'(x) + f'(x) - f'(x + \theta h)}{h} = \theta f''(x + \theta h).$$

Letting $h \to 0$, we have

$$f''(x) - \theta f''(x) = \theta f''(x),$$

i.e.

$$f''(x) = 2\theta f''(x).$$

If $\theta \neq \dfrac{1}{2}$, we know that $f''(x) = 0$, $\forall\, x \in \mathbb{R}$, thus $f(x)$ is a linear function. If $\theta = \dfrac{1}{2}$, (8.2) yields

$$f'(x+h) = f'\left(x + \frac{1}{2}h\right) + \frac{1}{2}hf''\left(x + \frac{1}{2}h\right).$$

Differentiating the above formula respect to h yields

$$f''(x+h) = f''\left(x + \frac{1}{2}h\right) + \frac{1}{4}hf'''\left(x + \frac{1}{2}h\right).$$

Hence

$$\frac{f''(x+h) - f''\left(x + \frac{1}{2}h\right)}{\frac{1}{2}h} = \frac{1}{2}hf'''\left(x + \frac{1}{2}h\right).$$

Letting $h \to 0$, we have

$$f'''(x) = \frac{1}{2}f'''(x),$$

i.e. $f'''(x) = 0$, $\forall\, x \in \mathbb{R}$, thus $f(x)$ is a quadratic function. □

Problem 8.13. *If f is defined on $(0, +\infty)$ and f', f'' exists , with $\lim\limits_{x\to+\infty} f(x)$ exists and f'' bounded, prove that $\lim\limits_{x\to+\infty} f'(x) = 0$.*

Proof. Without loss of generality, we assume that $\lim\limits_{x\to+\infty} f(x) = 0$ (otherwise, replace $f(x)$ by $f(x) - \lim\limits_{x\to+\infty} f(x)$, and those conditions are still satisfied). Then $\forall\, \varepsilon > 0$, $\exists\, a \in \mathbb{R}$ such that $\sup\limits_{x\in(a,\infty)} |f(x)| < \varepsilon$.

Let

$$\sup_{x\in(0,\infty)} |f''(x)| = M_2,$$

(exists finitely in $\mathbb{R}$ as f'' is bounded.) So, for a defined above, there is

$$\sup_{x\in(a,\infty)} |f''(x)| \leq M_2.$$

Taking $h > 0$, by Taylor's theorem we have

$$f'(x) = \frac{1}{2h}[f(x+2h) - f(x)] - hf''(\xi)$$

for some $\xi \in (x, x+2h)$. Hence

$$|f'(x)| \leq \frac{\varepsilon}{h} + hM_2.$$

Then we obtain

$$h^2M_2 - h\,|f'(x)| + \varepsilon \geq 0, \quad \forall\, x \in (a, \infty),$$

which is a quadratic in h, and since $M_2 > 0$, we have

$$|f'(x)|^2 \leq 4M_2\varepsilon, \quad \forall\, x \in (a, \infty).$$

Hence

$$\lim_{x\to\infty} |f'(x)|^2 = 0 \Longrightarrow \lim_{x\to\infty} |f'(x)| = 0 \Longrightarrow \lim_{x\to\infty} f'(x) = 0.$$

□

Problem 8.14. *Suppose that $f \in C^\infty(a, b)$, and $f^{(n)}(x) \geq 0$ for all $n \in \mathbb{N}_+$. If $|f(x)| \leq M$, prove that for every $x \in (a, b)$, $r > 0$, $x + r \in (a, b)$, there is*

$$f^{(n)}(x) \leq \frac{2Mn!}{r^n}, \quad \forall\, n \in \mathbb{N}_+.$$

Proof. By Taylor's formula, we have

$$\begin{aligned} f(x+r) &= f(x) + f'(x)r + \cdots + \frac{f^{(n)}(x)}{n!}r^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}r^{n+1} \\ &\geq f(x) + \frac{f^{(n)}(x)}{n!}r^n, \end{aligned}$$

which implies

$$f^{(n)}(x) \leq \frac{(f(x+r) - f(x))n!}{r^n} \leq \frac{2Mn!}{r^n}.$$

□

Problem 8.15 (Bernstein Theorem)**.** *Suppose that $f \in C^\infty(a, b)$, and $f^{(n)}(x) \geq 0$ for all $n \in \mathbb{N}_+$. Prove that for every $x_0 \in (a, b)$, there exists $r > 0$, such that $\forall\, x \in [x_0 - r, x_0 + r] \subset (a, b)$, there is*

$$f(x) = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

Proof. Choosing $r > 0$ small enough, such that $[x_0 - 3r, x_0 + 3r] \subset (a, b)$. Denote

$$M = \sup_{x\in[x_0-2r, x_0+2r]} |f(x)|.$$

By Problem 8.14, we have for every $x \in [x_0 - 2r, x_0 + 2r]$, there is $f^{(n)}(x) \leq \dfrac{2Mn!}{(2r)^n}$. Then, we have

$$\begin{aligned} \left| f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right| &= \frac{f^{(n+1)}(\xi)}{(n+1)!}|x - x_0|^{n+1} \\ &\leq \frac{f^{(n+1)}(\xi)}{(n+1)!}r^{n+1} \leq \frac{M}{2^n}. \end{aligned}$$

Hence, there is

$$f(x) = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

$\square$

Exercise 8.16 (5.12)**.** *Suppose that $f(x)$ is differentiable on (a,b), and $f'(x)$ is monotonic. Prove that $f'(x)$ is continuous on (a,b).*

Hint: Since $f'(x)$ is monotonic, we know that for any $x_0 \in (a,b)$, $\lim\limits_{x\to x_0+0} f'(x)$ and $\lim\limits_{x\to x_0-0} f'(x)$ exist. Then by Problem 7.18, we have that $f'(x)$ is continuous on (a,b). $\square$

Exercise 8.17. *Suppose that $f(x)$ is continuous on $[a,b]$, and twice differentiable on (a,b). If $|f''(x)| \geq m > 0$, and $f(a) = f(b) = 0$. Prove that $\max\limits_{x\in[a,b]} |f(x)| \geq \dfrac{m}{8}(b-a)^2$.*

Hint: Denote $|f(x_0)| = \max\limits_{x\in[a,b]} |f(x)|$. It's easy to see that $f'(x_0) = 0$. Then

$$f(a) = f(x_0) + f'(x_0)(a - x_0) + \frac{f''(\xi)}{2}(x_0 - a)^2 = f(x_0) + \frac{f''(\xi)}{2}(x_0 - a)^2,$$
$$f(b) = f(x_0) + f'(x_0)(b - x_0) + \frac{f''(\eta)}{2}(x_0 - b)^2 = f(x_0) + \frac{f''(\eta)}{2}(x_0 - b)^2.$$

Hence

$$|f(x_0)| \geq \frac{m}{2}(x_0 - a)^2 \geq \frac{m}{8}(b-a)^2, \quad x_0 \in \left[\frac{a+b}{2}, b\right],$$
$$|f(x_0)| \geq \frac{m}{2}(x_0 - b)^2 \geq \frac{m}{8}(b-a)^2, \quad x_0 \in \left[a, \frac{a+b}{2}\right].$$

$\square$

9. Week 12 (11.21)

Problem 9.1 (Schwarz Theorem)**.** *Define the generalized second order derivative as follows*

$$f^{[2]}(x) = \lim_{h\to 0+0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2}.$$

If $f \in C[a,b]$, and $f^{[2]}(x) = 0$ on (a,b), prove that f is a linear function.

Proof. For any $x \in [a,b]$ and $\forall\, \varepsilon > 0$, define

$$f_\varepsilon(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) + \varepsilon(x-a)(x-b).$$

Then $f_\varepsilon(a) = f_\varepsilon(b) = 0$. Note that

$$\begin{aligned}
f_\varepsilon^{[2]}(x) &= \lim_{h\to 0+0} \frac{f_\varepsilon(x+h)+f_\varepsilon(x-h)-2f_\varepsilon(x)}{h^2} \\
&= \lim_{h\to 0+0} \frac{1}{h^2}\Bigg(f(x+h) - f(a) - \frac{f(b)-f(a)}{b-a}(x+h-a) + \varepsilon(x+h-a)(x+h-b) \\
&\quad + f(x-h) - f(a) - \frac{f(b)-f(a)}{b-a}(x-h-a) + \varepsilon(x-h-a)(x-h-b) \\
&\quad - 2\left(f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) + \varepsilon(x-a)(x-b)\right) \\
&= \lim_{h\to 0+0} \frac{1}{h^2}\left(f(x+h)+f(x-h)-2f(x)+\varepsilon\cdot 2h^2\right) \\
&= f^{[2]}(x) + 2\varepsilon = 2\varepsilon.
\end{aligned}$$

We claim that $f_\varepsilon \le 0$, when $\varepsilon > 0$. Indeed, if there is $x_0 \in (a,b)$ such that $f_\varepsilon(x_0) > 0$, by $f_\varepsilon(a) = f_\varepsilon(b) = 0$ and the continuity of f_ε, we know that the maximum of f_ε is achieved in (a,b), say $f_\varepsilon(x^*) = \max\limits_{x\in[a,b]} f_\varepsilon(x)$. Hence, there is

$$f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h) \le 2f_\varepsilon(x^*).$$

Since $f_\varepsilon^{[2]}(x^*) = 2\varepsilon > 0$, we know that there is $h > 0$ such that

$$f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h) - 2f_\varepsilon(x^*) > 0.$$

Therefore, there is

$$f_\varepsilon(x^*) < \frac{1}{2}(f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h)) \le f_\varepsilon(x^*),$$

contradiction. Similarly, we can prove that $f_\varepsilon \ge 0$, when $\varepsilon < 0$. By the continuity of f_ε respect to ε , we know that

$$f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) = f_0(x) = 0,$$

i.e. f is a linear function. $\square$

Problem 9.2. *If $f'(x_0) > (<) 0$, can we say that f is monotonic on a small enough neighborhood of $x = x_0$?*

Solution. No. For example (6.59 in book), let

$$f(x) = \begin{cases} x + 2x^2 \sin \dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

It's easy to see that $f'(0) = 1 > 0$. However, by

$$f'(x) = 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x}, \quad x \neq 0,$$

we have

$$f'\left(\frac{1}{n\pi}\right) = 1 - 2(-1)^n,$$

which means that $f'(x)$ can not be always positve or negative on any neighboehoods of $x = 0$, hence f is not monotonic on any neighboehoods of $x = 0$. The graph of $f(x)$ is as follows:

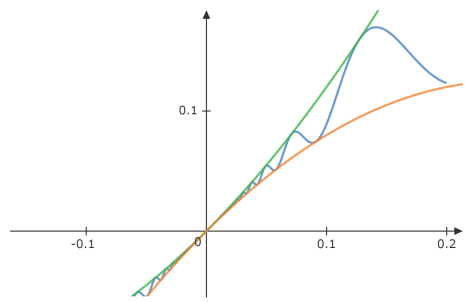

FIGURE 6. Graph of $f(x)$

□

Problem 9.3 (5.37)**.** *Suppose that $f(x)$ has nth-order derivative on $(a, +\infty)$. If $\lim\limits_{x \to +\infty} f(x) = A$, $\lim\limits_{x \to +\infty} f^{(n)}(x) = B$. Prove that $B = 0$.*

Proof. By L'Hospital's rule, we have

$$\begin{aligned} 0 &= \lim_{x \to +\infty} \frac{f(x)}{x^n} \\ &= \lim_{x \to +\infty} \frac{f'(x)}{nx^{n-1}} \\ &\vdots \end{aligned}$$

$$
\begin{aligned}
&= \lim_{x\to+\infty} \frac{f^{(n)}(x)}{n!} \\
&= \frac{B}{n!},
\end{aligned}
$$

which gives us $B = 0$. $\square$

Problem 9.4 (5.48)**.** *Suppose that $f(x)$ is defined on $[a, b]$ satisfying*

$$|f(x) - f(y)| \le k|x - y|^{1+\alpha}, \quad \forall\, x, y \in [a, b],\ \alpha > 0. \tag{9.1}$$

Prove that $f(x)$ is constant.

Proof. For $\forall\, x_0 \in [a, b]$, by (9.1), we have

$$\left|\frac{f(x) - f(x_0)}{x - x_0}\right| \le k|x - x_0|^{\alpha}, \quad \forall\, x \ne x_0.$$

Hence, letting $x \to x_0$, we know

$$f'(x_0) = 0, \quad \forall\, x_0 \in [a, b],$$

which means that $f(x)$ is constant. $\square$

Problem 9.5. *Find the extreme value of the following functions.*

(50(3)) $f(x) = \dfrac{(\ln x)^2}{x}$;

(50(4)) $f(x) = |x(x^2 - 1)|$;

(51(2)) $f(x) = 2\tan x - \tan^2 x$, $x \in [0, \pi/2)$;

(51(3)) $f(x) = \sqrt{x}\ln x$, $x \in (0, +\infty)$.

Solution. (50(3)) The derivative of $f(x)$ is

$$f'(x) = \frac{\ln x(2 - \ln x)}{x^2}.$$

Hence the local minimum is $f(1) = 0$ and the local maximum is $f(e^2) = 4/e^2$. The graph of $f(x)$ is as follows:

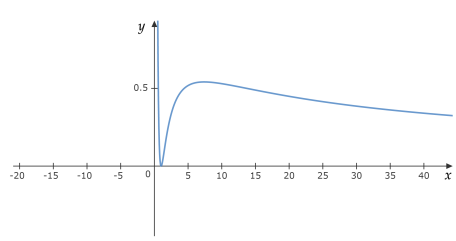

FIGURE 7. Graph of $f(x)$

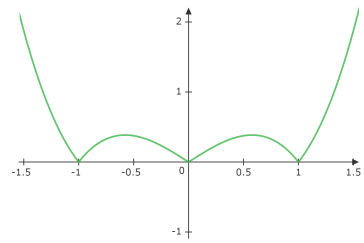

FIGURE 8. Graph of $f(x)$

(50(4))The graph of $f(x)$ is as follows: When $x = 0$, $x = \pm 1$, $f(x)$ has local minimun 0. When $x = \pm\frac{\sqrt{3}}{3}$, $f(x)$ has local maximun $\frac{2\sqrt{3}}{9}$.

(51(2)) The graph of $f(x)$ is as follows:

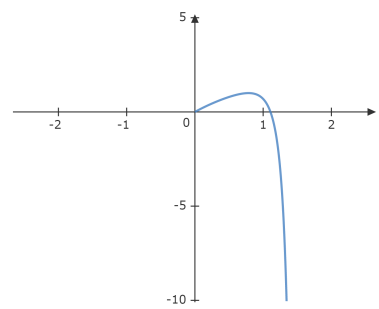

FIGURE 9. Graph of $f(x)$

(51(3)) The graph of $f(x)$ is as follows:

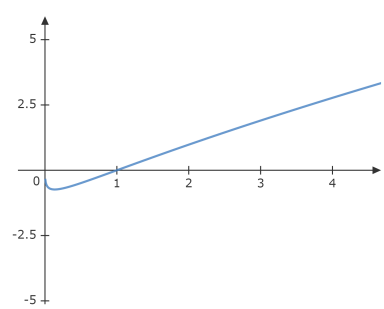

FIGURE 10. Graph of $f(x)$

□

Problem 9.6 (5.54(2)). *Suppose that $f(x)$ and $g(x)$ are differentiable on (a,b), and denote*

$$F(x) = f(x)g'(x) - f'(x)g(x), \quad x \in (a,b).$$

If $F(x) > 0$, $x \in (a,b)$. Prove that there must be zero points of $g(x)$ between the two zero points of $f(x)$.

Proof. Assume that x_1, $x_2 \in (a,b)$ are two zero points of $f(x)$. If $\forall\, x \in (x_1, x_2)$, there is $g(x) \neq 0$. Define

$$G(x) = \frac{f(x)}{g(x)}, \quad \forall\, x \in (x_1, x_2).$$

Hence, we have

$$G'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} < 0,$$

which gives us that $G(x)$ is monotonic decreasing on $[x_1, x_2]$. Then there is

$$0 = G(x_1) \geq G(x) \geq G(x_2) = 0,$$

i.e. $G(x) \equiv 0$, which contradicts with $F(x) > 0$. Hence we know that there must be zero points of $g(x)$ between x_1 and x_2. □

Problem 9.7. *Suppose that $f(x)$ is twice differentiable on $[a,b]$ satisfying*

$$f''(x) + b(x)f'(x) + c(x)f(x) = 0, \quad \forall\, x \in [a,b], \tag{9.2}$$

where $c(x) < 0$. Prove that

(1) *$f(x)$ can not admit a positive maximum or negative minimum in (a,b);*
(2) *If $f(a) = f(b) = 0$, then $f(x) \equiv 0$.*

Proof. (1) Prove by contradiction. Assume that $f(x)$ has a positive maximum in (a,b), i.e. there is $x_0 \in (a,b)$ such that $f(x_0) = \max\limits_{x\in[a,b]} f(x) > 0$. By the necessary condition for maximum value, we know that $f'(x_0) = 0$, $f''(x_0) \leq 0$. Then combining with $c(x_0) < 0$, we have

$$f''(x_0) + b(x_0)f'(x_0) + c(x_0)f(x_0) \leq c(x_0)f(x_0) < 0,$$

which contradicts with (9.2). Hence, we have that $f(x)$ can not admit a positive maximum in (a,b). Using the same method, we can prove that $f(x)$ can not admit a negative minimum in (a,b).

(2) By (1), we know that $\max\limits_{x\in[a,b]} f(x) \leq 0$ and $\min\limits_{x\in[a,b]} f(x) \geq 0$. Hence

$$0 \leq \min_{x\in[a,b]} f(x) \leq \max_{x\in[a,b]} f(x) \leq 0,$$

i.e. $f(x) \equiv 0$. □

Exercise 9.8 (5.67). *Suppose that the tangent line of the elliptic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ intersects the x-axis and y-axis at A and B, respectively.*

(1) *Find the minimum length of AB;*

(2) *Find the minimum area of the triangle formed by AB and the coordinate axis.*

Hint: (1) The tangent line at (x_0, y_0) is

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1.$$

Hence we know that $A\left(\frac{a^2}{x_0}, 0\right)$, $B\left(\frac{b^2}{y_0}, 0\right)$. Then

$$\begin{aligned} |AB| &= \sqrt{\frac{a^4}{x_0^2} + \frac{b^4}{y_0^2}} \\ &= \sqrt{\frac{a^2}{\frac{x_0^2}{a^2}} + \frac{b^2}{\frac{y_0^2}{b^2}}} \\ &\geq \sqrt{\frac{(a+b)^2}{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}}} \\ &= a + b. \end{aligned}$$

(2) By (1), we know

$$S_{\Delta AOB} = \frac{1}{2}\frac{a^2 b^2}{|x_0 y_0|} \geq \frac{ab}{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}} = ab.$$

The graph of the elliptic when $a = 2$, $b = \sqrt{3}$ is as follows:

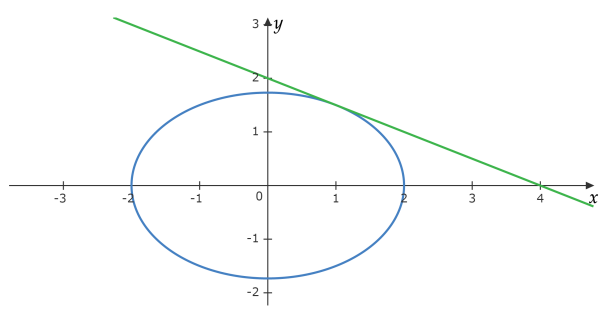

FIGURE 11. Graph of $\frac{x^2}{4} + \frac{y^2}{3} = 1$

□

Problem 9.9. *Suppose that $P(x)$ is a polynomial function. Prove that*

(1) *If $P'(x) + P(x) \geq 0$, then $P(x) \geq 0$;*

(2) *If $P(x) - P'(x) \geq 0$, then $P(x) \geq 0$;*

(3) *If* $P'''(x) - P''(x) - P'(x) + P(x) \geq 0$, *then* $P(x) \geq 0$.

Proof. (1) Let

$$F(x) = P(x)e^x.$$

Then there is

$$F'(x) = (P'(x) + P(x))e^x \geq 0.$$

Since $\lim\limits_{x\to-\infty} F(x) = 0$, we know that $F(x) \geq 0$, $\forall\, x \in \mathbb{R}$. Note that $e^x > 0$, $\forall\, x \in \mathbb{R}$, we have $P(x) \geq 0$.

(2) Let

$$G(x) = P(x)e^{-x}.$$

Then there is

$$G'(x) = (P'(x) - P(x))e^{-x} \leq 0.$$

Since $\lim\limits_{x\to+\infty} G(x) = 0$, we know that $G(x) \geq 0$, $\forall\, x \in \mathbb{R}$. Note that $e^{-x} > 0$, $\forall\, x \in \mathbb{R}$, we have $P(x) \geq 0$.

(3) Denote

$$Q(x) = P''(x) - P(x).$$

We have

$$Q'(x) - Q(x) \geq 0.$$

Hence by (2), we know that $Q(x) \leq 0$, i.e. $P(x) - P''(x) \geq 0$. Note that

$$P(x) - P''(x) = P(x) - P'(x) + P'(x) - P''(x),$$

we have by (1) that $P(x) - P'(x) \geq 0$. Then by (2) again, we have that $P(x) \geq 0$. □

Problem 9.10. *Suppose that* $f(x)$ *is a bounded convex function on* (a, b). *Prove that* $\lim\limits_{x\to a+0} f(x)$ *and* $\lim\limits_{x\to b-0} f(x)$ *exist.*

Proof. Since $f(x)$ is bounded, we may assume that $|f(x)| \leq M$, $\forall\, x \in (a, b)$. Let $x > x_1 > x_0$ be any points in (a, b). By the convexity of $f(x)$, we have $\dfrac{f(x) - f(x_0)}{x - x_0}$ is monotonic inreasing respect to x. Since

$$\frac{f(x) - f(x_0)}{x - x_0} \leq \frac{M - f(x_0)}{x_1 - x_0}, \quad \forall\, x > x_1 > x_0,$$

we have

$$\lim_{x\to b-0} \frac{f(x) - f(x_0)}{x - x_0} = A$$

by the monotone bounded convergence theorem. Then

$$\begin{aligned}\lim_{x\to b-0} f(x) &= \lim_{x\to b-0} \left[(x - x_0) \cdot \frac{f(x) - f(x_0)}{x - x_0} + f(x_0)\right] \\ &= A(b - x_0) + f(x_0),\end{aligned}$$

which is $\lim\limits_{x\to b-0} f(x)$ exists. Similarly, we can prove that $\lim\limits_{x\to a+0} f(x)$ exists. □

Problem 9.11. *Suppose that $f(x)$ is convex on (a,b). Prove that $\forall\,[c,d]\subset(a,b)$, $f(x)$ is Lipschitz continuous on $[c,d]$.*

Proof. Since $[c,d]\subset(a,b)$, we can choose $h>0$ small enough such that

$$[c-h,d+h]\subset(a,b).$$

Indeed, it suffices to choose $0<h<\min\{c-a,b-d\}$. Then $\forall\,x_1,x_2\in[c,d]$, if $x_1<x_2$, we take $x_3=x_2+h$. By the convexity of $f(x)$, we have

$$\frac{f(x_2)-f(x_1)}{x_2-x_1}\le\frac{f(x_3)-f(x_2)}{x_3-x_2}\le\frac{M-m}{h},$$

where $M=\sup\limits_{x\in[c-h,d+h]}f(x)$, $m=\inf\limits_{x\in[c-h,d+h]}f(x)$. If $x_2<x_1$, we take $x_3=x_2-h$. By the convexity of $f(x)$, we have

$$\frac{f(x_2)-f(x_3)}{x_2-x_3}\le\frac{f(x_1)-f(x_2)}{x_1-x_2}.$$

Then

$$\frac{f(x_2)-f(x_1)}{x_1-x_2}\le\frac{f(x_3)-f(x_2)}{x_2-x_3}\le\frac{M-m}{h}.$$

Hence, we have

$$f(x_2)-f(x_1)\le\frac{M-m}{h}|x_1-x_2|.$$

Switching x_1 and x_2, we know the above inequality is still valid. Hence, we know

$$|f(x_1)-f(x_2)|\le\frac{M-m}{h}|x_1-x_2|,\quad\forall\,x_1,x_2\in[c,d],$$

i.e. $f(x)$ is Lipschitz continuous on $[c,d]$. □

Problem 9.12. *Prove that the non-differentiable points of a convex function are countable.*

Proof. Let $f(x)$ be a convex function. For $\forall\,x<x_0<y$, by the convexity of f, we have

$$\frac{f(x)-f(x_0)}{x-x_0}\le\frac{f(y)-f(x_0)}{y-x_0}.$$

It's easy to see that $\dfrac{f(x)-f(x_0)}{x-x_0}$ is monotonic inreasing respect to x. Hence by the monotone bounded convergence theorem, we have

$$f'_-(x_0)\le f'_+(x_0).$$

If $f(x)$ is non-differentiable at $x=x_0$, there is $f'_-(x_0)<f'_+(x_0)$. Then every non-differentiable point x_0 is corresponding to a rational number in $\big(f'_-(x_0),f'_+(x_0)\big)$, and $\big\{\big(f'_-(x_0),f'_+(x_0)\big)\big\}$ is disjoint pairwise, thus the non-differentiable points of $f(x)$ are countable. □

Exercise 9.13. *Suppose that $f(x)$ is a bounded convex function on $\mathbb{R}$. Prove that f is a constant function.*

Hint: Prove by contradiction. Suppose that f is not a constant function. By Problem 9.12, we know that $f'_-(x)$ and $f'_+(x)$ exist. Then there is at least a x_0 such that $f'_-(x_0) \neq 0$ or $f'_+(x_0) \neq 0$. Without loss of generality, we assume that $f'_+(x_0) \neq 0$. Then by the convexity of f, we have

$$f(x) \geq f(x_0) + f'_+(x_0)(x - x_0),$$

which contradicts with f is bounded. $\square$

Exercise 9.14. *Suppose that $f \in C^2[0,1]$ is a nonegative function. Assume that $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, and $\forall\, x \in (0,1]$, $f(x) \neq x$. For any given $x_0 \in (0,1]$, define $x_{n+1} = f(x_n)$. Calculate the limitation $\lim\limits_{n\to\infty} nx_n$.*

Hint: By Taylor's formula, we have

$$f(x) = x - \frac{1}{2}x^2 + o(x^2).$$

Hence, for $x > 0$ small enough, we have $f(x) < x$. By $\forall\, x \in (0,1]$, $f(x) \neq x$ and the continuity of f, we know that $f(x) \leq x$. Hence, $\lim\limits_{n\to\infty} x_n$ exists and $\lim\limits_{n\to\infty} x_n = 0$. By Stolz theorem and Taylor's formula, we have

$$\begin{aligned}\lim_{n\to\infty} nx_n &= \lim_{n\to\infty} \frac{n}{\frac{1}{x_n}} = \lim_{n\to\infty} \frac{x_n x_{n+1}}{x_n - x_{n+1}} \\ &= \lim_{n\to\infty} \frac{x_n f(x_n)}{x_n - f(x_n)} = \lim_{n\to\infty} \frac{x_n^2}{\frac{1}{2}x_n^2} = 2.\end{aligned}$$

$\square$

Exercise 9.15. *Suppose that f has nth-order derivatives at $x = x_0$. Prove that*

$$f^{(n)}(x_0) = \lim_{h\to 0} \frac{1}{h^n} \sum_{k=0}^{n} (-1)^{n-k} C_n^k f(x_0 + kh).$$

Hint:

$$
\begin{aligned}
&\lim_{h\to 0}\frac{\sum_{k=0}^{n}(-1)^{n-k}\mathrm{C}_n^k f\left(x_0+kh\right)}{h^n}\\
&=\lim_{h\to 0}\frac{\sum_{k=0}^{n}(-1)^{n-k}\mathrm{C}_n^k\left(\sum_{m=0}^{n}\frac{f^{(m)}(x_0)}{m!}k^m h^m+o\left(h^n\right)\right)}{h^n}\\
&=\lim_{h\to 0}\frac{\sum_{k=0}^{n}(-1)^{n-k}C_n^k\sum_{m=0}^{n}\frac{f^{(m)}(x_0)}{m!}k^m h^m}{h^n}\\
&=\lim_{h\to 0}\frac{\sum_{k=0}^{n}\sum_{m=0}^{n}(-1)^{n-k}C_n^k\frac{f^{(m)}(x_0)}{m!}k^m h^m}{h^n}\\
&=\lim_{h\to 0}\frac{\sum_{m=0}^{n}\sum_{k=0}^{n}(-1)^{n-k}C_n^k\frac{f^{(m)}(x_0)}{m!}k^m h^m}{h^n}\\
&=\lim_{h\to 0}\frac{\sum_{m=0}^{n}\frac{f^{(m)}(x_0)}{m!}h^m\sum_{k=0}^{n}(-1)^{n-k}C_n^k k^m}{h^n}\\
&=\lim_{h\to 0}\frac{(-1)^n\cdot(-1)^n n!\frac{f^{(n)}(x_0)}{n!}h^n}{h^n}\\
&=f^{(n)}\left(x_0\right),
\end{aligned}
$$

where we used Problem 7.8 in the last equality. $\square$

10. Week 13 (11.28)

Problem 10.1. *Prove that $f(x)$ is a convex function on (a,b) if and only if $\forall\, x_1, x_2 \in (a,b)$, there is $\varphi(\lambda) = f(\lambda x_1 + (1-\lambda)x_2)$ is a convex function on $[0,1]$.*

Proof. "$\Rightarrow$" For $\forall\, \lambda_1, \lambda_2 \in [0,1]$, $\alpha \in [0,1]$, there is

$$\begin{aligned}\varphi(\alpha\lambda_1 + (1-\alpha)\lambda_2) &= f((\alpha\lambda_1 + (1-\alpha)\lambda_2)x_1 + (1-(\alpha\lambda_1 + (1-\alpha)\lambda_2))x_2)\\ &= f(\alpha(\lambda_1 x_1 + (1-\lambda_1)x_2) + (1-\alpha)(\lambda_2 x_1 + (1-\lambda_2)x_2))\\ &\le \alpha\varphi(\lambda_1) + (1-\alpha)\varphi(\lambda_2),\end{aligned}$$

where we used the convexity of f in the last inequality. Hence $\varphi(\lambda)$ is a convex function on $[0,1]$.
"$\Leftarrow$" It is easy to see that

$$\begin{aligned}f(\lambda x_1 + (1-\lambda)x_2) = \varphi(\lambda) &= \varphi(\lambda\cdot 1 + (1-\lambda)\cdot 0)\\ &\le \lambda\varphi(1) + (1-\lambda)\varphi(0)\\ &= \lambda f(x_1) + (1-\lambda)f(x_2),\end{aligned}$$

i.e. $f(x)$ is a convex function on (a,b). □

Problem 10.2. *Suppose that $f(x)$ is a strict convex function on I. Prove that if $f(x)$ has minimum $f(x_0)$, then $f(x_0)$ is unique, i.e. $\forall\, x \in I\backslash\{x_0\}$, there is $f(x) > f(x_0)$.*

Proof. Prove by contradiction. If there exists $x_1 \in I\backslash\{x_0\}$, such that $f(x_1) \le f(x_0)$. Then $\forall\, \lambda \in (0,1)$, there is

$$f(\lambda x_0 + (1-\lambda)x_1) < \lambda f(x_0) + (1-\lambda)f(x_1) \le f(x_0).$$

Then for any neighborhood of x_0, say $U(x_0,\delta)$ $(0 < \delta < |x_1 - x_0|)$, in I, we know that if we let $\lambda : 1-\lambda < \dfrac{\delta}{|x_1 - x_0|}$, and take $x = \lambda x_0 + (1-\lambda)x_1$, then there is $x \in U(x_0,\delta)$ and

$$f(x) = f(\lambda x_0 + (1-\lambda)x_1) < f(x_0),$$

which contradicts with $f(x_0)$ is the minimum of $f(x)$. □

Problem 10.3 (Challenge!)**.** *Suppose that $f(x) = \dfrac{1}{1+e^x}$.*

(i) *Prove that $f(x)$ is a convex function on $[0,+\infty)$. Moreover, there is $f(x)+f(y) \le f(0)+f(x+y)$, $\forall\, x,y \ge 0$.*

(ii) *Assume $n \ge 3$, determine the set $E = \left\{ \sum_{k=1}^{n} f(x_k) \,\middle|\, \sum_{k=1}^{n} x_k = 0,\ x_1,\cdots,x_n \in \mathbb{R} \right\}$.*

Proof. (i) Note that

$$f'(x) = -\frac{e^x}{(1+e^x)^2}, \quad f''(x) = \frac{e^x(e^x-1)}{(1+e^x)^3}.$$

When $x \geq 0$, there is $f''(x) \geq 0$, hence $f(x)$ is a convex function on $[0, +\infty)$. For $x, y \geq 0$, we define

$$g(x) = f(x+y) - f(x) - f(y) + f(0).$$

Then

$$g'(x) = f'(x+y) - f'(x) \geq 0,$$

since $f'(x)$ is increasing on $[0, +\infty)$. Hence $g(x) \geq g(0) = 0$, i.e. $f(x) + f(y) \leq f(0) + f(x+y)$.

(ii) By the continuity of f, it's easy to see that E is an interval. Hence, it suffices to find the infimum and supremum of E. Note that $x_1 + \cdots + x_n = 0$.

If $x_1 = x_2 = \cdots = x_n = 0$, then $\displaystyle\sum_{j=1}^{n} f(x_j) = \frac{n}{2}$.

If $x_1, \cdots, x_n$ are not all zero, we assume that the number of negative is k, and the number of nonnegative is m, then $k+m = n$, $1 \leq k \leq n-1$. Without loss of generality, we may assmue that $x_1, \cdots, x_m \geq 0$, $x_{m+1}, \cdots, x_n < 0$. Denote $y_1 = -x_{m+1}, \cdots, y_k = -x_n$, $x = x_1 + \cdots + x_m = y_1 + \cdots + y_k$. By (i), we have

$$f(y_1) + \cdots + f(y_k) \leq (k-1)f(0) + f(x).$$

Note that $mf\left(\dfrac{x}{m}\right) - f(x)$ is strictly decreasing on $[0, +\infty)$ and $f(x) + f(-x) = 1$, we have

$$\begin{aligned}
\sum_{j=1}^{n} f(x_j) &= \sum_{j=1}^{m} f(x_j) + k - \sum_{j=1}^{k} f(y_j) \\
&\geqslant mf\left(\frac{x}{m}\right) + k - ((k-1)f(0) + f(x)) \\
&> \lim_{u \to +\infty} \left[mf\left(\frac{u}{m}\right) + k - ((k-1)f(0) + f(u))\right] \\
&= \frac{k+1}{2} \geqslant 1,
\end{aligned}$$

which implies that $\inf E \geq 1$ and $1 \notin E$. On the other hand, for $u > 0$, let $x_1 = x_2 = \cdots = x_{n-1} = \frac{u}{n-1}$, $x_n = -u$, we have

$$\lim_{u \to +\infty} \sum_{j=1}^{n} f(x_j) = \lim_{u \to +\infty} \left((n-1)f\left(\frac{u}{n-1}\right) + 1 - f(u)\right) = 1,$$

thus $\inf E = 1$. Note that $f(-x) = 1 - f(x)$, we have

$$E = \{n - z | z \in E\}.$$

Hence, $\sup E = n-1$, and $n - 1 \notin E$. Therefore, we know that $E = (1, n-1)$. $\square$

Exercise 10.4 (Challenge!)**.** *Suppose that $f(x)$ is a concave function on $[a, b]$ satisfying $f(a) = 0$, $f(b) > 0$ and the right derivative of $f(x)$ at $x = a$ is non-zero. For $n \geq 2$,*

denote

$$S_n = \left\{ \sum_{k=1}^{n} k x_k : \sum_{k=1}^{n} k f(x_k) = f(b), x_k \in [a,b] \right\}.$$

(i) *Prove that for* $\forall\, \alpha \in (0, f(b))$, *there exists a unique* $x \in (a,b)$ *such that* $f(x) = \alpha$.

(ii) *Find* $\lim\limits_{n\to\infty} (\sup S_n - \inf S_n)$.

Hint: (i) Since $f(x)$ is continuous on $[a,b]$, we know that $\forall\, \alpha \in (0, f(b))$, there exists at least one point $\xi \in (a,b)$ such that $f(\xi) = \alpha$. Next, we prove that it is unique. Assume that there are $\xi, \eta \in (a,b)$ satisfying $\xi < \eta$ and $f(\xi) = f(\eta) = \alpha$. Then the point $(\eta, f(\eta)) = (\eta, \alpha)$ lies below in the segment connecting $(\xi, f(\xi)) = (\xi, \alpha)$ and $(b, f(b))$, contradicts with the concavity of f.

(ii) Denote

$$T_n = \left\{ (x_1, \cdots, x_n) : \sum_{k=1}^{n} k f(x_k) = f(b), x_k \in [a,b] \right\}, \quad n \geq 2.$$

For $\forall\, (x_1, \cdots, x_n) \in T_n$, by the concavity of $f(x)$, we have

$$\frac{2f(b)}{n(n+1)} = \frac{\sum_{k=1}^{n} k f(x_k)}{1+2+\cdots+n} \leq f\left(\frac{x_1 + 2x_2 + \cdots + n x_n}{1+2+\cdots+n} \right).$$

Hence

$$\frac{x_1 + 2x_2 + \cdots + n x_n}{1+2+\cdots+n} \geq f^{-1}\left(\frac{2f(b)}{n(n+1)} \right),$$

i.e.

$$\sum_{k=1}^{n} k x_k \geq \frac{n(n+1)}{2} f^{-1}\left(\frac{2f(b)}{n(n+1)} \right).$$

It is easy to see that "=" holds iff $x_1 = x_2 = \cdots = x_n = f^{-1}\left(\frac{2f(b)}{n(n+1)} \right)$. Note that $\left(f^{-1}\left(\frac{2f(b)}{n(n+1)} \right), \cdots, f^{-1}\left(\frac{2f(b)}{n(n+1)} \right) \right) \in T_n$, hence

$$\inf S_n = \frac{n(n+1)}{2} f^{-1}\left(\frac{2f(b)}{n(n+1)} \right).$$

On the other hand, by the concavity of $f(x)$, we have

$$\frac{f(b)}{b-a}(x-a) \leq f(x),$$

i.e

$$x \leq \frac{b-a}{f(b)} f(x) + a.$$

Hence

$$\sum_{k=1}^{n} k x_k \leq \frac{b-a}{f(b)} \sum_{k=1}^{n} k f(x_k) + \frac{n(n+1)}{2} a = b - a + \frac{n(n+1)}{2} a.$$

Note that "=" holds iff $x_1 = b, x_2 = x_3 = \cdots = x_n = a$, and $(b, a, a, \cdots, a) \in T_n$, then

$$\sup S_n = b - a + \frac{n(n+1)}{2}a.$$

Therefore, we have

$$\begin{aligned}
&\lim_{n\to\infty}(\sup S_n - \inf S_n) = b - a + \lim_{n\to\infty}\frac{n(n+1)}{2}\left(a - f^{-1}\left(\frac{2f(b)}{n(n+1)}\right)\right)\\
&= b - a + f(b)\lim_{n\to\infty}\frac{a - f^{-1}\left(\frac{2f(b)}{n(n+1)}\right)}{\frac{2f(b)}{n(n+1)}}\\
&= b - a + f(b)\lim_{x\to 0+0}\frac{a - f^{-1}(x)}{x} = b - a + f(b)\lim_{t\to a+0}\frac{a-t}{f(t)}\\
&= b - a - \frac{f(b)}{f'(a)}.
\end{aligned}$$

□

Problem 10.5. *Suppose that x, y, $z > 0$. Given $x + y + z = 1$, prove that:*

$$\frac{1}{x^2+y^2+z^2} + \frac{3}{xy+yz+zx} \geq 12.$$

Proof. First note that $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$, if we denote $t = xy+yz+zx$, we have

$$\frac{1}{x^2+y^2+z^2} + \frac{3}{xy+yz+zx} = \frac{1}{1-2t} + \frac{3}{t}$$

since $x+y+z=1$. By $xy+yz+zx \leq \frac{1}{3}(x+y+z)^2 = \frac{1}{3}$, we know that it suffices to minimize

$$f(t) := \frac{1}{1-2t} + \frac{3}{t}, \quad 0 < t \leq \frac{1}{3}.$$

Differentiating directly yields

$$f'(t) = \frac{2}{(1-2t)^2} - \frac{1}{t^2} = \frac{-10t^2+12t-3}{(1-2t)^2t^2} < 0, \quad 0 < t \leq \frac{1}{3}.$$

Hence we know $f(t) \geq f(\frac{1}{3}) = \frac{1}{1-\frac{2}{3}} + 9 = 12$. Then we are done! Finally, it is easy to see that " = " holds iff $x = y = z = \frac{1}{3}$. □

Problem 10.6. *If $a, b, c > 0$ and $2abc + 3(ab+ac+bc) = 27$. Prove that*

$$16(a^2+b^2+c^2) + 8abc \geq 135.$$

Proof. First by

$$a^2+b^2 \geq 2ab,\ b^2+c^2 \geq 2bc,\ c^2+a^2 \geq 2ca,$$

we get

$$a^2+b^2+c^2 \geq ab+bc+ca.$$

Then combining with $2abc + 3(ab + bc + ca) = 27$ yields

$$\begin{aligned}16(a^2+b^2+c^2)+8abc &= 16(a^2+b^2+c^2)+4(27-3(ab+bc+ca))\\ &= 16(a^2+b^2+c^2)-12(ab+bc+ca)+108\\ &\geq 4(a^2+b^2+c^2)+108.\end{aligned}$$

Hence it suffices to prove $a^2+b^2+c^2 \geq \frac{27}{4}$. Assume by contradiction that $a^2+b^2+c^2 < \frac{27}{4}$, and note that $a^2+b^2+c^2 \geq 3\sqrt[3]{a^2b^2c^2}$, we have

$$\begin{aligned}2abc+3(ab+bc+ca) &\leq 2\left(\frac{a^2+b^2+c^2}{3}\right)^{\frac{3}{2}}+3(a^2+b^2+c^2)\\ &< 2\left(\frac{9}{4}\right)^{\frac{3}{2}}+3\times\frac{27}{4}\\ &= 2\times\frac{27}{8}+3\times\frac{27}{4}=27\end{aligned}$$

contradict with $2abc + 3(ab + bc + ca) = 27$. Hence $a^2 + b^2 + c^2 \geq \frac{27}{4}$. Then we are done. Then it is easy to check that " $=$ " holds iff $(a, b, c) = (\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$. □

Remark 10.7. *We can also get* $a^2+b^2+c^2 \geq \frac{27}{4}$ *by solving the inequality* $2\left(\frac{a^2+b^2+c^2}{3}\right)^{\frac{3}{2}} + 3(a^2+b^2+c^2) \geq 27$.

Problem 10.8. *Prove the following inequalities:*

(1) $\displaystyle\sum_{k=1}^{n}\left(x_k+\frac{1}{x_k}\right)^{\alpha} \geq \frac{(n^2+1)^{\alpha}}{n^{\alpha-1}}$, $(\alpha > 1,\ x_1+\cdots+x_n = 1)$;

(2) $\displaystyle 1+\left(\sum_{k=1}^{n} p_k x_k\right)^{-1} \leq \prod_{k=1}^{n}\left(\frac{1+x_k}{x_k}\right)^{p_k}$, $(p_k > 0\, 0 < x_k < 1,\ p_1+\cdots+p_k = 1)$;

(3) $(\sin x)^{1-\cos 2x}+(\cos x)^{1+\cos 2x} \geq \sqrt{2}$, *where* $x \in (0, \pi/2)$;

(4) $2^n \geq 1+n\sqrt{2^{n-1}}$, $n \in \mathbb{N}$;

(5) $\displaystyle\left|\sum_{k=1}^{n}\frac{\sin kx}{k}\right| \leq 2\sqrt{2\pi}$, $x \in \mathbb{R}$, $n \in \mathbb{N}$.

Proof. (1) Since $f(x) = (x + 1/x)^{\alpha}$ is convex on $(0, +\infty)$, we have

$$\left(\frac{n^2+1}{n}\right)^{\alpha} = \left[\frac{1}{n}\sum_{k=1}^{n}x_k+\left(\frac{1}{n}\sum_{k=1}^{n}x_k\right)^{-1}\right]^{\alpha} \leq \frac{1}{n}\sum_{k=1}^{n}\left(x_k+\frac{1}{x_k}\right)^{\alpha}.$$

(2) Since $f(x) = \ln(1 + 1/x)$ is convex on $(0, +\infty)$, we have

$$\ln\left(1 + \left(\sum_{k=1}^{n} p_k x_k\right)^{-1}\right) \leq \sum_{k=1}^{n} p_k \cdot \ln\left(1 + \frac{1}{x_k}\right) = \ln\left(\prod_{k=1}^{n}\left(\frac{1 + x_k}{x_k}\right)^{p_k}\right).$$

(3) Note that

$$(\sin x)^{1-\cos 2x} + (\cos x)^{1+\cos 2x} = (\sin^2 x)^{sin^2 x} + (\cos^2 x)^{\cos^2 x}.$$

Since $f(x) = x^x$ is convex on $(0, +\infty)$, we have

$$\left(\frac{1}{2}\right)^{\frac{1}{2}} \leq \frac{1}{2}\left((\sin^2 x)^{sin^2 x} + (\cos^2 x)^{\cos^2 x}\right).$$

(4) Let

$$f(x) = 2^x - 1 - x\sqrt{2^{x-1}}, \quad (x \geq 1).$$

Then

$$f'(x) = 2^{\frac{x-1}{2}}\left(2^{\frac{x+1}{2}} \ln 2 - 1 - \frac{x}{2}\ln 2\right).$$

Let

$$g(x) = 2^{\frac{x+1}{2}} \ln 2 - 1 - \frac{x}{2}\ln 2, \quad (x \geq 1).$$

There is

$$g'(x) = 2^{\frac{x+1}{2}}(\ln 2)^2 \cdot \frac{1}{2} - \frac{1}{2}\ln 2 > 0, \quad (x \geq 1).$$

Hence g is increasing on $[1, +\infty)$, then $g(x) \geq g(1) = \frac{3}{2}\ln 2 - 1 > 0$. Therefore, we know that $f'(x) > 0$, i.e. f is also increasing on $[1, +\infty)$. Hence $f(x) \geq f(1) = 0$, i.e. $2^n \geq 1 + n\sqrt{2^{n-1}}$, $n \in \mathbb{N}$.

(5) It suffices to consider $x \in [0, \pi]$ since $f(x) = \left|\sum_{k=1}^{n} \frac{\sin kx}{k}\right|$ is an even function, and it has period, 2π. When $x = 0, \pi$, the inequality is clearly. Now, we assume that $0 < x < \pi$. We know that there must be some $m \in \mathbb{N}$ such that $m \leq \frac{\sqrt{2\pi}}{x} < m + 1$. Hence

$$\left|\sum_{k=1}^{n} \frac{\sin kx}{k}\right| = \sum_{k=1}^{m}\left|\frac{\sin kx}{k}\right| + \left|\sum_{k=m+1}^{n} \frac{\sin kx}{k}\right|.$$

When $m = 0$, the first formula of RHS is 0, when $m \geq n$, the second formula of RHS is 0. Note that

$$|\sin x| \leq |x| \quad \text{and} \quad \sin x > \frac{2}{\pi}x \ (0 < x < \pi/2).$$

We have

$$\sum_{k=1}^{m}\left|\frac{\sin kx}{k}\right| \leq \sum_{k=1}^{m} \frac{kx}{k} < \sqrt{2\pi},$$

and

$$\left|\sum_{k=m+1}^{n} \frac{\sin kx}{k}\right| = \left|\sum_{k=m+1}^{n-1} S_k \left(\frac{1}{k} - \frac{1}{k+1}\right) + S_n \cdot \frac{1}{n} - S_m \cdot \frac{1}{m+1}\right|$$
$$< \frac{1}{\left|\sin \frac{x}{2}\right|} \cdot \frac{2}{m+1} < \frac{\pi}{x} \cdot \frac{2x}{\sqrt{2\pi}} = \sqrt{2\pi},$$

where $S_k = \sin x + \sin 2x + \cdots + \sin kx$ and $|S_k| < 1/\left|\sin \frac{x}{2}\right|$ (Leave to the reader). Then we have

$$\left|\sum_{k=1}^{n} \frac{\sin kx}{k}\right| \leq 2\sqrt{2\pi}.$$

$\square$

11. Week 14 (12.5)

Problem 11.1. *Assume that f is twice diffentiable on $\mathbb{R}$, and such*

$$2f(x) + f''(x) = -xf'(x).$$

Prove that $f(x)$and $f'(x)$ are bounded on $\mathbb{R}$.

Proof. Define

$$g = f^2 + \frac{1}{2}(f')^2.$$

By definition, g is non-negative and differentiable; moreover,

$$g'(x) = f'(x) \cdot (2f(x) + f''(x)) = -x \cdot (f'(x))^2, \quad \forall x \in \mathbb{R}.$$

Therefore, g is increasing on $(-\infty, 0]$ and decreasing on $[0, +\infty)$, so $g(\mathbb{R}) \subset [0, g(0)]$. The conclusion follows. □

Problem 11.2. *Define $f \in C^2[a,b]$ satisfying $f''(x) = e^x f(x)$. Show that $f''(x) = e^x f(x)$ with $f(a) = f(b) = 0$ makes $f \equiv 0$ $\forall x \in [a,b]$.*

Proof. Assume that f is not identically zero on the interval, without loss of generality $f(x_0) > 0$ for some $x_0 \in (a,b)$. Then f attains its maximum $M > 0$ at some point $x_1 \in (a,b)$. At the maximum, we necessarily have

$$f'(x_1) = 0\,, \quad f''(x_1) \leq 0\,,$$

which is a contradiction to the assumption that

$$f''(x_1) = e^{x_1} f(x_1) = e^{x_1} M > 0\,.$$

□

Remark 11.3. *Actually, we can prove a general conclusion: if $f \in C^2[a,b]$ satisfying $f''(x) = g(x)f(x)$ where $g(x) \in C^0[a,b]$ satisfying $g(x) > 0$, and $f(a) = f(b) = 0$, we have $f \equiv 0$ $\forall x \in [a,b]$.*

Problem 11.4. *Suppose that there is equation*

$$x(1 - \ln(\varepsilon\sqrt{x})) = 1, \quad (x > 0,\, \varepsilon > 0). \tag{11.1}$$

Then

(i) *For small enough ε, (11.1) has two solutions (denote the small one as x_ε);*
(ii) $\lim_{\varepsilon \to 0+0} x_\varepsilon = 0$*;*
(iii) $\lim_{\varepsilon \to 0+0} \varepsilon^{-t} x_\varepsilon = +\infty (t > 0)$.

Proof. (i) For $x > 0$, we have $\varepsilon = e/\sqrt{x}e^{1/x} =: f(x)$. Denote $F(x) = \sqrt{x}e^{1/x} = e/f(x)$, there is

$$F'(x) = x^{-\frac{3}{2}} e^{\frac{1}{x}} \frac{x-2}{2}.$$

Hence $F(x)$ is strictly decreasing on $(0,2]$ and it is strictly increasing on $[2,+\infty)$. Note that

$$\lim_{x\to 0+0} F(x) = +\infty = \lim_{x\to+\infty} F(x), \quad F(2) > 0,$$

we have $\lim\limits_{x\to 0+0} f(x) = 0 = \lim\limits_{x\to+\infty} f(x)$ and $f(x)$ is strictly increasing on $(0,2]$ and it is strictly decreasing on $[2,+\infty)$, $f(2) = \sqrt{e/2}$. Let

$$f_1(x) = \begin{cases} f(x), & x \in (0,2), \\ 0, & x \in [2,+\infty), \end{cases} \quad f_2(x) = \begin{cases} 0, & x \in (0,2], \\ f(x), & x \in (2,+\infty). \end{cases}$$

Then for $0 < \varepsilon < \sqrt{e/2}$, (11.1) has two solutions: $x = f_1^{-1}(\varepsilon)$, $x = f_2^{-1}(\varepsilon)$. The small one is $x_\varepsilon = f_1^{-1}(\varepsilon)$.

(ii) Since f_1 is strictly increasing on $(0,2)$ and it is continuous, we know $\lim\limits_{\varepsilon\to 0+0} x_\varepsilon = 0$ by $f_1(0+0) = 0$.

(iii) For $t > 0$, we have

$$\varepsilon^{-t} x_\varepsilon = \left(\frac{\mathrm{e}}{\sqrt{x}\mathrm{e}^{1/x}}\right)^{-t} x = \mathrm{e}^{-t} x^{1+t/2} \mathrm{e}^{t/x} \to +\infty, \quad \text{as } x \to 0+0.$$

□

Problem 11.5. *Draw the graph of* $f(x) = |x+2|e^{-\frac{1}{x}}$.

Solution. Note that $f(x) \geq 0$ and $f(-2) = 0$. Hence 0 is the minimum of $f(x)$ and $x = -2$ is the minimal point. Rewrite $f(x)$ as

$$f(x) = \begin{cases} (x+2)e^{-\frac{1}{x}}, & x \in [-2,0) \cup (0,+\infty), \\ -(x+2)e^{-\frac{1}{x}}, & x \in (-\infty,-2). \end{cases}$$

When $x \in [-2,0) \cup (0,+\infty)$, there is

$$f'(x) = \frac{e^{-\frac{1}{x}}(x^2+x+2)}{x^2} > 0.$$

Hence $f(x)$ is strictly inreasing on $x \in [-2,0) \cup (0,+\infty)$. When $x \in (-\infty,-2)$, there is

$$f'(x) = -\frac{e^{-\frac{1}{x}}(x^2+x+2)}{x^2} < 0.$$

Hence $f(x)$ is strictly dereasing on $x \in (-\infty,-2)$.

It is easy to see that

$$\lim_{h\to 0+0} \frac{f(-2+h)-f(-2)}{h} = \lim_{h\to 0+0} \frac{he^{-\frac{1}{h-2}}}{h} = e^{\frac{1}{2}},$$

$$\lim_{h\to 0-0} \frac{f(-2+h)-f(-2)}{h} = \lim_{h\to 0-0} \frac{-he^{-\frac{1}{h-2}}}{h} = -e^{\frac{1}{2}},$$

which gives us that $f(x)$ is nondifferentiable at $x = -2$. Note that

$$\lim_{x\to 0+0} f(x) = \lim_{x\to 0+0} \frac{2+x}{e^{\frac{1}{x}}} = 0,$$

$$\lim_{x\to 0-0} f(x) = \lim_{x\to 0-0} \frac{2+x}{e^{\frac{1}{x}}} = +\infty.$$

Hence $x = 0$ is a vertical asymptote of $f(x)$. Note that

$$\lim_{x\to+\infty} \frac{f(x)}{x} = \lim_{x\to+\infty} \frac{x+2}{x} e^{-\frac{1}{x}} = 1,$$

$$\lim_{x\to+\infty} (f(x) - x) = \lim_{x\to+\infty} ((x+2)e^{-\frac{1}{x}} - x) = 1.$$

Hence $y = x + 1$ is a oblique asymptote of $f(x)$.

When $x \geq -2$, $f''(x) = \left(\frac{2}{x^4} - \frac{3}{x^3}\right) e^{-\frac{1}{x}}$. Hence we know that when $-2 \leq x \leq \frac{2}{3}$, $f(x)$ is convex; $x > \frac{2}{3}$, $f(x)$ is concave. When $x < -2$, $f''(x) = -\left(\frac{2}{x^4} - \frac{3}{x^3}\right) e^{-\frac{1}{x}} < 0$. Hence $f(x)$ is concave.

Combining above, we have the graph of f is as follows:

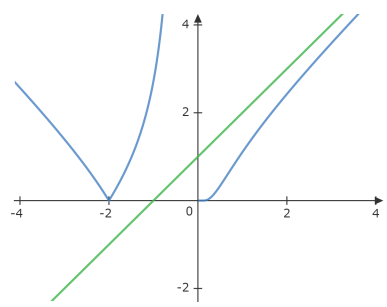

FIGURE 12. Graph of $f(x)$

□

Problem 11.6. *Calculate the following integrals.*

(1) $\displaystyle\int e^{\sqrt{x+1}}\, \mathrm{d}x$;

(2) $\displaystyle\int \frac{\mathrm{d}x}{x^2\sqrt{x^2+x-1}}$;

(3) $\displaystyle\int \frac{x \arctan x}{(1+x^2)^2}\, \mathrm{d}x$;

(4) $\displaystyle\int x \tan^2 x\, \mathrm{d}x$;

(5) $\displaystyle\int \frac{\ln(\sin x)}{\sin^2 x}\ \mathrm{d}x;$

(6) $\displaystyle\int \frac{\mathrm{d}x}{x\left(1+x^8\right)};$

(7) $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{e^x-1}};$

(8) $\displaystyle\int \frac{\sin x}{\sin x-\cos x}\ \mathrm{d}x;$

(9) $\displaystyle\int \frac{1+\cos x}{1+\sin x}\ \mathrm{d}x$

(10) $\displaystyle\int \frac{e^x(1+\sin x)}{1+\cos x}\ \mathrm{d}x;$

(11) $\displaystyle\int \frac{e^{3x}+e^x}{e^{4x}-e^{2x}+1}\ \mathrm{d}x;$

(12) $\displaystyle\int \frac{1-\ln x}{(x-\ln x)^2}\ \mathrm{d}x;$

(13) $\displaystyle\int \frac{x+\sin x\cos x}{(\cos x-x\sin x)^2}\ \mathrm{d}x;$

(14) $\displaystyle\int x^2e^x\cos 2x\ \mathrm{d}x;$

(15) $\displaystyle\int \frac{e^x\left(2-x^2\right)}{(1-x)\sqrt{1-x^2}}\ \mathrm{d}x;$

(16) $\displaystyle\int \frac{(1+x)\mathrm{d}x}{x^2e^x\left(1+xe^x\right)};$

(17) $\displaystyle\int \frac{x^2\sin^2 x}{(x+\sin x\cos x)^2}\ \mathrm{d}x;$

(18) $\displaystyle\int \frac{x^2}{(x\cos x-\sin x)(x\sin x+\cos x)}\mathrm{d}x;$

(19) $\displaystyle I=\int \frac{\sin^3 x}{\sin^3 x-\cos^3 x}\ \mathrm{d}x,\quad J=\int \frac{\cos^3 x}{\sin^3 x-\cos^3 x}\ \mathrm{d}x.$

Solution. (1) By changing of variable and integral by parts, we have

$$\begin{aligned}\int e^{\sqrt{x+1}}\ \mathrm{d}x \xlongequal{t=\sqrt{x+1}} &\int 2te^t\ \mathrm{d}t\\ =&2te^t-2\int e^t\ \mathrm{d}t\\ =&(2t-2)e^t+C\\ =&(2\sqrt{x+1}-2)e^{\sqrt{x+1}}+C.\end{aligned}$$

(2) By changing of variable, we have

$$\begin{aligned}\int \frac{\mathrm{d}x}{x^2\sqrt{x^2+x-1}} &= \int \frac{\mathrm{d}x}{x^3\sqrt{1+\frac{1}{x}-\frac{1}{x^2}}} \xlongequal{t=\frac{1}{x}} -\int \frac{t}{\sqrt{1+t-t^2}}\ \mathrm{d}t \\ &= -\int \frac{t-\frac{1}{2}}{\sqrt{\frac{5}{4}-(t-\frac{1}{2})^2}}\ \mathrm{d}t - \int \frac{\frac{1}{2}}{\sqrt{\frac{5}{4}-(t-\frac{1}{2})^2}}\ \mathrm{d}t \\ &= \sqrt{\frac{5}{4}-(t-\frac{1}{2})^2} - \frac{1}{2}\arcsin\frac{2}{\sqrt{5}}(t-\frac{1}{2}) + C \\ &= \frac{\sqrt{x^2+x-1}}{x} - \frac{1}{2}\arcsin\frac{2}{\sqrt{5}}(\frac{1}{x}-\frac{1}{2}) + C.\end{aligned}$$

(3) By integral by parts, we have

$$\begin{aligned}\int \frac{x\arctan x}{(1+x^2)^2}\ \mathrm{d}x &= -\frac{1}{2(1+x^2)}\arctan x + \frac{1}{2}\int \frac{1}{(1+x^2)^2}\ \mathrm{d}x \\ &= -\frac{1}{2(1+x^2)}\arctan x + \frac{1}{4}\int \frac{1+x^2+1-x^2}{(1+x^2)^2}\ \mathrm{d}x \\ &= -\frac{1}{2(1+x^2)}\arctan x + \frac{1}{4}\int \frac{1}{1+x^2}\ \mathrm{d}x + \frac{1}{4}\int \frac{1-x^2}{(1+x^2)^2}\ \mathrm{d}x \\ &= -\frac{1}{2(1+x^2)}\arctan x + \frac{1}{4}\arctan x + \frac{x}{4(1+x^2)} + C.\end{aligned}$$

(4) By integral by parts, we have

$$\begin{aligned}\int x\tan^2 x\ \mathrm{d}x &= \int x(\sec^2 x - 1)\ \mathrm{d}x \\ &= \int x\sec^2 x\ \mathrm{d}x - \frac{1}{2}x^2 \\ &= -\frac{1}{2}x^2 + x\tan x - \int \tan x\ \mathrm{d}x \\ &= -\frac{1}{2}x^2 + x\tan x + \ln\cos x + C.\end{aligned}$$

(5) By integral by parts, we have

$$\begin{aligned}\int \frac{\ln(\sin x)}{\sin^2 x}\ \mathrm{d}x &= \int \csc^2 x\ln(\sin x)\ \mathrm{d}x \\ &= -\cot x\ln(\sin x) + \int \cot^2 x\ \mathrm{d}x \\ &= -\cot x\ln(\sin x) + \int (\csc^2 x - 1)\ \mathrm{d}x \\ &= -x - \cot x - \cot x\ln(\sin x) + C.\end{aligned}$$

(6) By changing of variable, we have

$$\begin{aligned}\int \frac{\mathrm{d}x}{x\left(1+x^8\right)} &= \int \frac{x^7 \mathrm{d}x}{x^8\left(1+x^8\right)} \\ &\overset{t=x^8}{=\!=\!=} \frac{1}{8}\int \frac{\mathrm{d}t}{t\left(1+t\right)} \\ &= \frac{1}{8}\ln\left|\frac{t}{t+1}\right| + C \\ &= \frac{1}{8}\ln\left(\frac{x^8}{x^8+1}\right) + C.\end{aligned}$$

(7) For the integrand $\frac{1}{\sqrt{e^x-1}}$, substitute $u = e^x$ and $du = e^x dx$

$$\int \frac{\mathrm{d}x}{\sqrt{e^x-1}} = \int \frac{1}{\sqrt{u-1}u} du.$$

For the integrand $\frac{1}{\sqrt{u-1}u}$, substitute $s = u-1$ and $ds = du$, we have

$$\int \frac{1}{\sqrt{u-1}u} du = \int \frac{1}{\sqrt{s}(s+1)} ds.$$

For the integrand $\frac{1}{\sqrt{s}(s+1)}$, substitute $p = \sqrt{s}$ and $dp = \frac{1}{2\sqrt{s}} ds$,we know

$$\int \frac{1}{\sqrt{s}(s+1)} ds = 2\int \frac{1}{p^2+1} dp.$$

The integral of $\frac{1}{p^2+1}$ is $\tan^{-1}(p)$, then

$$2\int \frac{1}{p^2+1} dp = 2\tan^{-1}(p) + C.$$

Substitute back for $p = \sqrt{s}$,

$$\int \frac{1}{\sqrt{s}(s+1)} ds = 2\tan^{-1}(\sqrt{s}) + C.$$

Substitute back for $s = u-1$,

$$\int \frac{1}{\sqrt{u-1}u} du = 2\tan^{-1}(\sqrt{u-1}) + C.$$

Substitute back for $u = e^x$, we have Answer:

$$\int \frac{\mathrm{d}x}{\sqrt{e^x-1}} = 2\tan^{-1}\left(\sqrt{e^x-1}\right) + C.$$

(8)

$$\int \frac{\sin x}{\sin x - \cos x}\,\mathrm{d}x = \int \frac{\sin(x-\frac{\pi}{4}+\frac{\pi}{4})}{\sqrt{2}\sin(x-\frac{\pi}{4})}\,\mathrm{d}x$$

$$
\begin{aligned}
&= \int \frac{\frac{\sqrt{2}}{2}\sin(x-\frac{\pi}{4}) + \frac{\sqrt{2}}{2}\cos(x-\frac{\pi}{4})}{\sqrt{2}\sin(x-\frac{\pi}{4})}\ \mathrm{d}x \\
&= \frac{1}{2}x + \frac{1}{2}\ln\sin(x-\frac{\pi}{4}) + C.
\end{aligned}
$$

(9)

$$
\begin{aligned}
\int \frac{1+\cos x}{1+\sin x}\ \mathrm{d}x &= \int \frac{1}{1+\sin x}\ \mathrm{d}x + \int \frac{\cos x}{1+\sin x}\ \mathrm{d}x \\
&= \int \frac{1}{1+\cos(x-\frac{\pi}{2})}\ \mathrm{d}x + \ln(1+\sin x) \\
&= \int \frac{1}{2\cos^2(\frac{x}{2}-\frac{\pi}{4})} + \ln(1+\sin x) \\
&= \tan\left(\frac{x}{2}-\frac{\pi}{4}\right) + \ln(1+\sin x) + C.
\end{aligned}
$$

(10) By integral by parts, we have

$$
\begin{aligned}
\int \frac{e^x(1+\sin x)}{1+\cos x}\ \mathrm{d}x &= \int \frac{e^x(\sin\frac{x}{2}+\cos\frac{x}{2})^2}{2\cos^2\frac{x}{2}}\ \mathrm{d}x \\
&= \frac{1}{2}\int e^x\left(1+\tan\frac{x}{2}\right)^2\ \mathrm{d}x \\
&= \frac{1}{2}\int e^x\left(1+\tan^2\frac{x}{2}\right)\ \mathrm{d}x + \int e^x\tan\frac{x}{2}\ \mathrm{d}x \\
&= \frac{1}{2}\int e^x\sec^2\frac{x}{2}\ \mathrm{d}x + \int e^x\tan\frac{x}{2}\ \mathrm{d}x \\
&= e^x\tan\frac{x}{2} - \int e^x\tan\frac{x}{2}\ \mathrm{d}x + \int e^x\tan\frac{x}{2}\ \mathrm{d}x \\
&= e^x\tan\frac{x}{2} + C.
\end{aligned}
$$

(11)

$$
\begin{aligned}
\int \frac{e^{3x}+e^x}{e^{4x}-e^{2x}+1}\ \mathrm{d}x &= \int \frac{e^x+e^{-x}}{e^{2x}+e^{-2x}-1}\ \mathrm{d}x \\
&= \int \frac{\mathrm{d}(e^x-e^{-x})}{1+(e^x-e^{-x})^2} \\
&= \arctan(e^x-e^{-x}) + C.
\end{aligned}
$$

(12)

$$
\int \frac{1-\ln x}{(x-\ln x)^2}\ \mathrm{d}x = \int \frac{1-\ln x}{x^2}\frac{1}{(1-\frac{\ln x}{x})^2}\ \mathrm{d}x
$$

$$
\begin{aligned}
&= \int \frac{1}{(1-\frac{\ln x}{x})^2}\ \mathrm{d}\left(\frac{\ln x}{x}\right) \\
&= \frac{x}{x-\ln x} + C.
\end{aligned}
$$

(13)

$$
\begin{aligned}
\int \frac{x+\sin x\cos x}{(\cos x - x\sin x)^2}\ \mathrm{d}x &= \int \frac{x\sec^2 x + \tan x}{(1-x\tan x)^2}\ \mathrm{d}x \\
&= \int \frac{\mathrm{d}(x\tan x)}{(1-x\tan x)^2} \\
&= \frac{1}{1-x\tan x} + C.
\end{aligned}
$$

(14) Firstly, we have by integral by parts that

$$
\begin{aligned}
\int x^2 e^{(1+2i)x}\ \mathrm{d}x &= \frac{x^2}{1+2i}e^{(1+2i)x} - \frac{2}{1+2i}\int xe^{(1+2i)x}\ \mathrm{d}x \\
&= \frac{x^2}{1+2i}e^{(1+2i)x} - \frac{2x}{(1+2i)^2} + \frac{2}{(1+2i)^2}\int e^{(1+2i)x}\ \mathrm{d}x \\
&= \frac{x^2}{1+2i}e^{(1+2i)x} - \frac{2x}{(1+2i)^2}e^{(1+2i)x} + \frac{2}{(1+2i)^3}e^{(1+2i)x} + C \\
&= \frac{25x^2+30x-22+(-50x^2+40x+4)i}{125}e^{(1+2i)x} + C \\
&= \frac{1}{125}e^x(25x^2+30x-22+(-50x^2+40x+4)i)(\cos 2x + i\sin 2x) + C \\
&= \frac{1}{125}e^x\left((25x^2+30x-22)\cos 2x + (50x^2-40x-4)\sin 2x\right) \\
&\quad + \frac{i}{125}e^x\left((-50x^2+40x+4)\cos 2x + (25x^2+30x-22)\sin 2x\right) + C.
\end{aligned}
$$

Hence

$$
\int x^2 e^x \cos 2x\ \mathrm{d}x = \frac{1}{125}e^x\left((25x^2+30x-22)\cos 2x + (50x^2-40x-4)\sin 2x\right) + C.
$$

(15) Note that

$$
\mathrm{d}\left(e^x\sqrt{\frac{1+x}{1-x}}\right) = \frac{e^x\,(2-x^2)}{(1-x)\sqrt{1-x^2}}\ \mathrm{d}x,
$$

we have

$$
\int \frac{e^x\,(2-x^2)}{(1-x)\sqrt{1-x^2}}\ \mathrm{d}x = e^x\sqrt{\frac{1+x}{1-x}} + C.
$$

(16)

$$\begin{aligned}
\int \frac{(1+x)\mathrm{d}x}{x^2e^x\,(1+xe^x)} &= \int \frac{e^x(1+x)\mathrm{d}x}{x^2e^{2x}\,(1+xe^x)} \\
&= \int \frac{\mathrm{d}(xe^x)}{x^2e^{2x}\,(1+xe^x)} \\
&= \int \frac{\mathrm{d}(xe^x)}{x^2e^{2x}} - \int \frac{\mathrm{d}(xe^x)}{xe^x} + \int \frac{\mathrm{d}(xe^x)}{1+xe^x} \\
&= -\frac{1}{xe^x} - \ln(xe^x) + \ln(1+xe^x) + C.
\end{aligned}$$

(17)

$$\begin{aligned}
\int \frac{x^2\sin^2 x}{(x+\sin x\cos x)^2}\ \mathrm{d}x =& \frac{1}{2}\int \frac{x^2(1-\cos 2x)}{(x+\sin x\cos x)^2}\ \mathrm{d}x \\
=& \frac{1}{2}\int \frac{x^2(-1-\cos 2x)}{(x+\sin x\cos x)^2}\ \mathrm{d}x + \int \frac{x^2}{(x+\sin x\cos x)^2}\ \mathrm{d}x \\
=& \frac{1}{2}\cdot\frac{x^2}{x+\sin x\cos x} - \int \frac{x}{x+\sin x\cos x}\ \mathrm{d}x \\
&+ \int \frac{x^2}{(x+\sin x\cos x)^2}\ \mathrm{d}x \\
=& \frac{1}{2}\cdot\frac{x^2}{x+\sin x\cos x} + \int \frac{-x\sin x\cos x}{(x+\sin x\cos x)^2}\ \mathrm{d}x \\
=& \frac{1}{2}\cdot\frac{x^2}{x+\sin x\cos x} \\
&+ \int \frac{x\sin x\cos x}{1+\cos 2x}\cdot\frac{-1-\cos 2x}{(x+\sin x\cos x)^2}\ \mathrm{d}x \\
=& \frac{1}{2}\cdot\frac{x^2}{x+\sin x\cos x} + \frac{x\tan x}{2}\cdot\frac{1}{x+\sin x\cos x} \\
&- \frac{1}{2}\int (x\sec^2 x+\tan x)\cdot\frac{1}{x+\sin x\cos x}\ \mathrm{d}x \\
=& \frac{x^2}{2(x+\sin x\cos x)} + \frac{x\tan x}{2(x+\sin x\cos x)} - \frac{1}{2}\int \sec^2 x\ \mathrm{d}x \\
=& \frac{x^2}{2(x+\sin x\cos x)} + \frac{x\tan x}{2(x+\sin x\cos x)} - \frac{1}{2}\tan x + C \\
=& \frac{x^2-\sin^2 x}{2(x+\sin x\cos x)} + C.
\end{aligned}$$

(18)

$$\int \frac{x^2}{(x\cos x-\sin x)(x\sin x+\cos x)}\mathrm{d}x = \int \frac{x\cos x}{x\sin x+\cos x}\mathrm{d}x - \int \frac{-x\sin x}{x\cos x-\sin x}\mathrm{d}x$$
$$= \ln|x\sin x+\cos| - \ln|x\cos-\sin x| + C.$$

(19) It is easy to see that $I-J=x+C$. Note that

$$\begin{aligned}
I+J =& \int \frac{\sin^3 x+\cos^3 x}{\sin^3 x-\cos^3 x}\mathrm{d}x \\
=& \int \frac{(\sin x+\cos x)(\sin^2 x-\sin x\cos x+\cos^2 x)}{(\sin x-\cos x)(\sin^2 x+\sin x\cos x+\cos^2 x)}\mathrm{d}x \\
=& \int \frac{(\sin^2 x-\cos^2 x)(1-\frac{1}{2}\sin 2x)}{(\sin x-\cos x)^2(1+\frac{1}{2}\sin 2x)}\mathrm{d}x \\
=& \int \frac{-\cos 2x(1-\frac{1}{2}\sin 2x)}{(1-\sin 2x)(1+\frac{1}{2}\sin 2x)}\mathrm{d}x \\
=& \frac{1}{2}\int \frac{-\cos 2x}{1+\frac{1}{2}\sin 2x}\mathrm{d}x - \frac{1}{2}\int \frac{\cos 2x}{(1-\sin 2x)(1+\frac{1}{2}\sin 2x)}\mathrm{d}x \\
=& -\frac{1}{2}\ln(1+\frac{1}{2}\sin 2x) + \frac{1}{6}\ln(1-\sin 2x) - \frac{1}{6}\ln(1+\frac{1}{2}\sin 2x) + C \\
=& \frac{1}{6}\ln(1-\sin 2x) - \frac{2}{3}\ln(1+\frac{1}{2}\sin 2x) + C.
\end{aligned}$$

□

Problem 11.7. *Suppose that $F(x)$ ia a primitive function of $f(x)$ on $(0,+\infty)$, and $F(1)=\frac{\sqrt{2}\pi}{4}$. If there is*

$$f(x)F(x) = \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}, \quad x\in(0,+\infty),$$

find $f(x)$.

Proof. A direct integrating yields

$$\frac{1}{2}F(x)^2 = \int \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}\mathrm{d}x = (\arctan\sqrt{x})^2 + C.$$

By $F(1)=\frac{\sqrt{2}\pi}{4}$, we know that $C=0$. Hence $F(x)=\sqrt{2}\arctan\sqrt{x}$, which gives us

$$f(x) = \frac{\sqrt{2}}{2\sqrt{x}(1+x)}.$$

□

Exercise 11.8. *Use an elementary way to show that for positive integer n,*

$$\int \frac{\sin(nx)\sin x}{1-\cos x}\,dx = x + \frac{\sin(nx)}{n} + 2\sum_{k=1}^{n-1}\frac{\sin(kx)}{k}$$

Hint: Let

$$F(x) = x + \frac{\sin(nx)}{n} + 2\sum_{k=1}^{n-1}\frac{\sin(kx)}{k}.$$

Then

$$F'(x) = 1 + \cos(nx) + 2\sum_{k=1}^{n-1}\cos(kx).$$

Therefore, we only need to prove that

$$\frac{\sin(nx)\sin x}{1-\cos x} = 1 + \cos(nx) + 2\sum_{k=1}^{n-1}\cos(kx),$$

or equivalently that

$$\sin(nx)\sin(x) = (1-\cos(x))\left(1 + \cos(nx) + 2\sum_{k=1}^{n-1}\cos(kx)\right). \tag{11.2}$$

Note that

$$z = \cos(x) + i\sin(x)$$

and calculate

$$\begin{aligned}
1 + 2z + 2z^2 + .. + 2z^{n-1} + z^n &= 2(1 + z + z^2 + .. + z^{n-1} + z^n) - 1 - z^n \\
&= 2\frac{1-z^{n+1}}{1-z} - 1 - z^n \\
&= \frac{2 - z^{n+1} - z^n}{1-z} - 1 = \frac{(2 - z^{n+1} - z^n)(1-\bar{z})}{(1-\cos(x))^2 + \sin(x)^2} - 1 \\
&= \frac{2 - z^{n+1} - z^n - 2\bar{z} + z^n + z^{n-1}}{2 - 2\cos(x)} - 1 \\
&= \frac{2 - z^{n+1} - 2\bar{z} + z^{n-1}}{2 - 2\cos(x)} - 1.
\end{aligned}$$

By taking the real parts we get:

$$\begin{aligned}
\left(1 + \cos(nx) + 2\sum_{k=1}^{n-1}\cos(kx)\right) &= \frac{2 - \cos((n+1)x) - 2\cos(x) + \cos((n-1)x) - 2 + 2\cos(x)}{2 - 2\cos(x)} \\
&= \frac{-\cos((n+1)x) + \cos((n-1)x)}{2 - 2\cos(x)} \\
&= \frac{2\sin(nx)\sin(x)}{2 - 2\cos(x)},
\end{aligned}$$

which is exactly what we want. $\square$

Remark 11.9. *We can also try to prove* (11.2) *by writing*

$$(1-\cos(x))\left(\sum_{k=1}^{n-1}\cos(kx)\right) = 2\sin^2(\frac{x}{2})\left(\sum_{k=1}^{n-1}\cos(kx)\right),$$

and use the fact that

$$\sum_{k=1}^{n-1}\cos(kx)\sin(\frac{x}{2})$$

is telescopic.

12. Week 15 (12.12)

Problem 12.1 (7.3)**.** *Use the definition of integral to calculate limitations.*

(1) $\lim\limits_{n\to\infty}\left(\dfrac{1}{n+1}+\dfrac{1}{n+2}+\cdots+\dfrac{1}{n+n}\right)$;

(2) $\lim\limits_{n\to\infty}\dfrac{\sqrt[n]{n(n+1)\cdots(2n-1)}}{n}$;

(3) $\lim\limits_{n\to\infty}\dfrac{2\pi}{n}\sum\limits_{k=1}^{n}\left(2+\sin\dfrac{2k\pi}{n}\right)$.

Solution. (1) By definition, we have

$$\begin{aligned}&\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{n+n}\right)\\=&\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+\frac{i}{n}}=\int_0^1\frac{1}{1+x}\,\mathrm{d}x\\=&\ln(1+x)\Big|_0^1=\ln 2.\end{aligned}$$

(2) Take logarithm. By definition, we have

$$\begin{aligned}&\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\ln\left(1+\frac{i-1}{n}\right)=\int_0^1\ln(1+x)\,\mathrm{d}x\\=&((1+x)\ln(1+x)-x)\Big|_0^1=2\ln 2-1.\end{aligned}$$

Hence

$$\lim_{n\to\infty}\frac{\sqrt[n]{n(n+1)\cdots(2n-1)}}{n}=e^{2\ln 2-1}=\frac{4}{e}.$$

(3) By definition, we have

$$\begin{aligned}&\lim_{n\to\infty}\frac{2\pi}{n}\sum_{k=1}^{n}\left(2+\sin\frac{2k\pi}{n}\right)\\=&\int_0^{2\pi}(2+\sin x)\,\mathrm{d}x\\=&(2x-\cos x)\Big|_0^{2\pi}=4\pi.\end{aligned}$$

□

Problem 12.2 (7.5)**.** *Suppose that $f\in R[a,b]$, $g(x)$ is defined on (a,b) and $g(x)$ is different from $f(x)$ at only a finite number of points on (a,b). Prove that $g(x)$ is integrable on (a,b), and there is*

$$\int_a^b g(x)\,\mathrm{d}x=\int_a^b f(x)\,\mathrm{d}x.$$

Proof. Denote $I = \int_a^b f(x)\,\mathrm{d}x$. By definition, we know that $\forall\,\varepsilon > 0$, there exists $\delta' > 0$ such that for any $\Delta' : a = x_0 < x_1 < \cdots < x_n = b$ with $\lambda(\Delta') < \delta'$ and any $\xi_i \in [x_{i-1}, x_i]\,(i = 1, 2, \cdots, n)$, there is

$$\left|\sum_{i=1}^{n} f(\xi_i)\Delta x_i - I\right| < \frac{\varepsilon}{2},$$

where $\lambda(\Delta') = \max\limits_{1\le i\le n}\{x_i - x_{i-1}\}$. Since $f \in R[a,b]$ and $g(x)$ is different from $f(x)$ at only a finite number of points on (a,b), we know that $f(x)$ and $g(x)$ are both bounded. Assume that $\exists\, M > 0$ such that $|f(x)| \le M$ and $|g(x)| \le M$. Then, if we choose $\delta < \min\{\delta', \frac{\varepsilon}{4kM}\}$, where k is the number of points where $g(x)$ is different from $f(x)$, we have for any $\Delta : a = x_0 < x_1 < \cdots < x_n = b$ with $\lambda(\Delta) < \delta$ and any $\xi_i \in [x_{i-1}, x_i]\,(i = 1, 2, \cdots, n)$ that

$$\begin{aligned}\left|\sum_{i=1}^{n} g(\xi_i)\Delta x_i - I\right| &\le \left|\sum_{i=1}^{n} f(\xi_i)\Delta x_i - I\right| + 2kM\lambda(\Delta)\\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Hence $g(x)$ is integrable on (a,b), and

$$\int_a^b g(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x.$$

□

Problem 12.3 (7.6)**.** *Suppose that $f(x)$ is defined on $[a,b]$. Prove that $f(x) \in R[a,b]$ if and only if there exists $I \in \mathbb{R}$, for $\forall\,\varepsilon > 0$, $\exists\,\Delta : a = x_0 < x_1 < \cdots < x_n = b$ and for any $\xi_i \in [x_{i-1}, x_i]\,(i = 1, 2, \cdots, n)$, there is*

$$\left|\sum_{i=1}^{n} f(\xi_i)\Delta x_i - I\right| < \varepsilon.$$

Proof. "$\Rightarrow$" By the definition of intagral, it's trival.

"$\Leftarrow$" Denote $w_i = \sup\limits_{x\in[x_{i-1},x_i]} f(x) - \inf\limits_{x\in[x_{i-1},x_i]} f(x)$. It suffices to prove that for some partition $\Delta : a = x_0 < x_1 < \cdots < x_n = b$, there is

$$\sum_{i=1}^{n} w_i\Delta x_i < \varepsilon.$$

By assumption, we know that for $\forall\,\varepsilon > 0$, $\exists\,\Delta : a = x_0 < x_1 < \cdots < x_n = b$ and for any $\xi_i \in [x_{i-1}, x_i]\,(i = 1, 2, \cdots, n)$, there is

$$\left|\sum_{i=1}^{n} f(\xi_i)\Delta x_i - I\right| < \frac{\varepsilon}{4}.$$

By the definition of superemum and infimum, we have that there exist $\xi_i, \eta_i \in [x_{i-1}, x_i]$ such that

$$f(\xi_i) + \frac{\varepsilon}{4(b-a)} > \sup_{x\in[x_{i-1},x_i]} f(x) \quad \text{and} \quad f(\eta_i) - \frac{\varepsilon}{4(b-a)} < \inf_{x\in[x_{i-1},x_i]} f(x).$$

Hence, there is

$$\begin{aligned}
\sum_{i=1}^{n} w_i \Delta x_i &= \sum_{i=1}^{n} \left(\sup_{x\in[x_{i-1},x_i]} f(x) - \inf_{x\in[x_{i-1},x_i]} f(x) \right) \Delta x_i \\
&\le \left| \sum_{i=1}^{n} (f(\xi_i) - f(\eta_i)) \Delta x_i \right| + \frac{\varepsilon}{2} \\
&\le \left| \sum_{i=1}^{n} f(\xi_i) \Delta x_i - I \right| + \left| \sum_{i=1}^{n} f(\eta_i) \Delta x_i - I \right| + \frac{\varepsilon}{2} \\
&< \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{2} = \varepsilon,
\end{aligned}$$

which gives us that $f(x)$ is integrable on $[a, b]$. □

Problem 12.4 (7.16)**.** *Suppose that $f(x)$ is defined on $\mathbb{R}$, and $f(x)$ is integrable on every finite closed interval. Prove that for any closed interval $[a, b]$, there is*

$$\lim_{h\to 0} \int_a^b |f(x+h) - f(x)| \, \mathrm{d}x = 0. \tag{12.1}$$

Proof. Since $f(x)$ is integrable on $[a-1, b+1]$, we know that for $\forall\, \varepsilon > 0$, there is $g(x) \in C[a-1, b+1]$ such that

$$\int_{a-1}^{b+1} |f(x) - g(x)| \, \mathrm{d}x < \frac{\varepsilon}{3}.$$

(Leave to the reader.) Since $g(x)$ is continuous on $[a-1, b+1]$, we have by the Cantor theorem that $g(x)$ is uniformly continuous on $[a-1, b+1]$. Then there exists $\delta :\ 0 < \delta < 1$, such that $\forall\, h \in (-\delta, \delta)$, there is

$$|g(x+h) - g(x)| < \frac{\varepsilon}{3(b-a)}, \quad \forall\, x \in [a, b].$$

Therefore, we have for $\forall\, h \in (-\delta, \delta)$ that

$$\begin{aligned}
\int_a^b |f(x+h) - f(x)| \, \mathrm{d}x &\le \int_a^b |f(x+h) - g(x+h)| \, \mathrm{d}x + \int_a^b |g(x+h) - g(x)| \, \mathrm{d}x \\
&\quad + \int_a^b |g(x) - f(x)| \, \mathrm{d}x \\
&= \int_{a+h}^{b+h} |f(x) - g(x)| \, \mathrm{d}x + \int_a^b |g(x+h) - g(x)| \, \mathrm{d}x
\end{aligned}$$

$$
\begin{aligned}
&+\int_a^b |g(x)-f(x)|\,\mathrm{d}x\\
<&\frac{\varepsilon}{3}+\frac{\varepsilon}{3(b-a)}(b-a)+\frac{\varepsilon}{3}\\
=&\varepsilon,
\end{aligned}
$$

i.e.

$$
\lim_{h\to 0}\int_a^b |f(x+h)-f(x)|\,\mathrm{d}x=0.
$$

□

Remark 12.5. (12.1) *is called the absolute continuity of integrals.*

Problem 12.6 (7.18)**.** *Suppose that* $f(x)$, $g(x)\in R[a,b]$. *Prove the Cauchy-Schwarz inequality:*

$$
\left|\int_a^b f(x)g(x)\mathrm{d}x\right|\le\left[\int_a^b f^2(x)\mathrm{d}x\right]^{\frac{1}{2}}\left[\int_a^b g^2(x)\mathrm{d}x\right]^{\frac{1}{2}}. \tag{12.2}
$$

Proof. If $\int_a^b f(x)g(x)\mathrm{d}x=0$, (12.2) is clearly. Now, we assume that $\int_a^b f(x)g(x)\mathrm{d}x\neq 0$. It's obvious that $\int_a^b f^2(x)\mathrm{d}x\neq 0$ and $\int_a^b g^2(x)\mathrm{d}x\neq 0$ (Leave to the reader). Note that

$$
0\le\int_a^b (f(x)-tg(x))^2\,\mathrm{d}x=\int_a^b f^2(x)\mathrm{d}x-2t\int_a^b f(x)g(x)\mathrm{d}x+t^2\int_a^b g^2(x)\mathrm{d}x,\quad \forall\, t\in\mathbb{R}.
$$

Then there is

$$
4\left[\int_a^b f(x)g(x)\mathrm{d}x\right]^2-4\left[\int_a^b f^2(x)\mathrm{d}x\right]\left[\int_a^b g^2(x)\mathrm{d}x\right]\le 0,
$$

i.e.

$$
\left|\int_a^b f(x)g(x)\mathrm{d}x\right|\le\left[\int_a^b f^2(x)\mathrm{d}x\right]^{\frac{1}{2}}\left[\int_a^b g^2(x)\mathrm{d}x\right]^{\frac{1}{2}}.
$$

□

Problem 12.7 (7.19)**.** *Prove the following limitations.*

(1) $\lim\limits_{n\to\infty}\int_{-1}^1(1-x^2)^n\mathrm{d}x=0$;

(2) *Suppose that* $f(x)\in C[-1,1]$, *then*

$$
\lim_{n\to\infty}\frac{\int_{-1}^1 f(x)(1-x^2)^n\mathrm{d}x}{\int_{-1}^1(1-x^2)^n\mathrm{d}x}=f(0).
$$

Proof. (1) For any $\delta > 0$, we have

$$\begin{aligned}\int_{-1}^{1}(1-x^2)^n\mathrm{d}x &= \int_{|x|\le\delta}(1-x^2)^n\mathrm{d}x + \int_{\delta<|x|\le 1}(1-x^2)^n\mathrm{d}x \\ &\le 2\delta + 2(1-\delta^2)^n \\ &\to 2\delta \quad \text{as } n\to\infty.\end{aligned}$$

Since δ is arbitrary, we know that $\lim\limits_{n\to\infty}\int_{-1}^{1}(1-x^2)^n\mathrm{d}x = 0$.

(2) Since $f(x)\in C[-1,1]$, we have that $\forall\,\varepsilon>0$, there exists $\delta>0$, such that $\forall\,x$: $|x|<\delta$, there is

$$|f(x)-f(0)|\le\varepsilon.$$

Then

$$\begin{aligned}\left|\frac{\int_{-1}^{1}f(x)(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x} - f(0)\right| &= \left|\frac{\int_{-1}^{1}(f(x)-f(0))(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x}\right| \\ &\le \frac{\int_{|x|<\delta}|f(x)-f(0)|\,(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x} \\ &\quad + \frac{\int_{\delta\le|x|<1}|f(x)-f(0)|\,(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x} \\ &\le \varepsilon + 2M\cdot\frac{\int_{\delta\le|x|<1}(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x} \\ &\le \varepsilon + 2M\cdot\frac{\int_{\delta\le|x|<1}(1-x^2)^n\mathrm{d}x}{\int_{|x|\le\frac{\delta}{2}}(1-x^2)^n\mathrm{d}x} \\ &\le \varepsilon + 2M\frac{(1-\delta^2)^n}{(1-(\delta/2)^2)^n\delta} \\ &\to \varepsilon \quad \text{as } n\to\infty,\end{aligned}$$

where $M=\max\limits_{x\in[-1,1]}f(x)$. Since ε is arbitrary, we have

$$\lim_{n\to\infty}\frac{\int_{-1}^{1}f(x)(1-x^2)^n\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^n\mathrm{d}x} = f(0).$$

□

Exercise 12.8 (7.21(1))**.** *Suppose that $f(x)$ has continuous derivative on $[a,b]$. Prove that for any $x\in[a,b]$, there is*

$$|f(x)|\le\left|\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x\right| + \int_a^b|f'(x)|\mathrm{d}x.$$

Hint: By the mean value theorem for definite integrals, we know that there is $\xi \in (a, b)$ such that

$$f(\xi) = \frac{1}{b-a}\int_a^b f(x)\mathrm{d}x.$$

By the Newton-Leibniz formula, we have

$$|f(x) - f(\xi)| = \left|\int_\xi^x f'(t)\mathrm{d}t\right| \leq \int_a^b |f'(x)|\mathrm{d}x.$$

Hence

$$|f(x)| \leq |f(\xi)| + |f(x) - f(\xi)| \leq \left|\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x\right| + \int_a^b |f'(x)|\mathrm{d}x.$$

□

Problem 12.9 (7.22)**.** *Suppose that $f(x) \in C(\mathbb{R})$ and $f'(0)$ exists. Assume that $\forall\, x \in \mathbb{R}$, there is*

$$\int_0^x f(t)\mathrm{d}x = \frac{1}{2}xf(x). \tag{12.3}$$

Prove that $f(x) \equiv cx$, where $c = f'(0)$.

Proof. Denote $F(x) = \int_0^x f(t)\mathrm{d}t$. Since $f(x) \in C(\mathbb{R})$, we have that $F(x) \in C^1(\mathbb{R})$. Then by (12.3), we have $f(0) = 0$. Define

$$g(x) = \begin{cases} \dfrac{F(x)}{x^2}, & x \neq 0, \\ \dfrac{1}{2}f'(0), & x = 0. \end{cases}$$

Then by (12.3), we have

$$g'(x) = \frac{x^2 f(x) - 2xF(x)}{x^4} = 0, \quad \forall\, x \neq 0.$$

By definition, we know that

$$g'(0) = \lim_{x\to 0}\frac{g(x) - g(0)}{x} = \lim_{x\to 0}\frac{g'(\xi_x)}{x} = 0.$$

Hence $F(x) = \frac{f'(0)}{2}x^2$, $\forall\, x \in \mathbb{R}$. Then $f(x) = f'(0)x$, $\forall\, x \in \mathbb{R}$. □

Problem 12.10 (7.23)**.** *Suppose that $P_n(x)$ is a polynomial with degree $n \geq 1$, and $[a, b]$ is an closed interval. Prove that*

$$\int_a^b |P_n'(x)|\mathrm{d}x \leq 2n \max_{a\leq x\leq b}\{|P_n(x)|\}.$$

Proof. Since $P_n'(x)$ is a polynomial with degree $n-1$, we may assume that there are $a \le x_1 \le \cdots \le x_k \le b$ with $k \le n-1$, where $x_1, \cdots, x_{n-1}$ are zero points of $P_n'(x)$. Hence, we have

$$\begin{aligned}\int_a^b |P_n'(x)|\mathrm{d}x &= \left|\int_a^{x_1} P_n'(x)\mathrm{d}x\right| + \cdots + \left|\int_{x_k}^b P_n'(x)\mathrm{d}x\right| \\ &= |P_n(x_1) - P_n(a)| + \cdots + |P_n(b) - P_n(x_k)| \\ &\le 2(k+1)\max_{a\le x\le b}\{|P_n(x)|\} \\ &\le 2n\max_{a\le x\le b}\{|P_n(x)|\}.\end{aligned}$$

□

Problem 12.11 (7.28)**.** *Suppose that $f(x)$ is a periodic function with period 2π and $f(x) \in R[0, 2\pi]$. Prove that*

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T f(x)\mathrm{d}x = \frac{1}{2\pi}\int_0^{2\pi} f(x)\mathrm{d}x.$$

Proof. For any $T > 0$, we know that there exists $n \in \mathbb{N}$ such that $T = 2n\pi + r$, where $0 \le r < 2\pi$. Then

$$\begin{aligned}\int_0^T f(x)\mathrm{d}x &= \int_0^{2n\pi+r} f(x)\mathrm{d}x \\ &= \int_0^{2\pi} f(x)\mathrm{d}x + \cdots + \int_{2(n-1)\pi}^{2n\pi} f(x)\mathrm{d}x + \int_{2n\pi}^{2n\pi+r} f(x)\mathrm{d}x \\ &= n\int_0^{2\pi} f(x)\mathrm{d}x + \int_0^r f(x)\mathrm{d}x.\end{aligned}$$

Hence, we have

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T f(x)\mathrm{d}x = \lim_{T\to\infty}\frac{n}{2n\pi + r}\int_0^{2\pi} f(x)\mathrm{d}x + \lim_{T\to\infty}\frac{1}{T}\int_0^r f(x)\mathrm{d}x = \frac{1}{2\pi}\int_0^{2\pi} f(x)\mathrm{d}x.$$

□

Problem 12.12 (7.35)**.** *Suppose that $f(x) \in C[a, b]$ is nonnegative. Denote that $M = \sup_{a\le x\le b} f(x)$. Prove that*

$$\lim_{n\to\infty}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} = M.$$

Proof. Firstly, it is easy to see that

$$\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \le M(b-a)^{\frac{1}{n}}.$$

Then

$$\overline{\lim_{n\to\infty}}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \le M.$$

Secondly, since $f(x)$ is continuous, we know that there is $x_0 \in [a,b]$ such that $f(x_0) = M$. Then $\forall\, \varepsilon > 0$, there exists $\delta > 0$ such that $\forall\, x : |x - x_0| < \delta$, there is $|f(x) - f(x_0)| < \varepsilon$, thus $f(x) > M - \varepsilon$. Hence, we have

$$\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \ge \left[\int_{x_0-\delta}^{x_0+\delta} f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \ge (M-\varepsilon)(2\delta)^{\frac{1}{n}}.$$

Therefore, there is

$$\underline{\lim_{n\to\infty}}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \ge M - \varepsilon.$$

Since ε is arbitrary, we have

$$\underline{\lim_{n\to\infty}}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} \ge M \ge \overline{\lim_{n\to\infty}}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}},$$

which implies

$$\lim_{n\to\infty}\left[\int_a^b f^n(x)\mathrm{d}x\right]^{\frac{1}{n}} = M.$$

□

Exercise 12.13 (7.37)**.** *Suppose that $f(x)$ is monotonic on (a,b), $g(x)$ is a periodic function with period $T > 0$ on $\mathbb{R}$, and $\int_0^T g(x) = 0$. Prove that*

$$\lim_{\lambda\to\infty}\int_a^b f(x)g(\lambda x)\mathrm{d}x = 0.$$

Hint: By the second mean value theorem for definite integrals, we have that there exists $\xi \in (a,b)$ such that

$$\int_a^b f(x)g(\lambda x)\mathrm{d}x = f(a)\int_a^\xi g(\lambda x)\mathrm{d}x + f(b)\int_\xi^b g(\lambda x)\mathrm{d}x.$$

Note that

$$\int_a^\xi g(\lambda x)\mathrm{d}x = \frac{1}{\lambda}\int_{a\lambda}^{\xi\lambda} g(x)\mathrm{d}x = \frac{\xi}{\xi\lambda}\int_0^{\xi\lambda} g(x)\mathrm{d}x - \frac{a}{a\lambda}\int_0^{a\lambda} g(x)\mathrm{d}x.$$

Hence by Problem 12.11 and $\int_0^T g(x) = 0$, we have

$$\lim_{\lambda\to\infty}\int_a^\xi g(\lambda x)\mathrm{d}x = 0.$$

Similarly, we have

$$\lim_{\lambda\to\infty}\int_{\xi}^{b} g(\lambda x)\mathrm{d}x = 0.$$

Then we obtain that

$$\lim_{\lambda\to\infty}\int_{a}^{b} f(x)g(\lambda x)\mathrm{d}x = 0.$$

□

Remark 12.14. *In fact, we can remove the monotonicity of $f(x)$. It is the Riemann-Lebesgue lemma, which can be proved by real analysis' method, we leave it to the reader.*

Exercise 12.15. *Suppose that $f \in C[-1,1]$, Prove that*

$$\lim_{h\to 0^+}\int_{-1}^{1}\frac{h}{h^2+x^2}f(x)\,dx = \pi f(0).$$

Hint: Note that

$$\begin{aligned}
&\int_{-1}^{1}\frac{h}{h^2+x^2}f(x)\,dx - 2\arctan\frac{1}{h}f(0)\\
=&\int_{-1}^{1}\frac{h}{h^2+x^2}(f(x)-f(0))\,dx\\
=&\int_{|x|<\delta}\frac{h}{h^2+x^2}(f(x)-f(0))\,dx + \int_{1\geq|x|\geq\delta}\frac{h}{h^2+x^2}(f(x)-f(0))\,dx,
\end{aligned}$$

where $\delta > 0$ is such that $|f(x)-f(0)| < \varepsilon$ whence $|x| < \delta$ (Since $f(x)$ is continous). Then, we have

$$\begin{aligned}
&\left|\int_{-1}^{1}\frac{h}{h^2+x^2}f(x)\,dx - 2\arctan\frac{1}{h}f(0)\right|\\
\leq&\int_{|x|<\delta}\frac{h}{h^2+x^2}|f(x)-f(0)|\,dx + \int_{1\geq|x|\geq\delta}\frac{h}{h^2+x^2}|f(x)-f(0))|\,dx\\
\leq&\,\varepsilon\int_{-1}^{1}\frac{h}{h^2+x^2}\,dx + 2\sup|f|\int_{1\geq|x|\geq\delta}\frac{h}{h^2+x^2}\,dx\\
=&\,2\varepsilon\arctan\frac{1}{h} + 4\sup|f|\left(\arctan\frac{1}{h} - \arctan\frac{\delta}{h}\right)\\
\leq&\,\pi\varepsilon,\quad as\quad h\to 0.
\end{aligned}$$

Since ε is arbitrary, we know

$$\lim_{h\to 0}\int_{-1}^{1}\frac{h}{h^2+x^2}f(x)\,dx = \lim_{h\to 0}2\arctan\frac{1}{h}f(0) = \pi f(0).$$

□

Exercise 12.16. *Suppose that $f \in C[-1,1]$, given $\int_{-1}^{1} f(x)x^n dx = 0$ for $n = 0, 1, 2, \ldots$ then $f(x) = 0$, $\forall x \in [-1,1]$.*

Hint: Since

$$\limsup_{n\to\infty} |f(x) - p_n(x)| = 0,$$

we know for any $\epsilon > 0$, there is $N \in \mathbb{N}$ such that for $\forall n > N$

$$|f - p_n| < \frac{\epsilon}{2M},$$

where $M := \max_{[-1,1]} |f(x)|$.

Hence

$$\left| \int_{-1}^{1} f(x) \left(f(x) - p_n(x) \right) \, dx \right| < \frac{\epsilon}{2M} \int_{-1}^{1} |f(x)| \, dx \leq \epsilon, \quad \forall n > N.$$

□

Exercise 12.17 (Challenge!)**.** *Assume that $f(x) \in C[0, +\infty)$, and for all $a \geq 0$, we have*

$$\lim_{x\to\infty} (f(x+a) - f(x)) = 0. \tag{12.4}$$

Prove that there exists $g(x) \in C[0, +\infty)$ and $h(x) \in C^1[0, +\infty)$ such that $f(x) = g(x) + h(x)$, and such that they satisfy

$$\lim_{x\to\infty} g(x) = 0, \ \lim_{x\to\infty} h'(x) = 0.$$

Hint: By Exercise 4.23, we first know that $f(x)$ is uniformly continuous. Let's chose $a = 1$ and set $h(x) = \int_x^{x+1} f(t)\, dt$. We begin by writing

$$h(x) - f(x) = \int_0^1 (f(x+t) - f(x)) \, dt.$$

The integrand converges to 0 pointwise (from condition (12.4)), but this is not quite sufficient! We'll have to be a bit more careful and also use the uniform continuity of f. Let $\epsilon > 0$. Because f is uniformly continuous, there exists an integer $n > 0$ such that for all $x, y \geq 0$ with $|x - y| \leq \frac{1}{n}$, we have $|f(x) - f(y)| \leq \epsilon$.

Now we use condition (12.4) to get that for $1 \leq k \leq n$, there exists x_k such that for all $x \geq x_k$,

$$\left| f\left(x + \frac{k}{n}\right) - f(x) \right| \leq \varepsilon.$$

We set $x_0 = \max_{1\leq k\leq n} (x_k)$. Now, for $x \geq x_0$ we have

$$|h(x) - f(x)| = \left| \int_0^1 (f(x+t) - f(x)) \, dt \right|$$

$$
\begin{aligned}
&\leq \int_0^1 |f(x+t) - f(x)| \, dt \\
&\leq \sum_{k=1}^{n} \int_{\frac{k-1}{n}}^{\frac{k}{n}} \underbrace{\left| f(x+t) - f\left(x + \frac{k}{n}\right) \right|}_{\leq \varepsilon \quad \text{(from continuity)}} + \underbrace{\left| f\left(x + \frac{k}{n}\right) - f(x) \right|}_{\leq \varepsilon \quad \text{(from (12.4))}} \, dt \\
&\leq 2\varepsilon.
\end{aligned}
$$

□

School of Mathematical Sciences, Peking University, Beijing 100871, China.
Email address: lingwang@stu.pku.edu.cn